U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

[Effective from the Session 2009-10]

TEXTILE TECHNOLOGY

YEAR 2nd, SEMESTER- III

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	ETCEF301	Natural Textile Fibre	2	1	0	15	10	25	50	75	3
2	EAS301/ EOE031-EOE038	Mathematics-III /Science Based Open Elective	3	1	0	30	20	50	100	150	4
3	EHU301/ EHU302	Industrial Psychology/ Industrial Sociology	2	0	0	15	10	25	50	75	2
4	ETE302	Yarn Manufacture-I	3	1	0	30	20	50	100	150	4
5	ETE303	Fabric Manufacture-I	3	1	0	30	20	50	100	150	4
6	ETEE304	Wet Processing of Textiles-I	3	1	0	30	20	50	100	150	4
7	EHU-111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF301P	Identification of Fibres-I	0	0	3	0	20	20	30	50	1
9	ETE302P	Yarn Manufacture-I	0	0	3	0	20	20	30	50	1
10	ETE303P	Fabric Manufacture-I	0	0	3	0	20	20	30	50	1
11	ETEF304P	Wet Processing of Textiles-I	0	0	3	0	20	20	30	50	1
12	GP301	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								**1000**	**26**

*** Audit course**

Paper Code	**Science Based Open Electives:**
EOE031/EOE041	Introduction to Soft Computing (Neural Network, Fuzzy Logic and Genetic Algorithm
EOE032/EOE042	Nano Sciences
EOE033/EOE043	Laser Systems and Applications
EOE034/EoE044	Space Sciences
EOE035/EOE045	Polymer Science & Technology
EOE036/EOE046	Nuclear Science
EOE037/EOE047	Material Science
EOE038/EOE048	Discrete Mathematics

U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

TEXTILE TECHNOLOGY

[Effective from the Session 2009-10]

YEAR 2nd, SEMESTER- IV

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU402/ EHU401	Industrial Sociology/ Psychology	2	0	0	15	10	25	50	75	2
2	EOE041-EOE048/ EAS401	Science Based Open Elective/m Mathematics-III	3	1	0	30	20	50	100	150	4
3	ETE401	Yarn Manufacture-II	3	1	0	30	20	50	100	150	4
4	ETE402	Fabric Manufacture-II	3	1	0	30	20	50	100	150	4
5	ETEF403	Wet Processing of Textile II	3	1	0	30	20	50	100	150	4
6	ETEF 404	Colour & Design	2	1	0	15	10	25	50	75	3
7	EHE-111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETE 401P	Identification of Fibres-II	0	0	3	0	20	20	30	50	1
9	ETE 402P	Yarn Manufacture-II	0	0	3	0	20	20	30	50	1
10	ETE 403P	Fabric Manufacture-II	0	0	3	0	20	20	30	50	1
11	ETEF 404P	Wet Processing of Textiles-II	0	0	3	0	20	20	30	50	1
12	GP401	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								**1000**	**26**

Paper Code	**Science Based Open Electives:**
EOE031/EOE041	Introduction to Soft Computing (Neural Network, Fuzzy Logic and Genetic Algorithm
EOE032/EOE042	Nano Sciences
EOE033/EOE043	Laser Systems and Applications
EOE034/EoE044	Space Sciences
EOE035/EOE045	Polymer Science & Technology
EOE036/EOE046	Nuclear Science
EOE037/EOE047	Material Science
EOE038/EOE048	Discrete Mathematics

U.P TECHNICAL UNIVERSITY, LUCKNOW
STUDY & EVALUATION SCHEME
TEXTILE TECHNOLOGY
[Effective from the Session 2010-11]

YEAR 3rd, SEMESTER- V

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU501	Engineering & Managerial Economics	3	1	0	30	20	50	100	150	3
2	ETCEF501	Textile Testing-I	3	1	0	30	20	50	100	150	4
3	ETE502	Yarn Manufacture-III	3	1	0	30	20	50	100	150	4
4	ETE503	Fabric Manufacture-III	2	1	0	15	10	25	50	75	3
5	ETEF504	Fabric Structure	2	1	0	15	10	25	50	75	3
6	ETCEF505	Chemistry & Production of Fibres	3	1	0	30	20	50	100	150	4
7	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF501P	Textile Testing-I	0	0	3	-	20	20	30	50	1
9	ETE502P	Yarn Manufacture-III	0	0	3	-	20	20	30	50	1
10	ETE503P	Fabric Manufacture-III	0	0	3	-	20	20	30	50	1
11	ETEF504P	Fabric Analysis	0	0	3	-	20	20	30	50	1
12	GP501	General Proficiency	-	-	-	-	-	50	-	50	1
										1000	**26**

*** Audit course**

U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

TEXTILE TECHNOLOGY

[Effective from the Session 2010-11]

YEAR 3rd, SEMESTER- VI

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU 601	Industrial Management	3	0	0	30	20	50	100	150	3
2	ETEF011/ ETCE012	Departmental Elective-I	3	1	0	30	20	50	100	150	4
3	ETCEF601	Textile Testing-II	2	1	0	15	10	25	50	75	3
4	ETCEF602	Structure & Properties of Fibres	3	1	0	30	20	50	100	150	4
5	ETE603	Yarn Manufacture-IV	2	1	0	15	10	25	50	75	3
6	ETE604	Fabric Manufacture-IV	3	1	0	30	20	50	100	150	4
7	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF601P	Textile Testing-II	0	0	3	0	20	20	30	50	1
9	ETE602P	Yarn Manufacture-IV	0	0	3	0	20	20	30	50	1
10	ETE603P	Fabric Manufacture-IV	0	0	3	0	20	20	30	50	1
11	ETE604P	Yarn & Fabric Production	0	0	3	0	20	20	30	50	1
12	GP601	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1000**	**26**

*** Audit course**

Department Elective-1

Code	**Name of Subject**
ETEF011	Coating of Textiles
ETCE012	Post Spinning Operation

U.P TECHNICAL UNIVERSITY, LUCKNOW
STUDY & EVALUATION SCHEME
TEXTILE TECHNOLOGY
[Effective from the Session 2011-12]

YEAR 4th, SEMESTER- VII

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EOE071-074	Open Elective-I	3	1	0	30	20	50	100	150	4
2	ETCEF021/ ETCEF022	Departmental Elective-II	3	1	0	30	20	50	100	150	4
3	ETEF701	Mill Planning & Organization	3	1	0	30	20	50	100	150	4
4	ET702	Advance Spinning Technology	3	1	0	30	20	50	100	150	4
5	ETEF703	Knitting Technology	3	1	0	30	20	50	100	150	4
6	EHU111	**Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
7	ETEF703P	Knitting Technology	0	0	3	-	20	20	30	50	1
8	ETT751	Project	0	0	3	-	50	50	-	50	2
9	ETT752	Seminar	0	0	2	-	50	50	-	50	1
10	ETT753	Industrial Training Viva Voice	0	0	2	-	50	50	-	50	1
11	GP701	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**15**	**3**	**10**	**150**	**270**	**470**	**530**	**1000**	**26**

Paper Code	**Open Elective-I**
EOE071	Entrepreneurship Development
EOE072	Quality management
EOE073	Operation Research
EOE074	Introduction to Biotechnology
Paper Code	**Departmental Elective-II**
ETCEF021	Garment Manufacturing Technology
ETCEF022	Nonwoven

U.P TECHNICAL UNIVERSITY, LUCKNOW
STUDY & EVALUATION SCHEME
TEXTILE TECHNOLOGY
[Effective from the Session 2011-12]

YEAR 4th, SEMESTER- VIII

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EOE081-084	Open Elective-II	3	1	0	30	20	50	100	150	4
2	ETEF031/ ETEF032	Departmental Elective-III	3	1	0	30	20	50	100	150	4
3	ETEF041/ ETEF042	Departmental Elective-IV	3	1	0	30	20	50	100	150	4
4	ET801	Advance Weaving Technology	3	0	0	30	20	50	100	150	3
5	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
6	ETT851	Project	0	0	12	-	100	100	250	350	8
7	GP801	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**12**	**3**	**12**	**120**	**180**	**350**	**650**	**1000**	**24**

Paper Code	****Open Elective-II**
EOE081	Non Conventional Energy Resources
EOE082	Nonlinear Dynamic Systems
EOE083	Product Development
EOE084	Automation and Robotics

Paper Code	**Departmental Elective-III**
ETCEF031	Technical Textiles
ETEF032	Theory of Textile Structure

Paper Code	**Departmental Elective-IV**
ETEF041	Process Control in Spinning
ETEF042	Process Control in Weaving

3TH SEMESTER, B. TECH. TEXTILE TECHNOLOGY

EAS-301/EAS-401: MATHEMATICS –III

L	T	P
3	1	0

Unit – I : Function of Complex variable

Analytic function, C-R equations, Cauchy's integral theorem, Cauchy's integral formula for derivatives of analytic function, Taylor's and Laurent's series, singularities, Residue theorem, Evaluation of real integrals of the type $\int_0^{2\pi} f(\cos\theta, \sin\theta)\, d\theta$ and $\int_{-\infty}^{\infty} f(x)\, dx$ **10**

Unit – II : Statistical Techniques - I

Moments, Moment generating functions, Skewness, Kurtosis, Curve fitting, Method of least squares, Fitting of straight lines, Polynomials, Exponential curves etc., Correlation, Linear, non –linear and multiple regression analysis, Probability theory. **08**

Unit – III : Statistical Techniques - II

Binomial, Poisson and Normal distributions, Sampling theory (small and large), Tests of significations: Chi-square test, t-test, Analysis of variance (one way) , Application to engineering, medicine, agriculture etc.

Time series and forecasting (moving and semi-averages), Statistical quality control methods, Control charts, $\bar{X}$, R, p, np, and c charts. **08**

Unit – IV : Numerical Techniques – I

Zeroes of transcendental and polynomial equation using Bisection method, Regula-falsi method and Newton-Raphson method, Rate of convergence of above methods.

Interpolation: Finite differences, difference tables, Newton's forward and backward interpolation , Lagrange's and Newton's divided difference formula for unequal intervals. **08**

Unit – V : Numerical Techniques –II

Solution of system of linear equations, Gauss- Seidal method, Crout method. Numerical differentiation, Numerical integration , Trapezoidal , Simpson's one third and three-eight rules, Solution of ordinary differential (first order, second order and simultaneous) equations by Euler's, Picard's and forth-order Runge- Kutta mehthods. **08**

Test Books :-

1. Peter V. O'Neil, Advance Engineering Mathematics Thomson (Cengage) Learning, 2007.
2. Jain, Iyenger & Jain, Numerical Methods for Scientific and Engineering Computation, New Age International, New Delhi , 2003.
3. J.N. Kapur, Mathematical Statistics, S. Chand & company Ltd.,2000

Reference Books :-

1. R.K. Jain & S.R.K. Iyenger, Advance Engineering Mathematics, Narosa Publication House, 2002.
2. Chandrika Prasad, Advanced Mathematics for Engineers, Prasad Mudralaya, 1996.
3. E. Kreysig, Advanced Engineering Mathematics, John Wiley & Sons, 2005.
4. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 2005.

5. Devi Prasad, An introduction to Numerical Analysis, Narosa Publication house, New Delhi 2006.
6. T. Veerajan & T. Ramchandrandran, Theory & Problems in Numerical Methods, TMH, New Delhi, 2004.
7. S.P.Gupta, Statistical Methods, Sultan and Sons, New Delhi, 2004.
8. Devore, Probability and Statistics, Thomson(Cengage) Learning, 2007.
9. Walpole, Myers, Myers & Ye, Probability and Statistics for Engineers & Scientists, Pearson Education, 2003.

1. (Engineering Core: Interdisciplinary) Natural Textile Fibres: (ETCFE-301)

Unit (1) : Fibre, textile fibre (1). Classification of natural textile fibres (1), Essential and desirable properties of textile fibres (2), Grading of cotton (1), Physical properties of cotton fibre (1), Effect of acid & Alkalies on cotton fibre (1).
Total Lectures required = 07

Unit (2): Production, morphological structure, properties and end uses of: Flax (2), Jute (2), Ramie (2). Introduction to specialty natural fibres like banana, pineapple, spider etc
Total Lectures required = 08

Unit (3): Production of silk (raw) (2), Morphological structure of silk (1), Production of waste silk yarn (2), Chemical composition and physical properties of silk (1), Effect of acid and Alkalies (1), Various varieties of silk with brief description (1).
Total Lectures required =08

Unit (4): Morphological structure of wool (2), Composition of wool fibres (1), Properties of wool fibres (1), Varieties of wool fibre with brief description (2).
Total Lecture required = 06
Grand total of Lectures = 36

Books:

1. W.E. Morton & J.W.S. Hearle, Physical properties of textile fibres, Textile Institute, U.K.
2. Progress in textiles: Science and technology Vol.-2
   By Dr. V.K. Kothari, I.I.T. Delhi.
3. Hand book of textile fibres by J.Gordon Cook

2. Mathematics III/ Science Based Open Elective (EAS301/ EOE031-EOE038)

Syllabus to be decided by U.P.T.U.

3. Industrial Psychology/ Industrial Sociology (EHU301/EHU302)
Syllabus to be decided by UPTU

4. Yarn Manufacture –I (ETE-302)

Unit (1): Picking and pre-cleaning of cotton (3), objects of Ginning (1), Description & working of knife-roller (2), Macarthy and Saw Gins (1).

Total Lectures required = 07

Unit (2): Objects of mixing (1), different types of mixing & blending (2), Difference between mixing & blending (1), Principles underlying the relation of cotton for mixing (1), Different mixing methods (3), Their advantages and disadvantages.
Total Lectures required = 08

Unit (3): Objects of Blow room (1), Principles of opening and cleaning (2), Various types of openers, their construction and working (2), Selection of machinery according to type of cotton and their suitable combinations (1), Blow room line for synthetic fibres (2), Blending of man made fibres with cotton, (1), Modern blow room lines (1).
Total Lectures required = 10

Unit (4): Lap feed (1), Chute feed systems and their comparison (2), Lap rejection (2), Causes of lap defects and their remedies (1), Transport control and regulation of material blow room (1).
Total Lectures required = 07

Unit (5): Problem of contamination and their remedies (2), Performance assessment of blow room (2), Waste extracted at various openers and beaters (2), Cleaning efficiency of different machines (1), lint trash ratio, nep generation, fibre breakage and their control (1), Calculations pertaining to above subject matter (2).
Total Lectures required = 10
Grand Total Lectures required =42

Reference Book:

1. Spun yarn technology Vol. I – A. Ventaksubramaniam
2. Elements of Raw Cotton & Blowroom – Dr. A.R. Khare
3. A practical guide to Opening – W. Klein
4. Cotton Spinning – Taggart
5. Spun Yarn Technology – Oxtoby.
6. Spinning blow room and carding by Prof. K.R. Salhotra & Dr. R. Chattdadhya

5. Fabric Manufacture-I (ETE-303)

Unit (1) : Objects of winding, classification of winding machines (2), Study of slow, high and super high speed warp winding machines (2), Features of slow, high & super high speed winding machines (2), Machine traverse ratio, Angle of wind, conicity of warp and weft packages, Types of wound packages (2).
Total Lectures required = 08

Unit (2): Thread stop motion, tensioning devices (1), Mechanical and electronic slub-catchers (1), Clearing efficiency of winding m/c, Knot factor, various knots used in textiles (2), Latest developments in winding, Splicing (2).
Total Lectures required = 06

Unit (3): Ways of yarn traversing, ribbon formation and method of its elimination (2), Package density and winding parameters and knotting (2), Winding falls and their remedies (2).
Total Lectures required = 06

Unit (4): High speed weft winding machines (1), Detailed study of mechanism and setting of speed on weft winders (2), Automatic and non-automatic pirn winding machines (2), Bunch motion and its importance (1).
Total Lectures required =06

Unit (5): Yarn numbering systems (1), Resultant count method for various yarns (2), Estimation of productivity of pirn winding machines (2), brief idea of classmate test report (2).
Total Lectures required = 07
Grand total lectures required =33

Reference Books:

1. R. Sengupta, yarn preparation, the popular book depot Mumbai.
2. Cotton yarn weaving (ATIRA), 1980
3. M.K. Talukdar- An introduction to winding and warping.
4. P.K. Bannerjee- Yarn winding NCUTE, IIT Delhi.
5. Armored weaving technology and operation, Textile Institute U.K.

6. Wet Processing of Textiles -I (ETEF-304)

UNIT-1: Introduction to simple preparatory process, grey cloth singeing, hot plate, roller and gas singeing m/c. Desizing – its object and various types of desizing methods. Scouring – its objects and process involved. Various types of kier -its description and working. (8)

UNIT-2: Bleaching-its objects and chemistry involved, various types of bleaching agents – their preparation and uses ,bleaching of cotton fabrics, various faults in bleaching and their remedies and removal.(8)

UNIT-3: Merecrisation objects, physical and chemical aspects of mercerization , yarn mercerization and various methods of cloth mercerization ,various methods to test the efficiency of mercerization. (7)

UNIT-4: Finishing- objects, various types of physical and chemical finishes and their applications by various finishing equipments like padding mangle ,drying m/c's ,stenter ,calendar etc.(8)

UNIT-5: Introduction to various finishes like water proofing and water repellency, flame proofing and flame retardancy ,heat setting ,softening and stiffening. (7)

Grand Total Lectures Required =38

Books:

1. Chemical processing of cotton and p/c blends – ATIRA
2. A glimps on the chemical technology and textile fibres by R.R. Chackrawartty
3. Technology of bleaching and mercerization by V.A. Shenai
4. Technology of finishing by V.A. Shenai

7 *Human Values & Professional Ethics (EHU-111): Audit Course
Syllabus as provided by U.P.T.U.

U.P. TEXTILE TECHNOLOGY INSTITUTE, KANPUR
DETAILES SYALLUBUS

4TH SEMESTER, B. TECH. TEXTILE TECHNOLOGY

1. Industrial Psychology/ Industrial Sociology (EHU402/EHU401)
Syllabus to be decided by UPTU

2. Science Based Open Elective/ Mathematics III (EOE041-EOE048/EAS401)

Syllabus to be decided by U.P.T.U.

3. Yarn Manufacture –II (ETE-401)

Unit (1): Objects of carding, detailed study of revolving flat card, stripping and brushing action, Constructional features and working, details of licker-in, cylinder Doffer and flat for processing different materials.
Total Lecture required = 08

Unit (2): Flexible of metallic card clothing, grinding, processing parameters for different materials carding defects and their remedies. Fibre neps – assesment and control, carding calculations pertaining to draft and production.
Total Lectures required = 08

Unit (3): Tandem carding, Modern development in carding, Auto leveller used in carding, basic idea about woolen and jute flex carding.
Total Lectures required = 07

Unit (4): Object of draw frame, working principles of draw frame including constructional details, System of drafting, weighting in draw frame, Draft distribution, Drafting weaves, Drafting force, Coiling, over coiling, under coiling and bi-coiling and stop motion.
Total Lectures required = 08

Unit (5): Concept of ideal draft and formation of draft waves, Drafting irregularities and their causes, Principles of roller setting, Modern developments in draw frame, Auto leveling calculation pertaining to draft and production.
Total Lectures required = 08
Grand Total Lectures required = 39

Reference Books:

1. Advance technology of yarn production by R. Chattopadhadya
2. Mechanics of spinning m/c by R.S.Rangaswami
3. New spinning of by R.V. Mahendru Gowda
4. Technology of carding by R.Chattopadhahya
5. Spinning blow room and carding by Prof. K.R. Salhotra & Dr. R. Chattopadhahya.
6. Spinning drawing combing and robing by Dr. R. Chattopadhadya & R. Rangasami

4. Fabric Manufacture-II (ETE402)

Unit (1) : Objects of warping, Classification of warping m/cs, slow, high and super high speed warping m/c (2), Salient features of high and super high speed warping m/cs (2), Study and modifications of different types of creels (2), Tension devices and their setting
Total Lectures required = 07

Unit (2): Various stop motion used in warping m/cs (2), Qualities of good warping beam (1), Common faults in warper's beam and their causes and remedies (2), beam hardness and beam count (2)
Total Lectures required = 07

Unit (3): Sectional warping m/cs (2), Waxing attachment and computerized warping m/cs (2), Estimate of efficiency and productivity of various type of warping m/cs (2).
Total Lectures required = 06

Unit (4): Object of sizing and sizing terminology (1), Details of slasher sizing, multicylinder sizing (2), Hot air sizing, high pressure sizing, sizing ingredients used for cotton and synthetic warp (2), Detailed of study of various drying systems, classification of size percentage (1), Factors affecting size pick-up, causes and remedies of lapper and migration, shore hardness of squeeze roll (2).
Total Lectures required = 08

Unit (5): Slow speed motion, measuring and marking motion and slipping friction motion (2), Various controls used on a modern sizing m/c (2), Common faults occurring in sizing beam, their causes and remedies (2), Numerical problems concerning above (2).
Total Lectures required = 07
Grand total lectures required =35

Reference Books:

1. R. Sengupta, yarn preparation, the popular book depot Mumbai.
2. Cotton yarn weaving (ATIRA), 1980
3. M.K. Talukdar- An introduction to winding and warping.
4. P.K. Bannerjee- Yarn winding NCUTE, IIT Delhi.
5. Armored weaving technology and operation, Textile Institute U.K.

5. Wet Processing of Textiles II (ETEF-403)

UNIT -1: Classification of dyes according to there mode of application. Elementary knowledge of cotton dying with direct, vat, sulphur, azoic reactive and Solublised vat dyes. (9)

UNIT-2: dyeing of wool and silk with acid and basic and metal complex dyes (7)

UNIT-3: Dyeing of popular synthetic textile material like nylon, acrylic and polyester Fastness properties of dyeing material. (7)

UNIT-4: Printing –an introduction ,classification of methods of printing .Outline of various method like hand block printing .screen printing .roller printing ,transfer printing rotary screen printing etc. (9)

UNIT-5: Thickeners and their application .Various style of printing –direct style resist style and discharge style of printing. (8)
Grand Total Lectures Required =40

Reference Books:
1. Chemical processing of cotton and p/c blends – ATIRA
2. A glimplse on the chemical tech. of textile fibres by R.K. Charkrawarty
3. Technology of dyeing byV A. Shanai
4. Technology of printing byV A. Shanai

6. Colour & Design (ETEF-404)

Unit (1): Light and colour phenomena, physical basis of colour (2), Emission & absorption theory of light, Colour vision and light theory of colours (2), Complementary colours, Chromatic circle (2), Pigment theory of colours (2), Brewster circle (2), Attributes of the primary & secondary colours, Colour measurement, Primary, Secondary, Tertiary & compound colours (1).
Total Lectures required = 11
Unit (2): Biren's triangle (2), Modification of colours (2), Coloured greys, Colours in combination, Colour contrast, contrast in hue, contrast of tone, colour harmony (1), Relative spaces occupied by colours, divisional colours, Application of colours, Mixed colour effect (1), Composition of designs, Condition to be observed during ornamentation of fabrics (1), Mode shade, Harmony of succession, gradation of hue, Different stages of colouring of textile materials (1), Colour and weave effect and its classification (1).
Total Lectures required =09

Unit (3): Bases of Textile design (2), One third and one fourth drop design (2), Half drop and drop reverse design, Unit repeating design, Geometric ornamentation (2), Construction of symmetrical designs (2), Stripe and check effect designs (2), Sari border / vertical border design (2), Factor affecting the woven designs, reversing inclined figure, Diamond, Ogee, & diagonal waved line base (2).
Total Lecture required = 14

Unit (4): A compulsory question based on drawing sheet and application of colours will be asked from any topic of unit 1 to 3.
Total lectures required for unit 4 are distributed in unit 1, 2, 3.
Grand total Lectures required = 34
Reference Books:
1. W. Watson- Textile design and colour

7. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

U.P. TEXTILE TECHNOLOGY INSTITUTE, KANPUR
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5TH SEMESTER, B. TECH. TEXTILE TECHNOLOGY

1. **Engineering and Managerial Economics (EHU501): Syllabus to be provided by U.P.T.U.**

2. Textile Testing-I (ETCEF-501)

Unit (1): Introduction of textile testing (1), sampling, random sampling, biased sampling (1), sampling techniques, square, cut square, zoning technique (1), selection of sample for testing, grading of cotton fibre with respect to staple length (1), laboratory measurement of fibre length, span length, Baer sorter (1), Shirley photo electric staple, servo fibro graph (1), salient features of HVI (1).
Total Lectures Required = 7

Unit (2): Atmospheric conditions for testing, absolute & relative humidity, moisture regain & moisture contest (1), importance of moisture in textiles, measurement of moisture regain & content (1), effect of moisture on properties (physical & mechanical) of textile material, factors affecting the regain (2), correct numerical weight, oven dry weight (1), shrilly moisture meter (1).
Total Lectures Required = 6

Unit (3): Dry and wet bulb hygrometer, sling assmann, hair hygrometers (2), control of testing room atmosphere (1), AFIS (1), Napping potential (1), Nep count (1), rating of neps, maturity coefficient measurement by NaOH method, fibre fineness by airflow meter & Sheffield micronaire (2).
Total Lectures Required = 9

Unit (4): Fibre bundle strength by Pressley, stelometer (2), fibre quality index, linear density of man made fibres and strength (1), spin finish, crimp (1), Trash, Shirley trash analyzer (1). Yarn numbering system (1), wrapping test for lap, sliver, roving (1),
Total Lectures Required = 7

Unit (5): determination of yarn count, diameter (1), average & resultant count of folded yarn, relation between Ne, D, T, Nm (1). Instruments used for determination of count, quadrant balance, Knowles balance, bees lay balance and physical balance (2), Twist, classification of twist, twist measurement, direct counting method, continuous twist tester, twist-untwist method, (2), Twist tester, (2), R.B. twist tester, level of twist (1).
Total Lectures Required = 9
Grand total of lectures required = 36
Reference Books: -

1. Quality control and testing management by Dr. V.K. Kothari
2. Principle of textile testing by J.E. Booth
3. Physical testing of textiles by B.P. Savile.

3. Yarn Manufacture – III (ETE-502)

Unit (1): Object of combing (1), system of lap preparation and study of their lap, ribbon-lap and super-lap machine (3), configuration of fibers feed and its effect on the quality of product, noil percentage and fraction efficiency of comber (3).
Total Lecturers Required =7

Unit (2): Type of combers (2), distinguishing frameless of Hillman and Nasmith combers (3), detailed study of the Nasmith type comber (2), timing and setting of Nasmith comber for different classes of cotton (1), control of comber work (1), calculation pertaining to draft, production and noil percentage (1), recent developments in combing (1).
Total Lecturers Required =11

Unit (3): Objects of speed frame (1), conventional and modern roving processes (1), mechanism involved in drafting, twisting and winding (2), basic principle of designing of cone drum (2), differential motions and their working principles (2).
Total Lecturers Required =8

Unit (4): Builder motions their objects & types (1), working principles of any modern builder motion (1), drafting system (ordinary & high) (1), processing parameters for different hank roving (1).
Total Lecturers Required =4

Unit (5): Common defects in roving package their censes and remedies (2), calculations pertaining to gearing, contents, draft, TPI and production (2), twist multiplier and roving twist (1).
Total Lecturers Required =5
Grand Total of lectures required =35
Reference Book: -

1. Spun yarn technology Vol. I – A. Ventaksubramani
2. Elements of combing – Dr. A.R. Khare
3. A practical guide to combing & drawing – W. Klein
4. Cotton Spinning - Taggart
5. Spun Yarn Technology – Oxtoby.

4. Fabric Manufacture-III (ETE-503)

Unit (1): General description of plain power looms (2), their mechanical details (2), tuning and adjustment (1). Classification of fabric and weaving machinery (2), introduction to weaving process, primary, secondary and auxiliary (2) motion of looms (2)
Total Lectures Required = 9

Unit (2): Various ways of shedding (1), various types of sheds(1), over and under pick motion(1), beat up motion(1), tappet shedding,(1), idea of construction of tappet(1), under and over pick mechanism slay(1), beating up motion(1), early and late shedding,

temples and its utility, idea about healds count and reed count in different system(1), terry mechanism(1)
Total Lectures Required = 9

Unit (3): Negative and positive take up motion (1), negative and positive let-off motions (1), causes of shuttle flying and shuttle trapping (2), merits and demerits of negative of negative and positive take up and let of motion (2), 5 and 7 wheel take up motion. (2)
Total Lectures Required = 8

Unit (4): Warp protecting motion (1), side and center weft fork motion (1), negative and positive dobby (1), cross border dobby. Preparation of lattice, Development in dobby. Scope and limitations dobby (1), brief description of Cromption dobby, Knowles dobby (1), paper dobby (1), cross border dobby (1), pegging plan dobby faults and adjustment (1) Numerical problems on loom speed, production & efficiency (1).
Total Lectures Required = 9

Grand Total of lectures required = 35
Reference Books: -
1. Weaving mechanism by Fox.
2. Weaving mechanism by N.N. Bannerjee.
3. Weaving Calculation by R. Sengupta.
4. Weaving machine & mechanism by Talukdar.

5. Fabric Structure (ETEF-504)

Unit (1): Classification of various fabrics (1), construction of plain weave and its derivatives (2), ordinary twill right hand twill (1), warp faced, weft faced & balanced twills (2), Satin regular, irregular and their extension (1).
Total Lectures Required = 7

Unit (2): combined twills, end to end and pick-to-pick combination (1). Fancy twills-large diagonal shaded twills (1), sateen base diagonals and brained twills (1), elongated twill-steep and low twills (1), pointed, wave zig-zag, curved, broken, herring borne, transposed, corks screw twills (2), Diamond, mock leno, ordinary honeycomb, brighten honeycomb (2),
Total Lectures Required = 8

Unit (3): Huck-a-back and crepe weave (), derivatives of hopsack barley corn stitched hopsack and twilled hopsack (2), Simple and wadded bed ford cords (1), weft and piques (1), principle of figuring with extra material extra warp figuring (2) extra weft, limitation of extra thread (2).
Total Lectures Required = 9

Unit (4): Backed cloths (1), weft backed cloths, warp backed cloths with weeding threads (2), double cloths, center stitched, self stitched (2), inter changing double cloth, cut effect in interchanging double cloths (2).
Total Lectures Required = 7

Grand Total of lectures required = 31

Reference Book: -

1. W. Watson Textile Design & colour Longmans Greens Co. London.
2. Z.J Grosicki Watson's Textile design and colour Newnes Buter Worth, London.
3. Z.J. Grosicki, Advance Textile Design Newnes Butter Worth, London.
4. "Nishant" A Grammar of textile.

6 Chemistry and Production of Fibres (ETCEF-505)

Unit (1): Polyethylene Terephthalate Fibre – History of development, Polymer production by DMT & PTA route, Chips drying, Fibre manufacturing, Effect of process variable on properties of polyester fibre, some dope additives for specialty polyester fibre, Properties of polyester fibre.

Total Lectures required = 08

Unit (2): Polyamide Fibres – History of development, Different types of polyamide fibres, Nylon polymer production by continuous polymerization in VK Tube, Manufacturing of Nylon 6 fibre by melt spinning, Properties of nylon 6 fibre, Polymer production of Nylon 66, Nylon 66- fibre formation by melt spinning, Properties of Nylon 66 fibre, Brief introduction to Aramid fibres.

Total Lectures required =08

Unit (3): Introduction of polyolefin fibres, Polymerization of polyethylene, Polyethylene (PE) fibre formation, Properties of polyethylene fibre, Different type of polypropylene (PP), Polymerization of polypropylene, Polypropylene fibre formation, Properties of polypropylene fibre.

Total Lectures required =08

Unit (4): Introduction of vinyl fibres, Polyacrylonitrile (PAN) fibre, Use of ionic and neutral co-monomers, Polymerization of PAN, Acrylic fibre- formation by dry spinning, Dry –jet-wet spinning process, Effect of process variables on properties of PAN fibre, Dope formation, Properties of PAN fibre, Introduction to polyurethane fibre.

Total Lectures required = 08

Unit (5): Introduction of regenerated fibre, Concepts of regeneration of fibre, Raw material for viscose rayon, Manufacturing sequence of viscose fibre, Steeping and pressing, Cutting and shredding, Ageing, Xanthation of sodium cellulose, Mixing and filtration, Ripening, Wet spinning of viscose rayon, Formation of serrated edge cross-section of viscose rayon, Viscose fibre properties, Introduction of cuprammonium rayon in brief, Introduction of cellulose acetate rayon in brief, Introduction of lyocell fibre in brief.

Total Lectures required =08

Grand total lectures required = 40

Books:

1. Manufactured fibre technology by V.B. Gupta & V.K. Kothari
2. Essential fibre chemistry by M.E. Cartor
3. Synthetic fibres by Fourne

4. New fibre by T.Hongu, G.O. Phillips, Woodhead publications.
5. Fibre chemistry by M. Lewin, E.M. Pearce, Marcel & Dekkan Inc.
6. Regenerated Cellulosic fibres by C. Wooding, Woodhead publishing Ltd.

7. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

6TH SEMESTER, B. TECH. TEXTILE TECHNOLOGY

1. **Industrial Management (EHU 601) Syllabus as provided by U.P.T.U.**

2. **Departmental Elective-I:Coating of Textiles (ETEF011/EC601)/ 2.2 Post Spinning Operations (ETCE012/EF604)**

2.1 Coating of Textiles (ETEF011/EC602)

Unit-1: Polymeric materials for coating- (Rubbers: natural and synthetic, (3)(Polyvinyl chloride, Polyurethane, Acrylic polymers, Adhesive treatment (4)
Total lecture required=7

Unit-2: Coating Methods: Knife coating (1), Roll coating (1) transfer coating (1), Rotary screen printing (2) calendaring hot melt coating (2)
Total lecture required=7

Unit-3: Physical properties of coated fabric (2) Rheology of coating (1), Rheological Behaviors of fluids (2) pastes (1) hydrodynamic analysis of coating (2)
Total lecture required=8

Unit-4: Fabric for foul weather protection- Clothing comfort (1) Impermeable coating (1) breathable fabric (1) Non Apparel coating (1), Fabrics for Chemical protection (1) Thermo chromic Fabric (1) Temperature Adaptable Fabrics (1) Camouflage nets (1) Metal and conducting polymer-coated fabrics (2) Radiation cured coating (1)
Total lecture required=8

Unit-5: Test methods, Coating per Unit area, Degree of fusion/curing of coating (1) blocking Abrasion resistance (1) Test for colour- Fastness to dry and wet rubbing, Resistance to water penetration (3) Air permeability (1) water vapour permeability (1) low temperature bend test (1) low temperature impact test (1)
Total lecture required=9
Grand Total lecture required=39

Reference Books:
References:

1. Coating & Laminated Textiles by Water Fung
2. Coated Textile by A.K. Sen
3. Coated Fabric technology Vol 1-3 Technomic publication
4. Coated & laminated Fabric by AATCC symposium

2.2. Post spinning operation (ETCE012/EF604)

Unit (1): Concept of drawing ,drawing unit (3),Effect of drawing on orientation and crystallizations(3),Drawing of melt spun filaments, drawing of as-spun fibres through neck formation(3).
Total of lectures required=9

Unit (2): Heat setting ,nature of heat set(3),Influence of heat setting variables on structures &properties of fibres(3),Temporary, semi permanent and permanent heat setting, heat –setting equipments(3).
Total of lectures required=9

Unit (3): Method of producing bulk-false-twist texturing (3),effect of process variables, time, temperature, twist and tension on properties of false twist textured yarn,(3),edge crimping(3).
Total of lectures required=9

Unit (4): Air-jet texturing process(3),stuffer-box texturing process(3),effect of process variables on yarn properties of air jet & stuffer box textured yarns, Role of water in air jet texturing(4).
Total of lectures required=10

Unit (5): Tow to top conversion, stretch breaking (2),cutting method(1), Sydel stretch breaking(2),Pacific tow to top cutting system(2),Turbo stapler(2)
.**Total of lectures required=9**
Grand total lectures required =46.

Reference book:-
1. Manufactured fibre technology by V.B.Gupta & V.K.Kothari.
2. Synthetic fibre by Vaidhya
3. Textile fibre part-ll by V.K.Kothari.
4. Textured yarn technology by J.W.S.Hearle.

3. Textile Testing-II (ETCEF-601)

Unit (1): Textile properties of yarn and fabric(2), stress-strain curve(1), various methods for finding of yield point(1), methods for finding of various modulus(1), destination of tenacity(1), stiffness of fabric(1).
Total Lectures Required = 7

Unit (2): Yarn testing m/cs- single yarn strength tester, lea strength tester (2), fabric strength tester- impact tester, Graph test, fabric B.S. Test (3), Scott serigraph, Instron testing m/c (2).
Total Lectures Required = 7

Unit (3): Fabric abrasion & resistance to wear (1), Bursting strength test (1), Tear test, Wear (1), Brief introduction of FAST & KAWABATA (1), estimation of fabric

thickness, rigidity, air permeability (1), water repellency (1), drape handling (1), crimp, thermal transmission properties (1).
Total Lectures Required = 8

Unit (4): Cover factor of fabric (1), snagging-mace snagging test (1), weaver trials, advantages and disadvantages of weaver trial (1), lab test (1), measurement of lap, sliver, yarn irregularity (1) salient features of Uster evenness tester (2)
Total Lectures Required = 7
Grand Total of lectures required = 29
Reference Books: -

1. Physical testing of textiles by B.P. Saville.
2. Quality control and testing management by Dr. V.K. Kothari.
3. Principles of textile testing by J.E. Booth.
4. Quality control by V.K. Kothari.

4. Structure and Properties of Fibres (ETEF602/EC011))

Unit (1): Moisture absorption, heat of absorption (2), differential heat of absorption, integral heat of absorption (2), Quantitative theory of heat moisture absorption (2), Rate of moisture absorption (1)
Total Lectures Required = 7

Unit (2): Mechanical properties of fibres (2), Relation between structure and mechanical properties of fibres (2), Basic mechanical properties (tenacity elongation, modulus, work of rupture (2), Elastic recovery, time effects (2).
Total Lectures Required = 8

Unit (3): Thermal behavior of textile fibres by DSC (2), TGA, thermal mechanical analysis (2), Density gradient column (2), Preparation of density gradient coloumn (2).
Total Lectures Required = 8

Unit (4): Birefringence behavior, dielectric properties (2), fibre friction, fibre friction measurement (2), fibre yarn to surface friction measurement and static charge measurement (5)
Total Lectures Required = 7

Unit (5): Creep behavior (2), Optical properties of fibres (2), concept of moisture absorption by fibres (2), relation between fibre structure and fibre properties (2).
Total Lectures Required = 8
Grand Total of lectures required = 38
Reference Book: -

1. Manufactured fibre technology by V.B. Gupta, V.K. Kothari
2. Physical properties of fibre by J.W.S. Hearle
3. Thermal behavior of material by Turi
4. Modern yarn production by Ray
5. Textile fibres by ATIRA
6. ASTM Standard books

7. Absorberey chartering
8. Polymers by fibre & textiles encyclopedia
9. Advances in fibre source by S.K. Mukhopadhyaya

5. Yarn Manufacture – IV (ETE-603)

Unit (1): Principle and mechanism involved in drafting, twisting & winding (2), ordinary and high draft system (2), yarn twist: Terminology, twist level, concept of twist multiplier (1), propagation of twist (2),
Total Lecturers Required = 7

Unit (2): Builder motion (2), common package size, limitations to large package spinning (2), types of rings and travelers and their common uses (2), Rising and falling lappets, control rings, apron drafting system (2)
Total Lecturers Required = 8

Unit (3): System of waste collection of ring frame and types of spinning wastes (2), limitation in ring spinning and factors responsible for loss in efficiency (1), yarn faults and their remedies (2), Recent developments in ring spinning (1).
Total Lecturers Required = 7

Unit (4): Doubling: - Objects and terminology (1), study of ring doublers (1), fancy yarns (1), sewing thread and tyre Cord (1). Reeling: Objects and terminology, types of reeling construction and working of a reel (2), yarn bundling (1), calculation of draft, TPI and production of ring frame & doubling frame (2),
Total Lecturers Required = 9
Grand Total of lectures required =31
Reference Books: -

1. Elements of ring frame & doubling – Dr. A.R. Khare
2. The technology of short-shape staple spinning – W. Klein
3. Cotton spinning – Taggart
4. Spun yarn technology – Oxtoby

6. Fabric Manufacture-IV (ETE-604)

Unit (1): Jacquards shedding (1), types of jacquards and their principle of working (2), cross border jacquards (2). Harness mounting London and Norwich system (1), Card cutting (1), limitations of jacquards, Jacquards-driving study of following jacquard. (4),
(a). Single lift single cylinder
(b). Double lift single cylinder
(c). Double lift double cylinder
(d). Electronic jacquard.
Total Lectures Required = 11

Unit (2): Different systems of harness tieing. (2), terry mechanism (2), Recent development in jacquards (2)

Total Lectures Required = 6

Unit (3): Automatic looms – pirn (2), shuttle changing (2), Detailed study of various motions of automatic looms (2), cop changing loom (1), warp stop motions mechanical (1), electro-mechanical, electronic stop motion (1).
Total Lectures Required = 9

Unit (4): Multiple box motion, their types (2), principle of working of multiple box motion (1), two colours and four-colour drop box motion (1), brief description of pick-at will (1), pick and pick motion. On line process and quality control (1), estimation of productivity, snap study. (7)
Total Lectures Required = 7

Unit (5): Numerical problems concerning to above (5)
Total Lectures Required = 05
Grand Total of lectures required = 38

Reference Books: -

1. Weaving mechanism by Fox.
2. Weaving mechanism by N.N. Bannergy.
3. Weaving Calculation by R. Sengupta.
4. Weaving machine & mechanism by Talukdar

7. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

DETAILES SYALLUBUS

7TH SEMESTER, B. TECH. TEXTILE TECHNOLOGY

1. **Open Elective-I: (EOE071-074) Syllabus as provided by U.P.T.U.**

2. **Department Elective-II: Garment Manufacturing Technology (ETCEF021)/ Nonwoven (ETCEF022)**

2.1 Garment Manufacturing Technology (ETCEF021)

Unit (1): Introduction to Garment Manufacturing Technology, Sample Cutting, Fusing Technology, Sewing Technology, Pressing Technology, Finishing and Inspection, Sample Room, Communication, and organization.
Total Lectures required =6

Unit (2): Introduction to Measurement of Fabric Dimensional Properties, Fabric Comfort, Thermal Comfort, Objective Evaluation of Fabric, Low Stress Fabric Properties, Kawabawa System, Fabric Assurance by Sample Testing, Fabric Defects, Fabric Inspection and Feedback to back process.

Total lectures required =8

Unit (3): Introduction to Garment Cutting, Marker Planning, Efficiency of Marker Planning, Methods of Marker Planning and Marker Use, Spreading of the Fabric to form a lay, Spreading Requirements, Methods of Spreading, Fabric Packages, objective of Cuttings, Methods of Cuttings,
Total lectures required =8

Unit (4): Introduction to Seam, Stitch, Stitch Classification, Stitch Structure, Seam Formation, Joining Material, Surface Characteristics, Seam Appearance, Damages (Thermal and Mechanical), Seam Performance, Seam Degradation, Seam Failure and Seam Testings, and Seaming Machine.
Total lectures required =8

Unit (5): Importance of Garment Processing and Finishes, Types of Garments, Processing of Garments and Specialty Garment Finishes.
Total lectures required =6
Grand Total Lectures required = 36

Reference Books:

1. Introduction to Garment Manufacturing Technology By T Ramchandran
2. Garment Manufacturing Technology by By T Ramchandran
3. Practical Clothing Construction Part I & II by Mary Methews

2.2 Non Woven (ETCEF022)

Unit 1: Concept about felts and non woven (3), application of non-woven fabrics (2), fibre- characteristics needed for the manufacture of non-woven (2).
Total =7
Unit 2: Manufacturing passage for the felt preparation (4), web chrematistic and their influence properties (3)
Total =7

Unit 3: Process variable (2), and their effect on properties of non woven (4).
Total = 6

Unit 4: Various system of binding the web (5), and their merits (2).
Total = 7

Unit 5: Flocked fabric (2), Laminates (2), latest development in non-woven industry (3).
Total = 7
Grand total = 34

Reference Books & Text Books –

1. Non Woven – N.N. Banarjee
2. Non woven – NCUTE
3. Fibre to fabric – Corbman Bernard P.

3. Mill Planning & Organization (ETEF701)

Unit-1: Selection of site for a textile mill (2). Preparation of project report (2). Construction of building of a textile mill (2). Types of buildings, single and multistoried buildings. Fire hazards and their control (1). Prevention tours accidents (1)
Total of lectures required=9

Unit-2: Humidification of a textile mill (1). Humidifiers and dehumidifiers (1). Types of humidifiers (2). Different systems of ventilation system used in textile mill (1). System of lighting used in textile mill (2)
Total of lectures required=7

Unit-3: Balancing of machines for spinning and weaving machines or balancing of machines for bleaching (4). Dyeing, Printing and Finishing department. Layout of different machines of spinning, weaving process house (3).
Total of lectures required=7

Unit-4: Costing (1), elements of cost, fixed and variable cost (1). Knowledge of cost cancellation for spinning (2), weaving and processing department. Viability evaluation of a project (1). Break even point (1) Types of staff organization (2).
Total of lectures required=8

Unit-5: Different system wages used in textile industry (2), working of sales and purchase system (2). Material handling used in a textile mill (2). Maintenance of machines and safety devices used in a textile mill (2). Inventory controls (2) Resource Planning.
Total of lectures required=10
Total of lectures required=41

Reference Books

1. Industrial Engineering, Organization & management by Tarachand
2. Industrial Economics & Principle of Management by T.M. Chabra
3. Industrial Economics & Principle of Management by S. K Sharma

4. ADVANCE SPINNING TECHNOLOGY (ET702)

Unit-1: Limitation of ring spinning (1), Principles of unconventional method of yarn (1) Production open and spinning process(1), Advantages and Limitations of open and spinning process(1), Classification of new spinning yarn technology(1)

Unit-2: Rotor spinning- Objects of rotor spinning(1), Principle of operation(1), Raw material requirements(1), opening unit, yarn formation(1) Design of rotor, Navel and yarn withdrawal tube(1) Automation, yarn characteristics comparison of yarn properties with ring yarn(1) Rotor and ring yarn structure(1).

Unit-3: Friction spinning- Principle(1) DREF-2 and DREF-3, yarn formation , yarn quality, yarn structure(1) fibre specifications for optimum results, merits & limitations(1) Twist less spinning-TNO process and TWILLO process(1)

Unit-4: Air jet Spinning-Principle(1), concept of false twist(1), Fasciated yarn, Murata jet spinning, operation principle(1) Raw material requirement, Effect of process variables on yarn twist& (1) tenacity, yarn quality, limitation of air jet yarns(1) self twist process(2)

Unit-5: Bobtex ICS process(1), Wrap spinning(1), plyfil spinning(1),SIRO spinning(1),Electrostatic spinning(1), Core spinning(1) compact spinning(1)

Reference Book-

1. New Spinning Technology Vol. 4 & 5 –W.Klien

5. Knitting Technology (ETEF- 703)

Unit 1: Difference between knits & woven (1), knitting terms & definition (2), knit technology (2), knitting industries (1)
Total= 6

Unit 2: Classification of weft knitting machine(2), elements of knitting machine(2), technology of loop formation(2), geometry of loop structure(2) relation between yarn count, machine gauge and stitch density(2).
Total =10

Unit 3: Classification of knit-structures(2), loop formation on single jersey, Rib machines and inter look machines(2), knitting notation(2), socks knitting technology (3), warp knitting introduction(2), classification of warp knitting(2).
Total = 13

Unit 4: Special knit fibre and mechanics (4), electronics in knitting (4).
Total =8

Unit 5: Modern developments in knitting industry (3), calculations regarding production gsm, stitch density etc, set-up of knitting mills(3).
Total = 6
Grand Total = 46

Reference and Text Book-

1. Knitting technology – chamberlin
2. Knitting technology – W.J. Spencer
3. International Textile Journel – Knitting
4. Knitting Calculation – Chamberlin
5. Wet Knitting Vol. 1&2 –Published by IIT New Delhi.
6. Knitting – NCUTE

6. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

DETAILES SYALLUBUS

8TH SEMESTER, B. TECH. TEXTILE TECHNOLOGY

1. Open Elective-III: Syllabus as provided by UPTU

1. EOE081 Non Conventional Energy Resources/
2. EOE082 Introduction to Soft Computing/
3. EOE083 Product Development/
4. EOE084 Automation and Robotics

2. Departmental Elective III: Technical Textiles (ETCEF 031)/ Theory of Textile Structure (ETEF-032)

2.1 Technical Textiles (ETCEF 031)

Unit-1: Textiles in filtration, Filter fabric construction- Woven, needle felt & knitted filter fabric (2), Finishing treatment of filter fabric (2), Thermal and chemical properties of filter fabric (2), Essential requirements of good filter fabric (2).
Total = 8

Unit-2: Waterproof breathable fabrics- Densely woven fabric, membranes, coatings (3), Membranes- Micro porous membrane, Hydrophilic membrane (2), Coatings-Microporous coating, Hydrophilic coatings (2), Performance of waterproof breathable fabrics (2).
Total =9

Unit 3: Medical textiles, Fibres used (1), Classification of medical textiles – Nonimplantable materials-wound dressings, bandages, plasters etc. (2), Extracorporal devices – Artificial kidney, liver, lung (2), Implantable material- Sutures, Soft tissue implant, Orthopaedic implants, Cardiovascular implants.(2),Healthcare/ hygiene products (1).
Total =8

Unit 4: Smart textiles brief introduction of smart textiles, classification of smart textile(2), passive smart textiles(2) , active smart textiles , very smart or ultra smart textiles (2), application of phase change material in smart textiles(2), brief discussion of smart shirt, smart suit, musical jacket, space suit etc(2).
Total =10

Unit 5: Introduction of Geo Textile (2), Classification of Geo Textiles, Functions of Geo textiles-soil reinforcement, drainage (fluid transmission), filtration, Separation, erosion control/absorption (2), objects of geo textiles, manufacturing of geo textile (3).Essential properties of geo textiles- Mechanical determinants, Hydraulic determinants, durability determinants (2)
Total = 7
Grand Total 42

Reference Books:-

1. Hand book of technical textiles – AR Harrocks, SC Anand (Textile Institute, wood head publication).
2. Smart fibre, fabrics and clothing Tao X (Textile Institute, wood head publication).
3. Shears Handbook of Industrial Textiles.

2.2 Theory of Textile Structure (ETEF-032)

Unit (1): Classification of yarn (1), Yarn diameter & density (1), Twist in yarn (1), Geometry of twisted yarn (2), Contraction due to twist (2), Twist and fibre packing in yarn – open packing, hexagonal close packing, redial packing density in yarn (2)
Total lectures required = 09

Unit (2): Twist and strength relationship (1), Theoretical treatment of yarn irregularity (2), fibre extent & fibre migration (3), estimation of strength of blended yarn (3)
Total lectures required = 09

Unit (3): Spinnability of textile fibres (1), Elements of fabric geometry (3), Flexible & rigid threads model (2), Pierce's equation and latter modifications (4)
Total lectures required = 10

Unit (4): Relation of fabric properties to simple geometry (2), Crimp interchange in woven fabric (2), Cover factor (1),
Total lectures required = 05

Unit (5): Theoretical treatment of fabric deformation in tension, bending & shear (3) Cloth setting theories (1)

Reference Books: -
1. Textile yarns – B.C. Goswamoi, J.G. Martindle, F.L. Scardino.
2. Textile structure – J.W.S. Hearle, S.Backer, Grossberg.
3. Pierce Geometry – Textile institute

3. Department Elective IV: Process Control in Spinning (ETEF-041)/ Process Control in Weaving (ETEF042)

Process Control in Spinning (ETEF-041)

Unit-1: Importance and considerations for evolving a system for process control (1) Scope for process control in spinning, Key variable for process control, Establishing norms and standards, Taking corrective action. (Control of Mixing quality and Cost)- Instrumental evaluation of cotton, Control of mixing quality through fibre Characteristics, Simultaneous control of mixing cost and quality, Bale Management (1)

Unit-2: Control of waste and cleaning (1) in Blow room and carding-Determination of trash (1) control and cleaning efficiency, Cleaning and waste in Blow (1) room and card, Norms for waste and (1) cleaning in card & Blow room, Assessment of performance of Blow room (1). Optimizing cleaning at cards (1) Control of comber waste (1) Control of yarn realization (1)

Unit-3: Measurement and analysis of productivity definition of indices of productivity (1), Measurement of productivity (1), Productivity analysis, productivity and profitability (1), Means to Improve productivity (1) Maximizing machine efficiency in ring spinning (1) Controlling end breakage rate in ring spinning (1).
Reference book (1) process Control in spinning by ATIRA

Unit-4: Control of yarn quality(1)- Count, Strength and their variability(1), Study of with in bobbin and between bobbin count variation(1), Control of variability of lea strength(1), Single yarn strength and elongation(1) Control of yarn unevenness imperfection(1)

Unit-5: Control of yarn faults and package defects (1) - slubs, crackers, spinners doubles (2) bad piecing, slough of and hairiness (1). Machinery Audit- Impact of machine conditions on processing performance and yarn quality (1), test instruments for Machinery Audit (1)

Reference Books:

Process Control in Weaving (ETEF042)

Unit-1: Scope for process Control (1) Systems of process Control in wvg (2) Setting norms (2) and Schedule of Checks machinery audit (2)

Total lecture required=7

Unit-2: Process Control in wvg (2) optimizing quality of preparation. Control of quality of knot (2) producing good package (1) control of productivity (2)
Total lecture required=7

Unit-3: Process Control in warping (2) minimizing end wreak age in warping (2) quality of warping beam (2) control of productivity (2)
Total lecture required=8

Unit-4: Process Control in sizing (2) Choice of size recipe and size pick-up (2)
Control of size pick-up, Control of yarn stretch (2) quality of sized beams, Control of productivity (2)
Total lectures required=8

Unit-5: Care is use and Selection of healds (2) Care in use and selections of Reads (1)
Care in dressing of the beams for reducing incidence of cross ends (2) Causes and remedies of fabric defects (2)
Total lectures required=7

Grand total of lectures required=37

Reference Books:

1. Weaving tablets by ATIRA
2. Machine catalogues of various machines manufacture
3. Process control in weaving by ATIRA

4. Advance Waving Technology (ET801)

Unit 1: Principle of operation of shuttle less loom (2) Classification of shuttle less looms. Comparative study of various systems of weft inserting (2) advantage of shuttle less looms oven shuttle looms (2)
Total of lectures required=7

Unit-2: Basic mechanical principle of Sulzer projectile weaving machinery shedding, picking and beating (3), Selvedge formation (2) shuttle less weaving with more than one weft (2)
Total of lectures required=7

Unit-3: The Rapier system of weft insertion (1) Classification of Rapier looms and its description(2) flexible and rigid rapier (2) tip transfer and loop transfer(1) weft Control mechanism, loom timing etc(2)
Total lectures required=8

Unit-4: Working principle of Air-jet weaving with special reference the **maxoloom** (2) essential requirements of shedding, weft measuring, weft tensioning (2) beating-up, weft gripping, weft cutting (2) weft stop motion selvedge formation (2)
Total lectures required=8

Unit-5: Weft insertions of water-jet loom (2), general description of the water-jet loom (2) weft supply system, tension measuring device (1) operation of weft supply stream (1) fundamental problems of prydranlic weft insert (2)
Total lectures required=8

Grand total of lectures required=38

Reference books

1. Modern development in weaving machine by ra and Duxburg,
2. Weaving mecanism, machine and Management by Talukdar.
3. Weaving Machinism by Robinson.
4. Modern weaving machinery Ormerd.

5. *Human Values & Professional Ethics (Audit course for all Semester)
Syllabus as provided by U.P.T.U

U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

B. TECH. TEXTILE CHEMISTRY

[Effective from the Session 2009-10]

YEAR 2nd, SEMESTER- III

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	ETCEF301	Natural Textile Fiber	2	1	0	15	10	25	50	75	3
2	EAS301/ EOE031-EOE038	Mathematics III/ Science Based Open Elective	3	1	0	30	20	50	100	150	4
3	EHU301/ EHU302	Industrial Psychology/ Industrial Sociology	2	0	0	15	10	25	50	75	2
4	EC302	Industrial Chemistry	3	1	0	30	20	50	100	150	4
5	ECF303	Polymer Chemistry-	3	1	0	30	20	50	100	150	4
6	EC304	Principle of Yarn Manufacture	3	1	0	30	20	50	100	150	4
7	EHU111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF301P	Identification of Fibers-I	0	0	3	0	20	20	30	50	1
9	ETC302P	Industrial Chemistry lab	0	0	3	0	20	20	30	50	1
10	ETC303P	Polymer Chemistry Lab	0	0	3	0	20	20	30	50	1
11	EC304P	Yarn Manufacture	0	0	3	0	20	20	30	50	1
12	GP301	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								**1000**	**26**

*** Audit course**

Paper Code	**Science Based Open Elective**
EOE031/EOE041	Introduction to Soft Computing
EOE032/EOE042	Nano Sciences
EOE033/EOE043	Laser Systems and Applications
EOE034/EOE044	Space Sciences
EOE305/EoE405	Polymer Science & Technology
EOE036/EOE046	Nuclear Science
EOE037/EOE047	Material Science
EOE038/EOE0487	Discrete Mathematics

U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

B. TECH. TEXTILE CHEMISTRY

[Effective from the Session 2009-10]

YEAR 2nd, SEMESTER- IV

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU402/ EHU401	Industrial Sociology/ Psychology	2	0	0	15	10	25	50	75	2
2	EOE041-EOE408/ EAS401	Science Based Open Elective/ Mathematics III	3	1	0	30	20	50	100	150	4
3	EC401	Principles of Fabric Manufacture	2	1	0	15	10	25	50	75	3
4	EC402	Technology of Bleaching & Mercerization	3	1	0	30	20	50	100	150	4
5	EC403	Technology of Dyeing-I	3	1	0	30	20	50	100	150	4
6	EC404	Physical Chemistry of Dyeing	3	1	0	30	20	50	100	150	4
7	EHU111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF 401P	Identification of Fibre-II	0	0	3	0	20	20	30	50	1
9	EC402P	Fabric Manufacture	0	0	3	0	20	20	30	50	1
10	EC403P	Bleaching &Mercerization	0	0	3	0	20	20	30	50	1
11	EC404P	Experimental dyeing	0	0	3	0	20	20	30	50	1
12	GP301	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								**1000**	**26**

*** Audit course**

U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

B. TECH. TEXTILE CHEMISTRY

[Effective from the Session 2010-11]

YEAR 3rd, SEMESTER- V

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU501	Engineering & Managerial Economics	3	1	0	30	20	50	100	150	3
2	ETCEF501	Textile Testing-I	3	1	0	30	20	50	100	150	4
3	EC502	Chemistry of dyes & Color chemistry	2	1	0	15	10	25	50	75	3
4	EC503	Technology of dyeing-II	3	1	0	30	20	50	100	150	4
5	EC504	Textile Auxiliaries	2	1	0	15	10	25	50	75	3
6	ETCEF 505	Chemistry & Production of Fibers	3	1	0	30	20	50	100	150	4
7	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF 501P	Textile Testing-I	0	0	3	-	20	20	30	50	1
9	EC502P	Textile Analysis Lab	0	0	3	-	20	20	30	50	1
10	EC503P	Experimental Dyeing II	0	0	3	-	20	20	30	50	1
11	EC504P	Textile Chemical Testing	0	0	3	-	20	20	30	50	1
12	GP501	General Proficiency	-	-	-	-	-	50	-	50	1
										1000	**26**

*** Audit course**

U.P TECHNICAL UNIVERSITY, LUCKNOW
STUDY & EVALUATION SCHEME
B. TECH. TEXTILE CHEMISTRY
[Effective from the Session 2010-11]

YEAR 3rd, SEMESTER- VI

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU601	Industrial Management	3	0	0	30	20	50	100	150	3
2	EC011/ EC012	Department Elective-I	3	1	0	30	20	50	100	150	4
3	ETCEF601	Textile Testing-II	2	1	0	15	10	25	50	75	3
4	EC602/ ETEF011	Coating of Textiles	3	1	0	30	20	50	100	150	4
5	EC603	Technology of Printing-I	2	1	0	15	10	25	50	75	3
6	EC604/ EF012	Technology of Finishing- I	3	1	0	30	20	50	100	150	4
7	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF 601P	Textile Testing-II	0	0	3	0	20	20	30	50	1
9	EC602P	Coating of Textiles	0	0	3	0	20	20	30	50	1
10	ETC 603P	Printing Practice	0	0	3	0	20	20	30	50	1
11	ETC604P	Finishing of textiles-I	0	0	3	0	20	20	30	50	1
12	GP601	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1000**	**25**

*** Audit course**

Paper Code	**Department Elective-I**
EC011/ETEF603	Structure & Properties of Fibres
EC012/ EF604	Post Spinning Operation

U.P TECHNICAL UNIVERSITY, LUCKNOW
STUDY & EVALUATION SCHEME
B. TECH. TEXTILE CHEMISTRY
[Effective from the Session 2011-12]

YEAR 4th, SEMESTER- VII

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EOE071-EOE074	Open Elective-I	3	1	0	30	20	50	100	150	4
2	ETCEF021/ ETCEF022	Department Elective-II	3	1	0	30	20	50	100	150	4
3	EC701	Process House Planning	3	1	0	30	20	50	100	150	4
4	EC702	Technology of Finishing-II	3	1	0	30	20	50	100	150	4
5	EC703	Waste Management & Pollution Control	3	1	0	30	20	50	100	150	4
6	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
7	EC702	Finishing II	0	0	3	-	20	20	30	50	1
8	ETC751	Project	0	0	3	-	50	50	-	50	2
9	ETC752	Seminar	0	0	2	-	50	50	-	50	1
10	ETC753	Industrial Training Viva Voice	0	0	2	-	50	50	-	50	1
11	GP701	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**15**	**3**	**10**	**150**	**270**	**470**	**530**	**1000**	**26**

*** Audit course**

Paper Code	**Open Elective-I**
EOE071	Entrepreneurship Development
EOE072	Quality Management
EOE073	Operation Research
EOE074	Introduction to Biotechnology
Paper Code	**Department Elective-II**
ETCEF021	Garment Manufacturing Technology
ETCEF022	Nonwoven

U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

B. TECH. TEXTILE CHEMISTRY

[Effective from the Session 2011-12]

YEAR 4th, SEMESTER- VIII

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EOE801-EOE804	Open Elective-II	3	1	0	30	20	50	100	150	4
2	ETCEF031 /EC032	Departmental Elective-III	3	1	0	30	20	50	100	150	4
3	EC801	Technology of Printing-II	3	1	0	30	20	50	100	150	4
4	EC802	Developments in Wet processing	3	0	0	30	20	50	100	150	3
5	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
6	ETC851	Project	0	0	12	-	100	100	250	350	8
7	GP801	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**12**	**3**	**12**	**120**	**180**	**350**	**650**	**1000**	**24**

*** Audit course**

Paper Code	**Open Elective-II**
EOE081	Non Conventional Energy Resources
EOE082	Nonlinear Dynamic Systems
EOE083	Product Development
EOE084	Automation & Robotics
Paper Code	**Department Elective-III**
ETE801/ETCEF031	Technical Textiles
EF802/EC032	High Performance Fibres

3TH SEMESTER, B. TECH. TEXTILE CHEMISTRY

EAS-301/EAS-401: MATHEMATICS –III **L** **T** **P**

3 **1** **0**

Unit – I : Function of Complex variable

Analytic function, C-R equations, Cauchy's integral theorem, Cauchy's integral formula for derivatives of analytic function, Taylor's and Laurent's series, singularities, Residue theorem, Evaluation of real integrals of the type $\int_0^{2\pi} f(Cos\,\theta, Sin\,\theta)\,d\theta$ and $\int_{-\infty}^{+\infty} f(x)\,dx$ **10**

Unit – II : Statistical Techniques - I

Moments, Moment generating functions, Skewness, Kurtosis, Curve fitting, Method of least squares, Fitting of straight lines, Polynomials, Exponential curves etc., Correlation, Linear, non –linear and multiple regression analysis, Probability theory. **08**

Unit – III : Statistical Techniques - II

Binomial, Poisson and Normal distributions, Sampling theory (small and large), Tests of significations: Chi-square test, t-test, Analysis of variance (one way) , Application to engineering, medicine, agriculture etc.

Time series and forecasting (moving and semi-averages), Statistical quality control methods, Control charts, $\bar{X}$, R, p, np, and c charts. **08**

Unit – IV : Numerical Techniques – I

Zeroes of transcendental and polynomial equation using Bisection method, Regula-falsi method and Newton-Raphson method, Rate of convergence of above methods.

Interpolation: Finite differences, difference tables, Newton's forward and backward interpolation , Lagrange's and Newton's divided difference formula for unequal intervals. **08**

Unit – V : Numerical Techniques –II

Solution of system of linear equations, Gauss- Seidal method, Crout method. Numerical differentiation, Numerical integration , Trapezoidal , Simpson's one third and three-eight rules, Solution of ordinary differential (first order, second order and simultaneous) equations by Euler's, Picard's and forth-order Runge- Kutta mehthods. **08**

Test Books :-

1. Peter V. O'Neil, Advance Engineering Mathematics Thomson (Cengage) Learning, 2007.
2. Jain, Iyenger & Jain, Numerical Methods for Scientific and Engineering Computation, New Age International, New Delhi , 2003.

3. J.N. Kapur, Mathematical Statistics, S. Chand & company Ltd.,2000

Reference Books :-

1. R.K. Jain & S.R.K. Iyenger, Advance Engineering Mathematics, Narosa Publication House, 2002.
2. Chandrika Prasad, Advanced Mathematics for Engineers, Prasad Mudralaya, 1996.
3. E. Kreysig, Advanced Engineering Mathematics, John Wiley & Sons, 2005.
4. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 2005.
5. Devi Prasad, An introduction to Numerical Analysis, Narosa Publication house, New Delhi 2006.
6. T. Veerajan & T. Ramchandrandran, Theory & Problems in Numerical Methods, TMH, New Delhi, 2004.
7. S.P.Gupta, Statistical Methods, Sultan and Sons, New Delhi, 2004.
8. Devore, Probability and Statistics, Thomson(Cengage) Learning, 2007.
9. Walpole, Myers, Myers & Ye, Probability and Statistics for Engineers & Scientists, Pearson Education, 2003.

1. (Engineering Core: Interdisciplinary) Natural Textile Fibres: (ETCFE-301)

Unit (1) : Fibre, textile fibre (1). Classification of natural textile fibres (1), Essential and desirable properties of textile fibres (2), Grading of cotton (1), Physical properties of cotton fibre (1), Effect of acid & Alkalies on cotton fibre (1).
Total Lectures required = 07

Unit (2): Production, morphological structure, properties and end uses of: Flax (2), Jute (2), Ramie (2). Introduction to specialty natural fibres like banana, pineapple, spider etc
Total Lectures required = 08

Unit (3): Production of silk (raw) (2), Morphological structure of silk (1), Production of waste silk yarn (2), Chemical composition and physical properties of silk (1), Effect of acid and Alkalies (1), Various varieties of silk with brief description (1).
Total Lectures required =08

Unit (4): Morphological structure of wool (2), Composition of wool fibres (1), Properties of wool fibres (1), Varieties of wool fibre with brief description (2).
Total Lecture required = 06
Grand total of Lectures = 36

Books:

1. W.E. Morton & J.W.S. Hearle, Physical properties of textile fibres, Textile Institute, U.K.
2. Progress in textiles: Science and technology Vol.-2
   By Dr. V.K. Kothari, I.I.T. Delhi.
3. Hand book of textile fibres by J.Gordon Cook

2. Mathematics III/ Science Based Open Elective (EAS301/ EOE031-EOE038): Syllabus to be decided by U.P.T.U.

3. Industrial Psychology/ Industrial Sociology (EHU301/EHU302) Syllabus to be decided by UPTU

4. Industrial Chemistry (EC302)

Unit (1): Clausius- Claparyon equation and its applications (3), General Phase theory-Phase rule, phase equilibrium and P.T. diagram for one component system (2), Equilibrium in two and three component system (2)
Total Lectures required = 07

Unit (2): General introduction about Acids- (a)Hydrochloric Acid, Sulphuric Acid (b)Acetic Acid, Formic Acid;(3) Alcohols, Aldehydes, and Ketones (3)
Total Lectures required = 06

Unit (3): General methods of preparation and properties of – Monohydric alcohols, polyhydric alcohols e.g. Glycol and glycerol,(2) aldehydes-formaldehyde and ketones-acetone,(2) Dicarboxylic acids- oxalic acid, DMF (3)
Total Lectures required = 07

Unit (4): Introduction of Solution- saturated and unsaturated, ideal and non ideal solution (4) Different type of concentration units (2) Effect of temperature on concentration (1)
Total Lectures required = 07

Unit (5): Fractional distillation of coal tar and its products, and their use in textile industry (3), Isolation of Xylene, Benzene, Toluene, Naphthalene and Anthracene (5)
Total Lectures required = 08
Grand Total Lectures Required = 34
BOOKS:

1. *Industrial Chemistry by **B.K.Sharma***
2. *Synthetic Organic Chemistry by **Chatwal and Gurdeep***
3. *Chemistry of Dyes and Principles of Dying by **V.A.Shenai***
4. *Physical Chemistry of Surfaces by **A.W.Adamson***
5. *Chemical Principles of Synthetic Fibre Dying by **S.M.Burkinshaw***
6. *Physical Chemistry of Dying by **Vickerstaff***

5. Polymer Chemistry (ECF303)

Unit (1): Scope of Polymer Chemistry, (a) definition of monomers, homopolymers and copolymers (2), (b) Chemistry of important monomers (1), (c) Thermosets (d) Thermoplastic (e) Elastomers (f) Tg, Tm, and solution properties of polymers (3)
Total lectures required = 06

Unit (2): Condensation polymerization (1), (b) Mechanism, kinetic and molecular weight build up (3), (c) Bulk, Solution, Emulsion and suspension polymerization (3) (d) comparison of condensation and addition polymerization (1).
Total lectures required = 08

Unit (3): Addition polymerization (1), (b) Type of addition polymerization (1), (c) Radical, cationic and anionic polymerization (2) (d) Mechanism and kinetics of polymerization (2), (e) Copolymerization, Stereo-Regular polymerization, Block and Graft polymerization (2).
Total lectures required = 08

Unit (4): Characterization of molecular weight polymers (1), (b) End-Group analysis method (1), (c) Viscosity of measurement method (1), (d) G.P.C. Techniques for measurement (1).
Total lectures required = 04

Unit (5): Application of polymer to textiles (1): Fibre forming synthetic polymer (3) Properties of fibre forming synthetic polymer (3)
Total lectures required = 07

Grand Total Lectures required = 33

BOOKS:

1. *Text Book of Polymer science by* ***F.W. Billmeyer***
2. *Text Book of polymer Vol. I, II, III by* ***M.S. Bhatnagar***
3. *Polymer science by* ***Gawaskar, Vishwanathan, sreedhar and Jaydev***
4. *Polymer chemistry by* ***B.K. Sharma***

6. Principle of Yarn Manufacture (EC304)

Unit-1: Cotton ginning, Name of ginning machines. Pre ginning treatment and post ginning treatment and different types of mixing. Different machines of blow room department with the basic idea of each machines along with their technical parameters. Principles of opening and cleaning.
Total lectures required=06

Unit-2: Carding engine with feeding. Carding and doffing arrangements. Card clothing along with their deferment makes. Carding, striping and grinding actions. Different

carding engine setting and speed of different parts. Drafts (actual & mechanical) and draft constant, Quality of web and neps etc.
Total lectures required=06

Unit-3: Draw frame objects. Different types of roller drafting systems. Stop motion and their importance. Weighting system used in draw frame. Drafting wave and roller step in draw frame.
Total lectures required=06

Unit-04: Comber along with cylinder description and needling arrangements and its action, Fibre presentation and its effects on combing. Preparatory machines for comber and its working.
Total lectures required=05

Unit-5: Basic idea of speed frame. Twisting winding and making a suitable package for next process. Ring frame along with how twist is imparted in yarn. Weighing systems in ring frame. Use of aprons.
Total lectures required=07
Grand total lectures required=30

Reference Books:

1. Essential elements of practical cotton spinning by T.K. Pattabhiram
2. Cotton blow room,carding,ring frame by Gilbert R. Merrill
3. Cotton spinning by W.Taggart
4. Spun yarn technology by Eric Oxtoby

7 *Human Values & Professional Ethics (EHU-111): Audit Course
Syllabus as provided by U.P.T.U.

1. Industrial Psychology/ Industrial Sociology (EHU402/EHU401)
Syllabus to be decided by UPTU

2. Science Based Open Elective/ Mathematics III (EOE041-EOE048/EAS401); Syllabus to be decided by U.P.T.U.

3. Principle of Fabric Manufacture (EC401)

Unit1: study and working principles of automatic cone and cheese winders. Line and circular winding m/c. Weft and precision winding m/c. Winding faults and remedies.

Unit2: Study and working principles of Beam warping m/c. Sectional warping m/c Warper Beam defects—cause and remedies.

Unit:3: Sizing m/c (Slasher,multicylinder and hot air). Sizing ingredients commonly used for cotton and synthetic yarns. Ingredients commonly used for cotton and synthetic yarns. Mixing preparation method. Controls on sizing. Defects and remedies.

Unit:4 Study and working principles of Handloom,Powerloom and Automatic Loom Elementary idea of shuttle less loom and unconventional weaving m/c . Fabric faults and remedies. Elementary idea of shuttle less loom and unconventional weaving m/c. Fabric faults and remedies. Calculation concerning production and efficiency of above all

Reference Books:

1. R. Sengupta, yarn preparation, the popular book depot Mumbai.
2. Cotton yarn weaving (ATIRA), 1980
3. M.K. Talukdar- An introduction to winding and warping.
4. P.K. Bannerjee- Yarn winding NCUTE, IIT Delhi.
5. Armored weaving technology and operation, Textile Institute U.K.

4. Technology of Bleaching and Mercerization (EC402)

UNIT-1: Inspection of loom shed{gray} fabric ,mending ,shearing ,cropping. Natural and added impurities in grey fabric. Singeing-its object and various types of singeing. introduction to various preparatory process for cotton, wool, silk, nylon polyaster, acrylic and their blends.

Unit-2: Desizing- its object and various desizing methods with its advantages and disadvantages. Scouring, its object and chemistry involved. Kiers – various types of kiers and their working.

UNIT-3 : leaching –its object and introduction .verious typs of bleaching agent such as

NaOCl, CaOCl2 and H2O2, NaClO2. Chemistry and uses of these bleaching agent .mechanism used for batch wise, semi continuous and continuous bleaching process .

UNIT 4-: Methods used for determination of degradation of cotton during scouring and Bleaching such as copper no., methylene blue absorption method, cuprammonium fluidity etc. optical whiteners agent and their applications.

UNIT-5: History of mercerization, physical and chemical change in cotton due to mercerization. Methods and equipment for yarn and fabric mercerization .
Various methods of determination of efficiency of mercerization .

Reference Books:

1. Technology of bleaching Vol. 3 by V.A. Shenai
2. Textile Scouring & Bleaching by E.R. Trotman
3. Bleaching & mercerization by J.T. Marsh
4. Bleaching & mercerization by BTRA

5. Technology of Dyeing-I (EC403)

Unit – 1: Classification of dyes according to the mood of application (2). The Principles of dyeing (1), Mechanism of dyeing (2), Dye fibre interaction/bond. (2)
Total lectures required – 7

Unit – 2: The principles and Technology of dyeing of cellulosic fibres with direct (1), azoic(1) , reactive(1) . Vat (1) , sulphur(1), indigo dyes(1) and oxidation colours (1)
Total lectures required – 7

Unit – 3: Dyeing of protein fibres with acid (1), metal complex (2), mordant, basic (2) basic dyes (2)
Total lectures required -7

Unit – 4: Batch wise (2), Semicontinuous (2) and Continuous method of Application of dyes on Natural fibres and machinery involved therein (2).
Total lectures required -6

Unit – 5: Dyeing of synthetic fibres viz- polyester(2) ,acrylic(1), and their blends(2) Recent developments in dyeing techniques(2), problems associated with dyeing(1), common fault and their remedies.(2)
Total lecture required - 10
Grand Total of lectures=37

Reference Books

1. Chemical processing of cotton and p/c blends – ATIRA
2. A glimpse on the chemical technology and textile fibres by R.R. Chackrawartty
3. Technology of Dyeing by V.A. Shenai

6. Physical Chemistry of Dyeing (EC404)

Unit 1: Introduction to the theory of dyeing absorption- law of light absorption , rate of fibre structure in dyeing.

Unit 2: Absorption- Isotherms, chemical potential of dye, activity of dye in solution and in fibre, physio- chemical factors in dyeing.

Unit 3: Kinetic – diffusion of dyes inside the fibre rate of dyeing, thermodynamics – entropy of dyeing, temperature effects, heat of dyeing.

Unit 4: Simple relation for calculation of affinity, diffusion, coefficient of dyes into fibre.

Unit 5: Colour- atwbls and Nansels Colour system methods of Colour measurement, testing of dyed matinal grey scale f fastness tasts light fastness, washing fastness, Estimate of dyes with fibre. Processing of textiles, Chemistry of dyeing.

Text Book: -

1. Synthetic dye stuff – Cain & Thrope
2. Chemistry of dye agent principle of dying -V.K. Shenai

Reference Book:

1. Process of dye chemistry - Fierz, David & Ballengray
2. Dyeing and chemical technology of fibre - E.K. Frotran
3. Processing of textile fibres - Sadov
4. Chemistry of synthetic dyes - Venkataraman

7. *Human Values & Professional Ethics (EHU-111): Audit Course Syllabus as provided by U.P.T.U.

5TH SEMESTER, B. TECH. TEXTILE CHEMISTRY

1. Engineering and Managerial Economics (EHU501)
Syllabus to be provided by U.P.T.U.

2. Textile Testing-I (ETCEF-501)

Unit (1): Introduction of textile testing (1), sampling, random sampling, biased sampling (1), sampling techniques, square, cut square, zoning technique (1), selection of sample for testing, grading of cotton fibre with respect to staple length (1), laboratory measurement of fibre length, span length, Baer sorter (1), Shirley photo electric staple, servo fibro graph (1), salient features of HVI (1).
Total Lectures Required = 7

Unit (2): Atmospheric conditions for testing, absolute & relative humidity, moisture regain & moisture contest (1), importance of moisture in textiles, measurement of moisture regain & content (1), effect of moisture on properties (physical & mechanical) of textile material, factors affecting the regain (2), correct numerical weight, oven dry weight (1), shrilly moisture meter (1).
Total Lectures Required = 6

Unit (3): Dry and wet bulb hygrometer, sling assmann, hair hygrometers (2), control of testing room atmosphere (1), AFIS (1), Napping potential (1), Nep count (1), rating of neps, maturity coefficient measurement by NaOH method, fibre fineness by airflow meter & Sheffield micronaire (2).
Total Lectures Required = 9

Unit (4): Fibre bundle strength by Pressley, stelometer (2), fibre quality index, linear density of man made fibres and strength (1), spin finish, crimp (1), Trash, Shirley trash analyzer (1). Yarn numbering system (1), wrapping test for lap, sliver, roving (1),
Total Lectures Required = 7

Unit (5): determination of yarn count, diameter (1), average & resultant count of folded yarn, relation between Ne, D, T, Nm (1). Instruments used for determination of count, quadrant balance, Knowles balance, bees lay balance and physical balance (2), Twist, classification of twist, twist measurement, direct counting method, continuous twist tester, twist-untwist method, (2), Twist tester, (2), R.B. twist tester, level of twist (1).
Total Lectures Required = 9
Grand total of lectures required = 36
Reference Books: -

1. Quality control and testing management by Dr. V.K. Kothari
2. Principle of textile testing by J.E. Booth
3. Physical testing of textiles by B.P. Sarulle.

2. Chemistry of Dyes & Colour Chemistry (EC502)

Unit – 1 Unit organic process – sulphonation, nitration amination of mono and poly nuclear aromatic molecules which are listed in the preparation of the intermediates (4), Preparation of important intermediates – Napthionic acid, Koch acid, H-acid(4)
Total lectures required – 8

Unit – 2 Classification of dyes according to chemical constitution (2) Quantitative testing of dyes (3), Relation between colour and chemical constitution (3).
Total lecture required – 8

Unit – 3 Chemistry including the synthesis of Dyes: direct, acid & basic dyes.
Total lecture required – 6

Unit – 4 Chemistry of azoic colours, vat dyes, reactive dyes, disperses dyes.
Total lecture required – 8
Grand Total lecture required – 30

Reference Books:

1. Synthetic Dyes by Cain & Thorpe
2. Chemistry of Dyes Vol. 2 by V.A. Shenai
3. Chemistry of Synthetic Dyes Vol 1-8 by Venkatraman
4. Text Book of Dyes by M.G. Arora

4. Technology of Dyeing-II (EC503)

Unit – 1: Mass colouration of polyester (2), nylon (2), acrylics (2), polypropylene (2), viscose fibres (2).
Total lecture required – 10

Unit – 2: German Bann, its importance and acts, various pollutants caused by dyeing with various dyes.
Total lecture required – 6
Unit – 3: Development in dyeing m/cs- jet dyeing m/cs(2), soft flow m/cs(2), infra colour dyeing m/cs(2) closed jiggers(2).
Total lecture required – 8

Unit – 4: Natural dyes (1), their sources (1) and extraction (2). Different mordants used in dyeing (2), different methods – pre, post, simultaneous mordanting (4)
Total lecture required – 9

Unit – 5: Dyeing of cotton (2), wool (2), silk (2), with natural dyes Fastness properties of dyed goods (4).
Total lecture required - 10
Total no. of lectures =43

Reference & text Books:

1. Technology of Dyeing Vol 6 by V.A. Shenai
2. Dyeing & Chemical technology of textile Fibres by E.R. Trotman
3. Textile Chemistry Vol III by R.H. Petrers
4. Chemical Processing of Synthetic Fibres & Blends by Datye & Vaidhya

5. Textile Auxiliaries (EC504)

Unit-1: General Consideration and detailed Classification of textile auxiliaries, Mechanism of surface activity, Surface active agents wetting and detergency (8).

Unit-2: Leveling agents, dispensing agents, Requesting agents stripping agents nd their application (7).

Unit-3: General method of preparation of wetting agents, and soaps, Emulsion-its they and emulsifying agents H.L.B use of auxiliaries used in scouring , bleaching dyeing and printing(8).

Unit-4: Determining the effacing of wetting agents, Evaluation of detergency, evaluation of leveling agents, Evaluation of flame retarding agents, Estimation of available chlorine (8).

Total Lecture required= 31

Reference Books

1.	Textile auxiliaries	V.A. Shanai
2.	Textile auxiliaries finishing chemicals	A.A. Vaidaya trivedi
3.	Textile Scauring and bleading	U.A. Shanai

6. Chemistry and Production of Fibres (ETCEF-505)

Unit (1): Polyethylene Terephthalate Fibre – History of development, Polymer production by DMT & PTA route, Chips drying, Fibre manufacturing, Effect of process variable on properties of polyester fibre, some dope additives for specialty polyester fibre, Properties of polyester fibre.
Total Lectures required = 08

Unit (2): Polyamide Fibres – History of development, Different types of polyamide fibres, Nylon polymer production by continuous polymerization in VK Tube, Manufacturing of Nylon 6 fibre by melt spinning, Properties of nylon 6 fibre, Polymer production of Nylon 66, Nylon 66- fibre formation by melt spinning, Properties of Nylon 66 fibre, Brief introduction to Aramid fibres.
Total Lectures required =08

Unit (3): Introduction of polyolefin fibres, Polymerization of polyethylene, Polyethylene (PE) fibre formation, Properties of polyethylene fibre, Different type of polypropylene

(PP), Polymerization of polypropylene, Polypropylene fibre formation, Properties of polypropylene fibre.
Total Lectures required =08

Unit (4): Introduction of vinyl fibres, Polyacrylonitrile (PAN) fibre, Use of ionic and neutral co-monomers, Polymerization of PAN, Acrylic fibre- formation by dry spinning, Dry –jet-wet spinning process, Effect of process variables on properties of PAN fibre, Dope formation, Properties of PAN fibre, Introduction to polyurethane fibre.
Total Lectures required = 08
Unit (5): Introduction of regenerated fibre, Concepts of regeneration of fibre, Raw material for viscose rayon, Manufacturing sequence of viscose fibre, Steeping and pressing, Cutting and shredding, Ageing, Xanthation of sodium cellulose, Mixing and filtration, Ripening, Wet spinning of viscose rayon, Formation of serrated edge cross-section of viscose rayon, Viscose fibre properties, Introduction of cuprammonium rayon in brief, Introduction of cellulose acetate rayon in brief, Introduction of lyocell fibre in brief.
Total Lectures required =08

Grand total lectures required = 40
Books:

1. Manufactured fibre technology by V.B. Gupta & V.K. Kothari
2. Essential fibre chemistry by M.E. Cartor
3. Synthetic fibres by Fourne
4. New fibre by T.Hongu, G.O. Phillips, Woodhead publications.
5. Fibre chemistry by M. Lewin, E.M. Pearce, Marcel & Dekkan Inc.
6. Regenerated Cellulosic fibres by C. Wooding, Woodhead publishing Ltd.

7. *Human Values & Professional Ethics (EHU-111): Audit Course Syllabus as provided by U.P.T.U.

6TH SEMESTER, B. TECH. TEXTILE CHEMISTRY

1. Industrial Management (EHU601) Syllabus as provided by U.P.T.U.

2. Departmental Elective-I: Structure and Properties of Fibres/ Post Spinning Operations

2.1. Structure and Properties of Fibres (EC011/ETEF602)

Unit (1): Moisture absorption, heat of absorption (2), differential heat of absorption, integral heat of absorption (2), Quantitative theory of heat moisture absorption (2), Rate of moisture absorption (1)
Total Lectures Required = 7

Unit (2): Mechanical properties of fibres (2), Relation between structure and mechanical properties of fibres (2), Basic mechanical properties (tenacity elongation, modulus, work of rupture (2), Elastic recovery, time effects (2).
Total Lectures Required = 8

Unit (3): Thermal behavior of textile fibres by DSC (2), TGA, thermal mechanical analysis (2), Density gradient column (2), Preparation of density gradient coloumn (2).
Total Lectures Required = 8

Unit (4): Birefringence behavior, dielectric properties (2), fibre friction, fibre friction measurement (2), fibre yarn to surface friction measurement and static charge measurement (5)
Total Lectures Required = 7

Unit (5): Creep behavior (2), Optical properties of fibres (2), concept of moisture absorption by fibres (2), relation between fibre structure and fibre properties (2).
Total Lectures Required = 8
Grand Total of lectures required = 38
Reference Book: -

1. Manufactured fibre technology by V.B. Gupta, V.K. Kothari
2. Physical properties of fibre by J.W.S. Hearle
3. Thermal behavior of material by Turi
4. Modern yarn production by Ray
5. Textile fibres by ATIRA
6. ASTM Standard books
7. Absorberey chartering
8. Polymers by fibre & textiles encyclopedia
9. Advances in fibre source by S.K. Mukhopadhyaya

2.2. Post spinning operation (ETCE012/EF604)

Unit (1): Concept of drawing ,drawing unit (3),Effect of drawing on orientation and crystallizations(3),Drawing of melt spun filaments, drawing of as-spun fibres through neck formation(3).
Total of lectures required=9

Unit (2): Heat setting ,nature of heat set(3),Influence of heat setting variables on structures &properties of fibres(3),Temporary, semi permanent and permanent heat setting, heat –setting equipments(3).
Total of lectures required=9

Unit (3): Method of producing bulk-false-twist texturing (3),effect of process variables, time, temperature, twist and tension on properties of false twist textured yarn,(3),edge crimping(3).
Total of lectures required=9

Unit (4): Air-jet texturing process(3),stuffer-box texturing process(3),effect of process variables on yarn properties of air jet & stuffer box textured yarns, Role of water in air jet texturing(4).
Total of lectures required=10

Unit (5): Tow to top conversion, stretch breaking (2),cutting method(1), Sydel stretch breaking(2),Pacific tow to top cutting system(2),Turbo stapler(2).
.**Total of lectures required=9**
Grand total lectures required =46.
Reference book:-
1. Manufactured fibre technology by V.B.Gupta & V.K.Kothari.
2. Synthetic fibre by Vaidhya
3. Textile fibre part-ll by V.K.Kothari.
4. Textured yarn technology by J.W.S.Hearle.

3. Textile Testing-II (ETCEF-601)

Unit (1): Textile properties of yarn and fabric(2), stress-strain curve(1), various methods for finding of yield point(1), methods for finding of various modulus(1), destination of tenacity(1), stiffness of fabric(1).
Total Lectures Required = 7

Unit (2): Yarn testing m/cs- single yarn strength tester, lea strength tester (2), fabric strength tester- impact tester, Graph test, fabric B.S. Test (3), Scott serigraph, Instron testing m/c (2).
Total Lectures Required = 7

Unit (3): Fabric abrasion & resistance to wear (1), Bursting strength test (1), Tear test, Wear (1), Brief introduction of FAST & KAWABATA (1), estimation of fabric thickness, rigidity, air permeability (1), water repellency (1), drape handling (1), crimp, thermal transmission properties (1).
Total Lectures Required = 8

Unit (4): Cover factor of fabric (1), snagging-mace snagging test (1), weaver trials, advantages and disadvantages of weaver trial (1), lab test (1), measurement of lap, sliver, yarn irregularity (1) salient features of Uster evenness tester (2)
Total Lectures Required = 7
Grand Total of lectures required = 29

Reference Books: -

1. Physical testing of textiles by B.P. Saville.
2. Quality control and testing management by Dr. V.K. Kothari.
3. Principles of textile testing by J.E. Booth.
4. Quality control by V.K. Kothari.

4. Coating of Textiles (EC602/ETEF011)

Unit-1: Polymeric materials for coating- (Rubbers: natural and synthetic, (3)(Polyvinyl chloride, Polyurethane, Acrylic polymers, Adhesive treatment (4)
Total lecture required=7

Unit-2: Coating Methods: Knife coating (1), Roll coating (1) transfer coating (1), Rotary screen printing (2) calendaring hot melt coating (2)
Total lecture required=7

Unit-3: Physical properties of coated fabric (2) Rheology of coating (1), Rheological Behaviors of fluids (2) pastes (1) hydrodynamic analysis of coating (2)
Total lecture required=8

Unit-4: Fabric for foul weather protection- Clothing comfort (1) Impermeable coating (1) breathable fabric (1) Non Apparel coating (1), Fabrics for Chemical protection (1) Thermo chromic Fabric (1) Temperature Adaptable Fabrics (1) Camouflage nets (1) Metal and conducting polymer-coated fabrics (2) Radiation cured coating (1)
Total lecture required=8

Unit-5: Test methods, Coating per Unit area, Degree of fusion/curing of coating (1) blocking Abrasion resistance (1) Test for colour- Fastness to dry and wet rubbing, Resistance to water penetration (3) Air permeability (1) water vapour permeability (1) low temperature bend test (1) low temperature impact test (1)
Total lecture required=9
Grand Total lecture required=39

Reference Books:
1. Coating & Laminated Textiles by Water Fung
2. Coated Textile by A.K. Sen
3. Coated Fabric technology Vol 1-3 Technomic publication
4. Coated & laminated Fabric by AATCC symposium

5. Technology of Printing-I(EC603)

Unit (1): Introduction to printing (1), Natural printing thickener (1), classification of printing thickener (1) and methods of thickener paste preparation (1), emulsion thickness & synthetic thickener (1), and mechanism of viscosity build up in emulsion (1) and synthetic thickener (1)
Total Lectures Required = 7

Unit (2): Preparation of cloth for printing (2), preparation of colour for printing (2).
Total Lectures Required = 8

Unit (3): Methods of printing like block (2), spray (1), transfer (2), roller (3),
Total Lectures Required = 8
Unit (4): Manual and automatic flat bed screen printing (2), preparation of screen (& screen printing) (3), rotary screen printing (3).
Total Lectures Required = 8

Grand Total of lectures required = 31
Reference Books:-
1. Tech. of printing - V.A. Sherai
2. Printing - D.G. Kale
3. Tech. of textile printing – R.S. Prayag

6. Technology of Finishing-I (EC604/EF012)

Unit - 1: Principle of finishing of cotton, wool, silk (2), linen & jute (2). Classification of various finishing – (a) Mechanical (3)
(b) Chemical (3), Temporary and durable/permanent finishes. Semi permanent.
Total lectures-10

Unit – 2: Finishing machines – Mangle &their function (3), drying machines-(2), Stentering (2), Calandering (2), Samprising-(2), Decalising
Total lecture required - 11

Unit – 3: Beetling of linen (1), Crimping of Silk &rayan, Finishing of wool1), Softening, Stiffining (3).
Total lecture required – 8

Unit – 4: Easy care finish (3) , minimum application technique including foam technology (5)
Total lecture required – 8

Unit – 5: Finishing of woolen fabrics Dry Wet decatising, Felting, Milling, Permanent Setting, Shrink Proofing.
Total lecture required – 8
Total no. of lectures =45

Reference & Text Books:
1. Introduction to textile finishing by J.T. marsh
2. Chemical Processing of Synthetic Fibres & Blends by Datye & Vaidhya
3. Textile Finishing by V.A. Shenai

7. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

7TH SEMESTER, B. TECH. TEXTILE CHEMISTRY

1. 1. Open Elective-I: (EOE071-074) Syllabus as provided by U.P.T.U.

2. Department Elective-II: Garment Manufacturing Technology (ETCEF021)/ Nonwoven (ETCEF022)

2.1 Garment Manufacturing Technology (ETCEF021)

Unit (1): Introduction to Garment Manufacturing Technology, Sample Cutting, Fusing Technology, Sewing Technology, Pressing Technology, Finishing and Inspection, Sample Room, Communication, and organization.
Total Lectures required =6
Unit (2): Introduction to Measurement of Fabric Dimensional Properties, Fabric Comfort, Thermal Comfort, Objective Evaluation of Fabric, Low Stress Fabric Properties, Kawabawa System, Fabric Assurance by Sample Testing, Fabric Defects, Fabric Inspection and Feedback to back process.
Total lectures required =8

Unit (3): Introduction to Garment Cutting, Marker Planning, Efficiency of Marker Planning, Methods of Marker Planning and Marker Use, Spreading of the Fabric to form a lay, Spreading Requirements, Methods of Spreading, Fabric Packages, objective of Cuttings, Methods of Cuttings,
Total lectures required =8
Unit (4): Introduction to Seam, Stitch, Stitch Classification, Stitch Structure, Seam Formation, Joining Material, Surface Characteristics, Seam Appearance, Damages (Thermal and Mechanical), Seam Performance, Seam Degradation, Seam Failure and Seam Testings, and Seaming Machine.
Total lectures required =8
Unit (5): Importance of Garment Processing and Finishes, Types of Garments, Processing of Garments and Specialty Garment Finishes.
Total lectures required =6

Grand Total Lectures required = 36
Reference Books:

1. Introduction to Garment Manufacturing Technology By T Ramchandran
2. Garment Manufacturing Technology by By T Ramchandran
3. Practical Clothing Construction Part I & II by Mary Methews

2.2 Non Woven (ETCEF022)

Unit 1: Concept about felts and non woven (3), application of non-woven fabrics (2), fibre- characteristics needed for the manufacture of non-woven (2).
Total =7
Unit 2: Manufacturing passage for the felt preparation (4), web chrematistic and their influence properties (3)
Total =7

Unit 3: Process variable (2), and their effect on properties of non woven (4).
Total = 6

Unit 4: Various system of binding the web (5), and their merits (2).
Total = 7

Unit 5: Flocked fabric (2), Laminates (2), latest development in non-woven industry (3).
Total = 7
Grand total = 34

Reference Book & Text Book –

1. Non Woven – N.N. Banarjee
2. Non woven – NCUTE
3. Fibre to fabric – Corbman Bernard P.

3. Process House Planning (EC701)

UNIT 1: 1. Plan layout, concept of Plan layout, types of Layout (process, product and combination type), factors affecting plan layout.
2. Production, types of production-Mass production, job production and batch production
Materials planning and allocation, Process planning and allocation, Record keeping regarding men, materials and machine.
Inventory control: need of inventory control, levels in inventory control, Duty & responsibility of shift in charge

UNIT 2: Maintenance, Objective and importance of maintenance, Types of maintenance-procedures and advantages,
2. Materials Handling, Importance of materials handling in a process house,Handling of dyes & chemicals –methods & precautions, Benefited systems of handling of processed goods.

UNIT 3: Accidents and safety measures, Types of accidents-fire, mechanical & chemical accidents, Common sources of different types of accidents and their prevention -Methods of minimizing the accidents in a process house,
2. Cost Estimation, Introduction and function of cost estimation, Estimate of procedure, Elements of cost

Unit 4: Environment Protection, Importance effulgent characteristics – their effect in environment, Tolerance limit enforced by state pollution Boards & its purpose.

-Characteristics of process waste streams-process, process, possible pollutants & nature of waste water.
-Methods of disposal of industrial waste (from dye house & print house specially)
-Various methods of effluent treatment
-Design layout & functioning of an effluent treatment plant
-Red listed dyes & chemicals.
2. Water energy (steam) Source & its conservation
-Steam and water consumption
-Reutilization of water
-Recovery of chemicals from waste water
-Methods of minimizing water & steam consumption
.
UNIT 5: Need & scope of suitable ventilation & lightening system in a process house
2. Standard and codes-National and international codes .ISO 9000-concept and its implications.

Reference Books:

1- Art of Dyeing by B.S. Chauhan
2-Health hazards in Textile mills by NITRA
3-Dye house management, Colour Publication, Bombay
4-Modern TextileManagementJ.BRattan, Abhishek Publication, Chandigarh.
5-Water and effluents in textile by ATIRA.
6-Economy, energy and environment in textile wet processing by S.S. Trived.
7-Occupational Health and Safety inn Textile mills by Dr. V.A.Shenai,Sevak Publication, Mumbai
8-Energy Conservation in Textile WET Processing by Dr.M.L. Gulrajani, Mahajan publication, Ahemdabad.

4. Technology of Finishing-II (EC702)

Unit – 1: Water proofing and water repelling, testing of water repellancy, mechanism of water repelling, area of use
Total lecture required - 8

Unit – 2: Principle & mechanism of flame retardency, Flame retardency of cotton, polyester and P/C blends. Testing methods for evaluation of flame retardency.
Total lecture required - 8

Unit – 3: Finishing of woolen fabrics- Moth proofing, permanent set& testing.
Total lecture required -6

Unit – 4: Finishing of Synthetic fibres/fabric – Heat setting (2), Delusturising (2), antistatic (2), soil resistance finishes (2).
Total lecture required - 8

Unit – 5: Finishing of synthetics and Lenin fabrics eg. 100% Polyester, Nylons, acrylics and their blends with cotton, viscose, wool.
Total lecture required -8
Total lectures=38

Reference Books:
1. Introduction to textile finishing by J.T. marsh
2. Chemical Processing of Synthetic Fibres & Blends by Datye & Vaidhya
3. Textile Finishing by V.A. Shenai

5. Waste Management & Pollution Control (EC703)

Unit (1): Toxicity of intermediates, dyes, processing aids -bleaching, dyeing, printing and finishing auxiliaries etc. .Analytical methods for various pollutants. Formaldehyde, Pentachlorophenol, Biological Oxygen Demand(BOD), Chemical Oxygen Demand(COD).

Total lectures required = 08

Unit (2):. Environmental Impact assessment, Definition & Need, Introduction to environmental impact assessment methodology. unit process: waste minimization and recycling

Total lectures required = 06

Unit (3): Textile effluent and their characterization. Methods of effluent treatment , Disposal of effluents, Reuse of water in a process house, Fiber and Polymer Waste, Recovery and recycling of monomer. Modification of polymer waste its utilization.

Total lectures required = 06

Unit (4): Source of water: Factors contributing water pollution and their effect, water pollution parameters, physical, biological, chemical standards for quality of treated water. Effluent treatment methods and control, basic principles- Unit operations (sedimentation, precipitation, filtration, and incineration), specific pollutants:. (8)

Total lectures required = 08

Unit (5): Pollution of air, causes, effect, monitoring and control, Source of noise pollution, its effect and control .Legislation- salient provisions of water act, Air act, Environment protection act, . Environment Impact Assessment: Basic principles, purpose, components, methodology and constraints.

Total lectures required = 07

Grand total lectures required = 36

Reference Books:
1. Basic course in environmental studies. – S. Deswal & Anupama Deswal.
2. Environment impact Assessment by Mc. Graw Hill by Caeter L.W.

3. Environment Pollution & Control by H.S> Bhatia
4. Environmental Engineering & Management by S.K. Dhameja
5. Textile management by V.D.Dudeja.
6. Water and effluent in textile mills by P.B.Jhala.

6. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

8TH SEMESTER, B. TECH. TEXTILE CHEMISTRY

1. Open Elective-III: Syllabus as provided by UPTU

1. EOE081 Non Conventional Energy Resources/
2. EOE082 Introduction to Soft Computing/
3. EOE083 Product Development/
4. EOE084 Automation and Robotics

2. Departmental Elective III: Technical Textiles/ High Performance Fibres

2.1 Departmental Elective III: Technical Textiles (ETCEF 031)

Unit-1: Textiles in filtration, Filter fabric construction- Woven, needle felt & knitted filter fabric (2), Finishing treatment of filter fabric (2), Thermal and chemical properties of filter fabric (2), Essential requirements of good filter fabric (2).
Total = 8

Unit-2: Waterproof breathable fabrics- Densely woven fabric, membranes, coatings (3), Membranes- Micro porous membrane, Hydrophilic membrane (2), Coatings-Microporous coating, Hydrophilic coatings (2), Performance of waterproof breathable fabrics (2).
Total =9

Unit 3: Medical textiles, Fibres used (1), Classification of medical textiles – Nonimplantable materials-wound dressings, bandages, plasters etc. (2), Extracorporal devices – Artificial kidney, liver, lung (2), Implantable material- Sutures, Soft tissue implant, Orthopaedic implants, Cardiovascular implants.(2),Healthcare/ hygiene products (1).
Total =8

Unit 4: Smart textiles brief introduction of smart textiles, classification of smart textile(2), passive smart textiles(2) , active smart textiles , very smart or ultra smart textiles (2), application of phase change material in smart textiles(2), brief discussion of smart shirt, smart suit, musical jacket, space suit etc(2).
Total =10

Unit 5: Introduction of Geo Textile (2), Classification of Geo Textiles, Functions of Geo textiles-soil reinforcement, drainage (fluid transmission), filtration, Separation, erosion control/absorption (2), objects of geo textiles, manufacturing of geo textile (3).Essential properties of geo textiles- Mechanical determinants, Hydraulic determinants, durability determinants (2)
Total 7

Grand Total 42

Reference Books:-

1. Hand book of Technical textiles – AR Harrocks, SC Anand (Textile Institute, wood head publication).
2. Smart fibre, Fabrics and Clothing- Tao X (Textile Institute, Wood Head Publication).
3. Shears Handbook of Industrial Textiles.

2.2 High Performance Fibres (EC032/EF801)

Unit – I; Introduction- Definition, molecular dimensionality mechanical properties, Fibre markets, Hi -Performance Gelspun Polyethylene fibres- Manufacture, fibres characteristics, properties & applications.
Total lectures required=8

Unit- II: Aramids- Introduction, polymer preparation, Spinning, Structure & properties, applications, Fibres based on liquid crystalline polymer (PPTA fibre)
Total lectures required=7

Unit- III: Carbon Fibres- Physical properties, PAN bases Carbon fibres, Pitch based Carbon fibres, Vapour grown Carbon fibres, Applications
Total lectures required= 7

Unit-IV: Glass Fibres- Glass for Fibres, Fibre manufacture, fibre finish, fibre properties& application, optical fibres
Total lectures required=6

Unit –V: Vectran (Melt spun wholly aromatic polyester fibre), Fibre production, properties & application, PBO (Polyphenylene benzobisoxazole) fibres- Fibre production, properties & application. PEEK Fibres -Fibre production, properties & application
Total lectures required=8
Grand total of lectures required= 36

Reference Books
1. High Performance Fibree by J.W.S. Hearle

3. Technology of Printing-II (EC801)

Unit-1: Various styles of printing (1) including resist (1) & discharge style of printing (1) Printing of cotton) wool and silk with different dye classes (4).
Total lecture required=7

Unit-2: Printing of polyester (2) nylon (2) acrylic (2) and polyester (2) cotton blends with different dye classes.
Total lecture required=8

Unit-3: Pigment printing of cotton (3), Other styles of printing like raised (1) crepon(1) damask(1) metal (1) and flock(1)
Total lecture required = 8

Unit-4:), Faults in printing & their prevention (2), Methods of print fixation (2) after treatment after printing (3)
Total lecture required =7

Unit-5: Various machines used for drying (1) steaming (1) washing (1) ageing (1) and curing (1) Transfer printing (1), methods. Sublimation (1) transfer printing of polyester Digital or inkjet printing (1)
Total lecture required=6
Grand Total lecture required=36
Reference & text Books:

1. Technology of Printing by V.A. Shenai
2. Technology of Printing by G.D. Kale
3. Chemical Processing of Synthetic Fibres & Blends by Datye & Vaidhya

4. Development in Wet Processing (EC802)

Unit – 1: Water proofing and water repelling, testing of water repellancy, mechanism of water repelling, area of use
Total lecture required - 8

Unit – 2: Principle & mechanism of flame retardency, Flame retardency of cotton, polyester and P/C blends. Testing methods for evaluation of flame retardency.
Total lecture required - 8

Unit – 3: Finishing of woolen fabrics- Moth proofing, permanent set& testing.
Total lecture required -6
Unit – 4: Finishing of Synthetic fibres/fabric – Heat setting (2), Delusturising (2), antistatic (2), soil resistance finishes (2).
Total lecture required - 8

Unit – 5: Finishing of synthetics and Lenin fabrics eg. 100% Polyester, Nylons, acrylics and their blends with cotton, viscose, wool.
Total lecture required -8
Total lectures=38

Reference Books:

1. Technology of Printing by V.A. Shenai
2. Chemical Processing of Synthetic Fibres & Blends by Datye & Vaidhya
3. Introduction to textile finishing by J.T. marsh

5. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U.:

U.P TECHNICAL UNIVERSITY, LUCKNOW
Study and Evaluation
[Effective from the Session 2009-10]
TEXTILE ENGINEERING

YEAR 2nd, SEMESTER- III

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	ETCEF301	Natural Textile Fibre	2	1	0	15	10	25	50	75	3
2	EAS301/ EOE031-EOE038	Mathematics-III /Science Based Open Elective	3	1	0	30	20	50	100	150	4
3	EHU301/ EHU302	Industrial Psychology/ Industrial Sociology	2	0	0	15	10	25	50	75	2
4	ETE302	Yarn Manufacture-I	3	1	0	30	20	50	100	150	4
5	ETE303	Fabric Manufacture-I	3	1	0	30	20	50	100	150	4
6	ETEE304	Wet Processing of Textiles-I	3	1	0	30	20	50	100	150	4
7	EHU-111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF301P	Identification of Fibres-I	0	0	3	0	20	20	30	50	1
9	ETE302P	Yarn Manufacture-I	0	0	3	0	20	20	30	50	1
10	ETE303P	Fabric Manufacture-I	0	0	3	0	20	20	30	50	1
11	ETEF304P	Wet Processing of Textiles-I	0	0	3	0	20	20	30	50	1
12	GP301	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								**1000**	**26**

*** Audit course**

Paper Code	**Science Based Open Electives:**
EOE031/EOE041	Introduction to Soft Computing (Neural Network, Fuzzy Logic and Genetic Algorithm
EOE032/EOE042	Nano Sciences
EOE033/EOE043	Laser Systems and Applications
EOE034/EoE044	Space Sciences
EOE035/EOE045	Polymer Science & Technology
EOE036/EOE046	Nuclear Science
EOE037/EOE047	Material Science
EOE038/EOE048	Discrete Mathematics

U.P TECHNICAL UNIVERSITY, LUCKNOW
STUDY & EVALUATION SCHEME
TEXTILE ENGINEERING
[Effective from the Session 2009-10]

YEAR 2nd, SEMESTER- IV

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU402/ EHU401	Industrial Sociology/ Psychology	2	0	0	15	10	25	50	75	2
2	EOE041-EOE048/ EAS401	Science Based Open Elective/m Mathematics-III	3	1	0	30	20	50	100	150	4
3	ETE401	Yarn Manufacture-II	3	1	0	30	20	50	100	150	4
4	ETE402	Fabric Manufacture-II	3	1	0	30	20	50	100	150	4
5	ETEF403	Wet Processing of Textile II	3	1	0	30	20	50	100	150	4
6	ETEF 404	Colour & Design	2	1	0	15	10	25	50	75	3
7	EHE-111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETE 401P	Identification of Fibres-II	0	0	3	0	20	20	30	50	1
9	ETE 402P	Yarn Manufacture-II	0	0	3	0	20	20	30	50	1
10	ETE 403P	Fabric Manufacture-II	0	0	3	0	20	20	30	50	1
11	ETEF 404P	Wet Processing of Textiles-II	0	0	3	0	20	20	30	50	1
12	GP401	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								**1000**	26

Paper Code	**Science Based Open Electives:**
EOE031/EOE041	Introduction to Soft Computing (Neural Network, Fuzzy Logic and Genetic Algorithm
EOE032/EOE042	Nano Sciences
EOE033/EOE043	Laser Systems and Applications
EOE034/EoE044	Space Sciences
EOE035/EOE045	Polymer Science & Technology
EOE036/EOE046	Nuclear Science
EOE037/EOE047	Material Science
EOE038/EOE048	Discrete Mathematics

U.P TECHNICAL UNIVERSITY, LUCKNOW
STUDY & EVALUATION SCHEME
TEXTILE ENGINEERING
[Effective from the Session 2010-11]

YEAR 3rd, SEMESTER- V

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU501	Engineering & Managerial Economics	3	1	0	30	20	50	100	150	3
2	ETCEF501	Textile Testing-I	3	1	0	30	20	50	100	150	4
3	ETE502	Yarn Manufacture-III	3	1	0	30	20	50	100	150	4
4	ETE503	Fabric Manufacture-III	2	1	0	15	10	25	50	75	3
5	ETEF504	Fabric Structure	2	1	0	15	10	25	50	75	3
6	ETCEF505	Chemistry & Production of Fibres	3	1	0	30	20	50	100	150	4
7	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF501P	Textile Testing-I	0	0	3	-	20	20	30	50	1
9	ETE502P	Yarn Manufacture-III	0	0	3	-	20	20	30	50	1
10	ETE503P	Fabric Manufacture-III	0	0	3	-	20	20	30	50	1
11	ETEF504P	Fabric Analysis	0	0	3	-	20	20	30	50	1
12	GP501	General Proficiency	-	-	-	-	-	50	-	50	1
										1000	**26**

*** Audit course**

U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

TEXTILE ENGINEERING

[Effective from the Session 2010-11]

YEAR 3rd, SEMESTER- VI

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU 601	Industrial Management	3	0	0	30	20	50	100	150	3
2	ETEF011/ ETCE012	Departmental Elective-I	3	1	0	30	20	50	100	150	4
3	ETCEF601	Textile Testing-II	2	1	0	15	10	25	50	75	3
4	ETCEF602	Structure & Properties of Fibres	3	1	0	30	20	50	100	150	4
5	ETE603	Yarn Manufacture-IV	2	1	0	15	10	25	50	75	3
6	ETE604	Fabric Manufacture-IV	3	1	0	30	20	50	100	150	4
7	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
8	ETCEF601P	Textile Testing-II	0	0	3	0	20	20	30	50	1
9	ETE602P	Yarn Manufacture-IV	0	0	3	0	20	20	30	50	1
10	ETE603P	Fabric Manufacture-IV	0	0	3	0	20	20	30	50	1
11	ETE604P	Yarn & Fabric Production	0	0	3	0	20	20	30	50	1
12	GP601	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1000**	**26**

*** Audit course**

Department Elective-1

Code	**Name of Subject**
ETEF011	Coating of Textiles
ETCE012	Post Spinning Operation

U.P TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHEME

TEXTILE ENGINEERING

[Effective from the Session 2011-12]

YEAR 4th, SEMESTER- VII

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EOE071-074	Open Elective-I	3	1	0	30	20	50	100	150	4
2	ETCEF021/ ETCEF022	Departmental Elective-II	3	1	0	30	20	50	100	150	4
3	ETEF701	Mill Planning & Organization	3	1	0	30	20	50	100	150	4
4	EE702	Principles & Designs of Spinning Machines	3	1	0	30	20	50	100	150	4
5	ETEF703	Knitting Technology	3	1	0	30	20	50	100	150	4
6	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
7	ETFE703P	Knitting Technology	0	0	3	-	20	20	30	50	1
8	ETE751	Project	0	0	3	-	50	50	-	50	2
9	ETE752	Seminar	0	0	2	-	50	50	-	50	1
10	ETE753	Industrial Training Viva Voce	0	0	2	-	50	50	-	50	1
11	GP701	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**15**	**3**	**10**	**150**	**270**	**470**	**530**	**1000**	**26**

*** Audit course**

Paper Code	**Open Elective-I**
EOE071	Entrepreneurship Development
EOE072	Quality management
EOE073	Operation Research
EOE074	Introduction to Biotechnology
Paper Code	**Departmental Elective-II**
ETCEF021	Garment manufacturing Technology
ETCEF022	Nonwoven

U.P TECHNICAL UNIVERSITY, LUCKNOW
STUDY & EVALUATION SCHEME
TEXTILE ENGINEERING
[Effective from the Session 2011-12]

YEAR 4th, SEMESTER- VIII

S. No	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EOE0881-084	Open Elective-II	3	1	0	30	20	50	100	150	4
2	ETCFE031/ ETEF032	Departmental Elective-III	3	1	0	30	20	50	100	150	4
3	ETEF041/ ETEF042	Departmental Elective-IV	3	1	0	30	20	50	100	150	4
4	EE801	Principles & Designs of Weaving m/cs	3	0	0	30	20	50	100	150	3
5	EHU111	*Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/ DESIGN/DRAWING											
6	ETE851	Project	0	0	12	-	100	100	250	350	8
7	GP801	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**12**	**3**	**12**	**120**	**180**	**350**	**650**	**1000**	**24**

*** Audit course**

Paper Code	****Open Elective-II**
EOE081	Non Conventional Energy Resources
EOE08	Nonlinear Dynamic Systems
EOE083	Product Development
EOE084	Automation and Robotics
Paper Code	**Departmental Elective-III**
ETCFE031	Technical Textiles
ETFE032	Theory of Textile Structure
Paper Code	**Departmental Elective-IV**
ETFE041	Process Control in Spinning
ETFE042	Process Control in Weaving

3TH SEMESTER, B. TECH. TEXTILE ENGINEERING

EAS-301/EAS-401: MATHEMATICS –III

L	T	P
3	1	0

Unit – I : Function of Complex variable

Analytic function, C-R equations, Cauchy's integral theorem, Cauchy's integral formula for derivatives of analytic function, Taylor's and Laurent's series, singularities, Residue theorem, Evaluation of real integrals of the type $\int_0^{2\pi} f(Cos\,\theta, Sin\,\theta)\,d\theta$ and $\int_{-\infty}^{\infty} f(x)\,dx$ **10**

Unit – II : Statistical Techniques - I

Moments, Moment generating functions, Skewness, Kurtosis, Curve fitting, Method of least squares, Fitting of straight lines, Polynomials, Exponential curves etc., Correlation, Linear, non –linear and multiple regression analysis, Probability theory. **08**

Unit – III : Statistical Techniques - II

Binomial, Poisson and Normal distributions, Sampling theory (small and large), Tests of significations: Chi-square test, t-test, Analysis of variance (one way) , Application to engineering, medicine, agriculture etc.

Time series and forecasting (moving and semi-averages), Statistical quality control methods, Control charts, $\bar{X}$, R, p, np, and c charts. **08**

Unit – IV : Numerical Techniques – I

Zeroes of transcendental and polynomial equation using Bisection method, Regula-falsi method and Newton-Raphson method, Rate of convergence of above methods.

Interpolation: Finite differences, difference tables, Newton's forward and backward interpolation , Lagrange's and Newton's divided difference formula for unequal intervals. **08**

Unit – V : Numerical Techniques –II

Solution of system of linear equations, Gauss- Seidal method, Crout method. Numerical differentiation, Numerical integration , Trapezoidal , Simpson's one third and three-eight rules, Solution of ordinary differential (first order, second order and simultaneous) equations by Euler's, Picard's and forth-order Runge- Kutta mehthods. **08**

Test Books :-

1. Peter V. O'Neil, Advance Engineering Mathematics Thomson (Cengage) Learning, 2007.
2. Jain, Iyenger & Jain, Numerical Methods for Scientific and Engineering Computation, New Age International, New Delhi , 2003.
3. J.N. Kapur, Mathematical Statistics, S. Chand & company Ltd.,2000

Reference Books :-

1. R.K. Jain & S.R.K. Iyenger, Advance Engineering Mathematics, Narosa Publication House, 2002.
2. Chandrika Prasad, Advanced Mathematics for Engineers, Prasad Mudralaya, 1996.

3. E. Kreysig, Advanced Engineering Mathematics, John Wiley & Sons, 2005.
4. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 2005.
5. Devi Prasad, An introduction to Numerical Analysis, Narosa Publication house, New Delhi 2006.
6. T. Veerajan & T. Ramchandrandran, Theory & Problems in Numerical Methods, TMH, New Delhi, 2004.
7. S.P.Gupta, Statistical Methods, Sultan and Sons, New Delhi, 2004.
8. Devore, Probability and Statistics, Thomson(Cengage) Learning, 2007.
9. Walpole, Myers, Myers & Ye, Probability and Statistics for Engineers & Scientists, Pearson Education, 2003.

1. (Engineering Core: Interdisciplinary) Natural Textile Fibres: (ETCFE-301)

Unit (1) : Fibre, textile fibre (1). Classification of natural textile fibres (1), Essential and desirable properties of textile fibres (2), Grading of cotton (1), Physical properties of cotton fibre (1), Effect of acid & Alkalies on cotton fibre (1).
Total Lectures required = 07

Unit (2): Production, morphological structure, properties and end uses of: Flax (2), Jute (2), Ramie (2). Introduction to specialty natural fibres like banana, pineapple, spider etc
Total Lectures required = 08

Unit (3): Production of silk (raw) (2), Morphological structure of silk (1), Production of waste silk yarn (2), Chemical composition and physical properties of silk (1), Effect of acid and Alkalies (1), Various varieties of silk with brief description (1).
Total Lectures required =08

Unit (4): Morphological structure of wool (2), Composition of wool fibres (1), Properties of wool fibres (1), Varieties of wool fibre with brief description (2).
Total Lecture required = 06
Grand total of Lectures = 36

Books:

1. W.E. Morton & J.W.S. Hearle, Physical properties of textile fibres, Textile Institute, U.K.
2. Progress in textiles: Science and technology Vol.-2
   By Dr. V.K. Kothari, I.I.T. Delhi.
3. Hand book of textile fibres by J.Gordon Cook

2. Mathematics III/ Science Based Open Elective (EAS301/ EOE031-EOE038)

Syllabus to be decided by U.P.T.U.

3. Industrial Psychology/ Industrial Sociology (EHU301/EHU302)

Syllabus to be decided by UPTU

4. Yarn Manufacture –I (ETE-302)

Unit (1): Picking and pre-cleaning of cotton (3), objects of Ginning (1), Description & working of knife-roller (2), Macarthy and Saw Gins (1).

Total Lectures required = 07

Unit (2): Objects of mixing (1), different types of mixing & blending (2), Difference between mixing & blending (1), Principles underlying the relation of cotton for mixing (1), Different mixing methods (3), Their advantages and disadvantages.
Total Lectures required = 08

Unit (3): Objects of Blow room (1), Principles of opening and cleaning (2), Various types of openers, their construction and working (2), Selection of machinery according to type of cotton and their suitable combinations (1), Blow room line for synthetic fibres (2), Blending of man made fibres with cotton, (1), Modern blow room lines (1).
Total Lectures required = 10

Unit (4): Lap feed (1), Chute feed systems and their comparison (2), Lap rejection (2), Causes of lap defects and their remedies (1), Transport control and regulation of material blow room (1).
Total Lectures required = 07

Unit (5): Problem of contamination and their remedies (2), Performance assessment of blow room (2), Waste extracted at various openers and beaters (2), Cleaning efficiency of different machines (1), lint trash ratio, nep generation, fibre breakage and their control (1), Calculations pertaining to above subject matter (2).
Total Lectures required = 10
Grand Total Lectures required =42

Reference Book:

1. Spun yarn technology Vol. I – A. Ventaksubramaniam
2. Elements of Raw Cotton & Blowroom – Dr. A.R. Khare
3. A practical guide to Opening – W. Klein
4. Cotton Spinning – Taggart
5. Spun Yarn Technology – Oxtoby.
6. Spinning blow room and carding by Prof. K.R. Salhotra & Dr. R. Chattdadhya

5. Fabric Manufacture-I (ETE-303)

Unit (1) : Objects of winding, classification of winding machines (2), Study of slow, high and super high speed warp winding machines (2), Features of slow, high & super high speed winding machines (2), Machine traverse ratio, Angle of wind, conicity of warp and weft packages, Types of wound packages (2).
Total Lectures required = 08

Unit (2): Thread stop motion, tensioning devices (1), Mechanical and electronic slub-catchers (1), Clearing efficiency of winding m/c, Knot factor, various knots used in textiles (2), Latest developments in winding, Splicing (2).
Total Lectures required = 06

Unit (3): Ways of yarn traversing, ribbon formation and method of its elimination (2), Package density and winding parameters and knotting (2), Winding falls and their remedies (2).
Total Lectures required = 06

Unit (4): High speed weft winding machines (1), Detailed study of mechanism and setting of speed on weft winders (2), Automatic and non-automatic pirn winding machines (2), Bunch motion and its importance (1).
Total Lectures required =06

Unit (5): Yarn numbering systems (1), Resultant count method for various yarns (2), Estimation of productivity of pirn winding machines (2), brief idea of classmate test report (2).
Total Lectures required = 07
Grand total lectures required =33

Reference Books:

1. R. Sengupta, yarn preparation, the popular book depot Mumbai.
2. Cotton yarn weaving (ATIRA), 1980
3. M.K. Talukdar- An introduction to winding and warping.
4. P.K. Bannerjee- Yarn winding NCUTE, IIT Delhi.
5. Armored weaving technology and operation, Textile Institute U.K.

6. Wet Processing of Textiles -I (ETEF-304)

UNIT-1: Introduction to simple preparatory process, grey cloth singeing, hot plate, roller and gas singeing m/c. Desizing – its object and various types of desizing methods. Scouring – its objects and process involved. Various types of kier -its description and working. (8)

UNIT-2: Bleaching-its objects and chemistry involved, various types of bleaching agents – their preparation and uses ,bleaching of cotton fabrics, various faults in bleaching and their remedies and removal.(8)

UNIT-3: Merecrisation objects, physical and chemical aspects of mercerization , yarn mercerization and various methods of cloth mercerization ,various methods to test the efficiency of mercerization. (7)

UNIT-4: Finishing- objects, various types of physical and chemical finishes and their applications by various finishing equipments like padding mangle ,drying m/c's ,stenter ,calendar etc.(8)

UNIT-5: Introduction to various finishes like water proofing and water repellency, flame proofing and flame retardancy ,heat setting ,softening and stiffening. (7)

Grand Total Lectures Required =38

Books:

1. Chemical processing of cotton and p/c blends – ATIRA
2. A glimps on the chemical technology and textile fibres by R.R. Chackrawartty
3. Technology of bleaching and mercerization by V.A. Shenai
4. Technology of finishing by V.A. Shenai

7 *Human Values & Professional Ethics (EHU-111): Audit Course
Syllabus as provided by U.P.T.U.

4TH SEMESTER, B. TECH. TEXTILE ENGINEERING

1. Industrial Psychology/ Industrial Sociology (EHU402/EHU401)
Syllabus to be decided by UPTU

2. Science Based Open Elective/ Mathematics III (EOE041-EOE048/EAS401)

Syllabus to be decided by U.P.T.U.

3. Yarn Manufacture –II (ETE-401)

Unit (1): Objects of carding, detailed study of revolving flat card, stripping and brushing action, Constructional features and working, details of licker-in, cylinder Doffer and flat for processing different materials.
Total Lecture required = 08

Unit (2): Flexible of metallic card clothing, grinding, processing parameters for different materials carding defects and their remedies. Fibre neps – assesment and control, carding calculations pertaining to draft and production.
Total Lectures required = 08

Unit (3): Tandem carding, Modern development in carding, Auto leveller used in carding, basic idea about woolen and jute flex carding.
Total Lectures required = 07

Unit (4): Object of draw frame, working principles of draw frame including constructional details, System of drafting, weighting in draw frame, Draft distribution, Drafting weaves, Drafting force, Coiling, over coiling, under coiling and bi-coiling and stop motion.
Total Lectures required = 08

Unit (5): Concept of ideal draft and formation of draft waves, Drafting irregularities and their causes, Principles of roller setting, Modern developments in draw frame, Auto leveling calculation pertaining to draft and production.
Total Lectures required = 08
Grand Total Lectures required = 39

Reference Books:

1. Advance technology of yarn production by R. Chattopadhadya
2. Mechanics of spinning m/c by R.S.Rangaswami
3. New spinning of by R.V. Mahendru Gowda
4. Technology of carding by R.Chattopadhahya
5. Spinning blow room and carding by Prof. K.R. Salhotra & Dr. R. Chattopadhahya.
6. Spinning drawing combing and robing by Dr. R. Chattopadhadya & R. Rangasami

4. Fabric Manufacture-II (ETE402)

Unit (1) : Objects of warping, Classification of warping m/cs, slow, high and super high speed warping m/c (2), Salient features of high and super high speed warping m/cs (2), Study and modifications of different types of creels (2), Tension devices and their setting
Total Lectures required = 07

Unit (2): Various stop motion used in warping m/cs (2), Qualities of good warping beam (1), Common faults in warper's beam and their causes and remedies (2), beam hardness and beam count (2)
Total Lectures required = 07

Unit (3): Sectional warping m/cs (2), Waxing attachment and computerized warping m/cs (2), Estimate of efficiency and productivity of various type of warping m/cs (2).
Total Lectures required = 06

Unit (4): Object of sizing and sizing terminology (1), Details of slasher sizing, multicylinder sizing (2), Hot air sizing, high pressure sizing, sizing ingredients used for cotton and synthetic warp (2), Detailed of study of various drying systems, classification of size percentage (1), Factors affecting size pick-up, causes and remedies of lapper and migration, shore hardness of squeeze roll (2).
Total Lectures required = 08

Unit (5): Slow speed motion, measuring and marking motion and slipping friction motion (2), Various controls used on a modern sizing m/c (2), Common faults occurring in sizing beam, their causes and remedies (2), Numerical problems concerning above (2).
Total Lectures required = 07
Grand total lectures required =35

Reference Books:

1. R. Sengupta, yarn preparation, the popular book depot Mumbai.
2. Cotton yarn weaving (ATIRA), 1980
3. M.K. Talukdar- An introduction to winding and warping.
4. P.K. Bannerjee- Yarn winding NCUTE, IIT Delhi.
5. Armored weaving technology and operation, Textile Institute U.K.

5. Wet Processing of Textiles II (ETEF-403)

UNIT -1: Classification of dyes according to there mode of application. Elementary knowledge of cotton dying with direct, vat, sulphur, azoic reactive and Solublised vat dyes. (9)

UNIT-2: dyeing of wool and silk with acid and basic and metal complex dyes (7)

UNIT-3: Dyeing of popular synthetic textile material like nylon ,acrylic and polyester Fastness properties of dyeing material. (7)

UNIT-4: Printing –an introduction ,classification of methods of printing .Outline of various method like hand block printing .screen printing .roller printing ,transfer printing rotary screen printing etc. (9)

UNIT-5: Thickeners and their application .Various style of printing –direct style resist style and discharge style of printing. (8)
Grand Total Lectures Required =40

Reference Books:
1. Chemical processing of cotton and p/c blends – ATIRA
2. A glimplse on the chemical tech. of textile fibres by R.K. Charkrawarty
3. Technology of dyeing byV A. Shanai
4. Technology of printing byV A. Shanai

6. Colour & Design (ETEF-404)

Unit (1): Light and colour phenomena, physical basis of colour (2), Emission & absorption theory of light, Colour vision and light theory of colours (2), Complementary colours, Chromatic circle (2), Pigment theory of colours (2), Brewster circle (2), Attributes of the primary & secondary colours, Colour measurement, Primary, Secondary, Tertiary & compound colours (1).
Total Lectures required = 11

Unit (2): Biren's triangle (2), Modification of colours (2), Coloured greys, Colours in combination, Colour contrast, contrast in hue, contrast of tone, colour harmony (1), Relative spaces occupied by colours, divisional colours, Application of colours, Mixed colour effect (1), Composition of designs, Condition to be observed during ornamentation of fabrics (1), Mode shade, Harmony of succession, gradation of hue, Different stages of colouring of textile materials (1), Colour and weave effect and its classification (1).
Total Lectures required =09

Unit (3): Bases of Textile design (2), One third and one fourth drop design (2), Half drop and drop reverse design, Unit repeating design, Geometric ornamentation (2), Construction of symmetrical designs (2), Stripe and check effect designs (2), Sari border / vertical border design (2), Factor affecting the woven designs, reversing inclined figure, Diamond, Ogee, & diagonal waved line base (2).
Total Lecture required = 14

Unit (4): A compulsory question based on drawing sheet and application of colours will be asked from any topic of unit 1 to 3.
Total lectures required for unit 4 are distributed in unit 1, 2, 3.
Grand total Lectures required = 34
Reference Books:
1. W. Watson- Textile design and colour

7. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

5TH SEMESTER, B. TECH. TEXTILE ENGINEERING

1. **Engineering and Managerial Economics (EHU501): Syllabus to be provided by U.P.T.U.**

2. Textile Testing-I (ETCEF-501)

Unit (1): Introduction of textile testing (1), sampling, random sampling, biased sampling (1), sampling techniques, square, cut square, zoning technique (1), selection of sample for testing, grading of cotton fibre with respect to staple length (1), laboratory measurement of fibre length, span length, Baer sorter (1), Shirley photo electric staple, servo fibro graph (1), salient features of HVI (1).
Total Lectures Required = 7

Unit (2): Atmospheric conditions for testing, absolute & relative humidity, moisture regain & moisture contest (1), importance of moisture in textiles, measurement of moisture regain & content (1), effect of moisture on properties (physical & mechanical) of textile material, factors affecting the regain (2), correct numerical weight, oven dry weight (1), shrilly moisture meter (1).
Total Lectures Required = 6

Unit (3): Dry and wet bulb hygrometer, sling assmann, hair hygrometers (2), control of testing room atmosphere (1), AFIS (1), Napping potential (1), Nep count (1), rating of neps, maturity coefficient measurement by NaOH method, fibre fineness by airflow meter & Sheffield micronaire (2).
Total Lectures Required = 9

Unit (4): Fibre bundle strength by Pressley, stelometer (2), fibre quality index, linear density of man made fibres and strength (1), spin finish, crimp (1), Trash, Shirley trash analyzer (1). Yarn numbering system (1), wrapping test for lap, sliver, roving (1),
Total Lectures Required = 7

Unit (5): determination of yarn count, diameter (1), average & resultant count of folded yarn, relation between Ne, D, T, Nm (1). Instruments used for determination of count, quadrant balance, Knowles balance, bees lay balance and physical balance (2), Twist, classification of twist, twist measurement, direct counting method, continuous twist tester, twist-untwist method, (2), Twist tester, (2), R.B. twist tester, level of twist (1).
Total Lectures Required = 9
Grand total of lectures required = 36
Reference Books: -

1. Quality control and testing management by Dr. V.K. Kothari
2. Principle of textile testing by J.E. Booth
3. Physical testing of textiles by B.P. Savile.

3. Yarn Manufacture – III (ETE-502)

Unit (1): Object of combing (1), system of lap preparation and study of their lap, ribbon-lap and super-lap machine (3), configuration of fibers feed and its effect on the quality of product, noil percentage and fraction efficiency of comber (3).
Total Lecturers Required =7

Unit (2): Type of combers (2), distinguishing frameless of Hillman and Nasmith combers (3), detailed study of the Nasmith type comber (2), timing and setting of Nasmith comber for different classes of cotton (1), control of comber work (1), calculation pertaining to draft, production and noil percentage (1), recent developments in combing (1).
Total Lecturers Required =11

Unit (3): Objects of speed frame (1), conventional and modern roving processes (1), mechanism involved in drafting, twisting and winding (2), basic principle of designing of cone drum (2), differential motions and their working principles (2).
Total Lecturers Required =8

Unit (4): Builder motions their objects & types (1), working principles of any modern builder motion (1), drafting system (ordinary & high) (1), processing parameters for different hank roving (1).
Total Lecturers Required =4

Unit (5): Common defects in roving package their censes and remedies (2), calculations pertaining to gearing, contents, draft, TPI and production (2), twist multiplier and roving twist (1).
Total Lecturers Required =5
Grand Total of lectures required =35
Reference Book: -

1. Spun yarn technology Vol. I – A. Ventaksubramani
2. Elements of combing – Dr. A.R. Khare
3. A practical guide to combing & drawing – W. Klein
4. Cotton Spinning - Taggart
5. Spun Yarn Technology – Oxtoby.

4. Fabric Manufacture-III (ETE-503)

Unit (1): General description of plain power looms (2), their mechanical details (2), tuning and adjustment (1). Classification of fabric and weaving machinery (2), introduction to weaving process, primary, secondary and auxiliary (2) motion of looms (2)
Total Lectures Required = 9

Unit (2): Various ways of shedding (1), various types of sheds(1), over and under pick motion(1), beat up motion(1), tappet shedding,(1), idea of construction of tappet(1), under and over pick mechanism slay(1), beating up motion(1), early and late shedding,

temples and its utility, idea about healds count and reed count in different system(1), terry mechanism(1)
Total Lectures Required = 9

Unit (3): Negative and positive take up motion (1), negative and positive let-off motions (1), causes of shuttle flying and shuttle trapping (2), merits and demerits of negative of negative and positive take up and let of motion (2), 5 and 7 wheel take up motion. (2)
Total Lectures Required = 8

Unit (4): Warp protecting motion (1), side and center weft fork motion (1), negative and positive dobby (1), cross border dobby. Preparation of lattice, Development in dobby. Scope and limitations dobby (1), brief description of Cromption dobby, Knowles dobby (1), paper dobby (1), cross border dobby (1), pegging plan dobby faults and adjustment (1) Numerical problems on loom speed, production & efficiency (1).
Total Lectures Required = 9

Grand Total of lectures required = 35
Reference Books: -

1. Weaving mechanism by Fox.
2. Weaving mechanism by N.N. Bannerjee.
3. Weaving Calculation by R. Sengupta.
4. Weaving machine & mechanism by Talukdar.

5. Fabric Structure (ETEF-504)

Unit (1): Classification of various fabrics (1), construction of plain weave and its derivatives (2), ordinary twill right hand twill (1), warp faced, weft faced & balanced twills (2), Satin regular, irregular and their extension (1).
Total Lectures Required = 7

Unit (2): combined twills, end to end and pick-to-pick combination (1). Fancy twills-large diagonal shaded twills (1), sateen base diagonals and brained twills (1), elongated twill-steep and low twills (1), pointed, wave zig-zag, curved, broken, herring borne, transposed, corks screw twills (2), Diamond, mock leno, ordinary honeycomb, brighten honeycomb (2),
Total Lectures Required = 8

Unit (3): Huck-a-back and crepe weave (), derivatives of hopsack barley corn stitched hopsack and twilled hopsack (2), Simple and wadded bed ford cords (1), weft and piques (1), principle of figuring with extra material extra warp figuring (2) extra weft, limitation of extra thread (2).
Total Lectures Required = 9

Unit (4): Backed cloths (1), weft backed cloths, warp backed cloths with weeding threads (2), double cloths, center stitched, self stitched (2), inter changing double cloth, cut effect in interchanging double cloths (2).
Total Lectures Required = 7

Grand Total of lectures required = 31

Reference Book: -

1. W. Watson Textile Design & colour Longmans Greens Co. London.
2. Z.J Grosicki Watson's Textile design and colour Newnes Buter Worth, London.
3. Z.J. Grosicki, Advance Textile Design Newnes Butter Worth, London.
4. "Nishant" A Grammar of textile.

6 Chemistry and Production of Fibres (ETCEF-505)

Unit (1): Polyethylene Terephthalate Fibre – History of development, Polymer production by DMT & PTA route, Chips drying, Fibre manufacturing, Effect of process variable on properties of polyester fibre, some dope additives for specialty polyester fibre, Properties of polyester fibre.
Total Lectures required = 08

Unit (2): Polyamide Fibres – History of development, Different types of polyamide fibres, Nylon polymer production by continuous polymerization in VK Tube, Manufacturing of Nylon 6 fibre by melt spinning, Properties of nylon 6 fibre, Polymer production of Nylon 66, Nylon 66- fibre formation by melt spinning, Properties of Nylon 66 fibre, Brief introduction to Aramid fibres.
Total Lectures required =08

Unit (3): Introduction of polyolefin fibres, Polymerization of polyethylene, Polyethylene (PE) fibre formation, Properties of polyethylene fibre, Different type of polypropylene (PP), Polymerization of polypropylene, Polypropylene fibre formation, Properties of polypropylene fibre.
Total Lectures required =08

Unit (4): Introduction of vinyl fibres, Polyacrylonitrile (PAN) fibre, Use of ionic and neutral co-monomers, Polymerization of PAN, Acrylic fibre- formation by dry spinning, Dry –jet-wet spinning process, Effect of process variables on properties of PAN fibre, Dope formation, Properties of PAN fibre, Introduction to polyurethane fibre.
Total Lectures required = 08

Unit (5): Introduction of regenerated fibre, Concepts of regeneration of fibre, Raw material for viscose rayon, Manufacturing sequence of viscose fibre, Steeping and pressing, Cutting and shredding, Ageing, Xanthation of sodium cellulose, Mixing and filtration, Ripening, Wet spinning of viscose rayon, Formation of serrated edge cross-section of viscose rayon, Viscose fibre properties, Introduction of cuprammonium rayon in brief, Introduction of cellulose acetate rayon in brief, Introduction of lyocell fibre in brief.
Total Lectures required =08
Grand total lectures required = 40
Books:

1. Manufactured fibre technology by V.B. Gupta & V.K. Kothari
2. Essential fibre chemistry by M.E. Cartor

3. Synthetic fibres by Fourne
4. New fibre by T.Hongu, G.O. Phillips, Woodhead publications.
5. Fibre chemistry by M. Lewin, E.M. Pearce, Marcel & Dekkan Inc.
6. Regenerated Cellulosic fibres by C. Wooding, Woodhead publishing Ltd.

7. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

6TH SEMESTER, B. TECH. TEXTILE ENGINEERING

1. **Industrial Management (EHU601) Syllabus as provided by U.P.T.U.**

2. **Departmental Elective-I: Coating of Textiles (ETEF011/EC601)/ 2.2 Post Spinning Operations (ETCE012/EF604)**

2.1 Coating of Textiles (ETEF011/EC602)

Unit-1: Polymeric materials for coating- (Rubbers: natural and synthetic, (3)(Polyvinyl chloride, Polyurethane, Acrylic polymers, Adhesive treatment (4)
Total lecture required=7

Unit-2: Coating Methods: Knife coating (1), Roll coating (1) transfer coating (1), Rotary screen printing (2) calendaring hot melt coating (2)
Total lecture required=7

Unit-3: Physical properties of coated fabric (2) Rheology of coating (1), Rheological Behaviors of fluids (2) pastes (1) hydrodynamic analysis of coating (2)
Total lecture required=8

Unit-4: Fabric for foul weather protection- Clothing comfort (1) Impermeable coating (1) breathable fabric (1) Non Apparel coating (1), Fabrics for Chemical protection (1) Thermo chromic Fabric (1) Temperature Adaptable Fabrics (1) Camouflage nets (1) Metal and conducting polymer-coated fabrics (2) Radiation cured coating (1)
Total lecture required=8

Unit-5: Test methods, Coating per Unit area, Degree of fusion/curing of coating (1) blocking Abrasion resistance (1) Test for colour- Fastness to dry and wet rubbing, Resistance to water penetration (3) Air permeability (1) water vapour permeability (1) low temperature bend test (1) low temperature impact test (1)
Total lecture required=9
Grand Total lecture required=39

Reference Books:
References:

1. Coating & Laminated Textiles by Water Fung
2. Coated Textile by A.K. Sen
3. Coated Fabric technology Vol 1-3 Technomic publication
4. Coated & laminated Fabric by AATCC symposium

2.2. Post spinning operation (ETCE012/EF604)

Unit (1): Concept of drawing ,drawing unit (3),Effect of drawing on orientation and crystallizations(3),Drawing of melt spun filaments, drawing of as-spun fibres through neck formation(3).
Total of lectures required=9

Unit (2): Heat setting ,nature of heat set(3),Influence of heat setting variables on structures &properties of fibres(3),Temporary, semi permanent and permanent heat setting, heat –setting equipments(3).
Total of lectures required=9

Unit (3): Method of producing bulk-false-twist texturing (3),effect of process variables, time, temperature, twist and tension on properties of false twist textured yarn,(3),edge crimping(3).
Total of lectures required=9

Unit (4): Air-jet texturing process(3),stuffer-box texturing process(3),effect of process variables on yarn properties of air jet & stuffer box textured yarns, Role of water in air jet texturing(4).
Total of lectures required=10

Unit (5): Tow to top conversion, stretch breaking (2),cutting method(1), Sydel stretch breaking(2),Pacific tow to top cutting system(2),Turbo stapler(2)
.**Total of lectures required=9**
Grand total lectures required =46.

Reference book:-
1. Manufactured fibre technology by V.B.Gupta & V.K.Kothari.
2. Synthetic fibre by Vaidhya
3. Textile fibre part-ll by V.K.Kothari.
4. Textured yarn technology by J.W.S.Hearle.

3. Textile Testing-II (ETCEF-601)

Unit (1): Textile properties of yarn and fabric(2), stress-strain curve(1), various methods for finding of yield point(1), methods for finding of various modulus(1), destination of tenacity(1), stiffness of fabric(1).
Total Lectures Required = 7

Unit (2): Yarn testing m/cs- single yarn strength tester, lea strength tester (2), fabric strength tester- impact tester, Graph test, fabric B.S. Test (3), Scott serigraph, Instron testing m/c (2).
Total Lectures Required = 7
Unit (3): Fabric abrasion & resistance to wear (1), Bursting strength test (1), Tear test, Wear (1), Brief introduction of FAST & KAWABATA (1), estimation of fabric thickness, rigidity, air permeability (1), water repellency (1), drape handling (1), crimp, thermal transmission properties (1).

Total Lectures Required = 8

Unit (4): Cover factor of fabric (1), snagging-mace snagging test (1), weaver trials, advantages and disadvantages of weaver trial (1), lab test (1), measurement of lap, sliver, yarn irregularity (1) salient features of Uster evenness tester (2)
Total Lectures Required = 7
Grand Total of lectures required = 29

Book: -

1. Physical testing of textiles by B.P. Saville.
2. Quality control and testing management by Dr. V.K. Kothari.
3. Principles of textile testing by J.E. Booth.
4. Quality control by V.K. Kothari.

4. Structure and Properties of Fibres (ETEF602/EC011))

Unit (1): Moisture absorption, heat of absorption (2), differential heat of absorption, integral heat of absorption (2), Quantitative theory of heat moisture absorption (2), Rate of moisture absorption (1)
Total Lectures Required = 7

Unit (2): Mechanical properties of fibres (2), Relation between structure and mechanical properties of fibres (2), Basic mechanical properties (tenacity elongation, modulus, work of rupture (2), Elastic recovery, time effects (2).
Total Lectures Required = 8

Unit (3): Thermal behavior of textile fibres by DSC (2), TGA, thermal mechanical analysis (2), Density gradient column (2), Preparation of density gradient coloumn (2).
Total Lectures Required = 8

Unit (4): Birefringence behavior, dielectric properties (2), fibre friction, fibre friction measurement (2), fibre yarn to surface friction measurement and static charge measurement (5)
Total Lectures Required = 7

Unit (5): Creep behavior (2), Optical properties of fibres (2), concept of moisture absorption by fibres (2), relation between fibre structure and fibre properties (2).
Total Lectures Required = 8
Grand Total of lectures required = 38
Reference Book: -

1. Manufactured fibre technology by V.B. Gupta, V.K. Kothari
2. Physical properties of fibre by J.W.S. Hearle
3. Thermal behavior of material by Turi
4. Modern yarn production by Ray
5. Textile fibres by ATIRA
6. ASTM Standard books
7. Absorberey chartering

8. Polymers by fibre & textiles encyclopedia
9. Advances in fibre source by S.K. Mukhopadhyaya

5. Yarn Manufacture – IV (ETE-603)

Unit (1): Principle and mechanism involved in drafting, twisting & winding (2), ordinary and high draft system (2), yarn twist: Terminology, twist level, concept of twist multiplier (1), propagation of twist (2),
Total Lecturers Required = 7

Unit (2): Builder motion (2), common package size, limitations to large package spinning (2), types of rings and travelers and their common uses (2), Rising and falling lappets, control rings, apron drafting system (2)
Total Lecturers Required = 8

Unit (3): System of waste collection of ring frame and types of spinning wastes (2), limitation in ring spinning and factors responsible for loss in efficiency (1), yarn faults and their remedies (2), Recent developments in ring spinning (1).
Total Lecturers Required = 7

Unit (4): Doubling: - Objects and terminology (1), study of ring doublers (1), fancy yarns (1), sewing thread and tyre Cord (1). Reeling: Objects and terminology, types of reeling construction and working of a reel (2), yarn bundling (1), calculation of draft, TPI and production of ring frame & doubling frame (2),
Total Lecturers Required = 9
Grand Total of lectures required =31

Reference Books: -

1. Elements of ring frame & doubling – Dr. A.R. Khare
2. The technology of short-shape staple spinning – W. Klein
3. Cotton spinning – Taggart
4. Spun yarn technology – Oxtoby

6. Fabric Manufacture-IV (ETE-604)

Unit (1): Jacquards shedding (1), types of jacquards and their principle of working (2), cross border jacquards (2). Harness mounting London and Norwich system (1), Card cutting (1), limitations of jacquards, Jacquards-driving study of following jacquard. (4),
(a). Single lift single cylinder
(b). Double lift single cylinder
(c). Double lift double cylinder
(d). Electronic jacquard.
Total Lectures Required = 11

Unit (2): Different systems of harness tieing. (2), terry mechanism (2), Recent development in jacquards (2)

Total Lectures Required = 6

Unit (3): Automatic looms – pirn (2), shuttle changing (2), Detailed study of various motions of automatic looms (2), cop changing loom (1), warp stop motions mechanical (1), electro-mechanical, electronic stop motion (1).
Total Lectures Required = 9

Unit (4): Multiple box motion, their types (2), principle of working of multiple box motion (1), two colours and four-colour drop box motion (1), brief description of pick-at will (1), pick and pick motion. On line process and quality control (1), estimation of productivity, snap study. (7)
Total Lectures Required = 7

Unit (5): Numerical problems concerning to above (5)
Total Lectures Required = 05
Grand Total of lectures required = 38

Reference Books: -

1. Weaving mechanism by Fox.
2. Weaving mechanism by N.N. Bannergy.
3. Weaving Calculation by R. Sengupta.
4. Weaving machine & mechanism by Talukdar

7. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U

7TH SEMESTER, B. TECH. TEXTILE ENGINEERING

1. **Open Elective-I: (EOE071-074) Syllabus as provided by U.P.T.U.**

2. **Department Elective-II: Garment Manufacturing Technology (ETE021)/ Nonwoven (ETE022)**

2.1 Garment Manufacturing Technology (ETE021)

Unit (1): Introduction to Garment Manufacturing Technology, Sample Cutting, Fusing Technology, Sewing Technology, Pressing Technology, Finishing and Inspection, Sample Room, Communication, and organization.
Total Lectures required =6

Unit (2): Introduction to Measurement of Fabric Dimensional Properties, Fabric Comfort, Thermal Comfort, Objective Evaluation of Fabric, Low Stress Fabric Properties, Kawabawa System, Fabric Assurance by Sample Testing, Fabric Defects, Fabric Inspection and Feedback to back process.

Total lectures required =8

Unit (3): Introduction to Garment Cutting, Marker Planning, Efficiency of Marker Planning, Methods of Marker Planning and Marker Use, Spreading of the Fabric to form a lay, Spreading Requirements, Methods of Spreading, Fabric Packages, objective of Cuttings, Methods of Cuttings,
Total lectures required =8

Unit (4): Introduction to Seam, Stitch, Stitch Classification, Stitch Structure, Seam Formation, Joining Material, Surface Characteristics, Seam Appearance, Damages (Thermal and Mechanical), Seam Performance, Seam Degradation, Seam Failure and Seam Testings, and Seaming Machine.
Total lectures required =8

Unit (5): Importance of Garment Processing and Finishes, Types of Garments, Processing of Garments and Specialty Garment Finishes.
Total lectures required =6
Grand Total Lectures required = 36

Reference Books:

1. Introduction to Garment Manufacturing Technology By T Ramchandran
2. Garment Manufacturing Technology by By T Ramchandran
3. Practical Clothing Construction Part I & II by Mary Methews

2.2 Non Woven (ETE022)

Unit 1: Concept about felts and non woven (3), application of non-woven fabrics (2), fibre- characteristics needed for the manufacture of non-woven (2).
Total =7
Unit 2: Manufacturing passage for the felt preparation (4), web chrematistic and their influence properties (3)
Total =7

Unit 3: Process variable (2), and their effect on properties of non woven (4).
Total = 6

Unit 4: Various system of binding the web (5), and their merits (2).
Total = 7

Unit 5: Flocked fabric (2), Laminates (2), latest development in non-woven industry (3).
Total = 7
Grand total = 34

Reference Book & Text Book –

1. Non Woven – N.N. Banarjee
2. Non woven – NCUTE
3. Fibre to fabric – Corbman Bernard P.

3. Mill Planning & Organization (ETEF703)

Unit-1: Selection of site for a textile mill (2). Preparation of project report (2). Construction of building of a textile mill (2). Types of buildings, single and multistoried buildings. Fire hazards and their control (1). Prevention tours accidents (1)
Total of lectures required=9

Unit-2: Humidification of a textile mill (1). Humidifiers and dehumidifiers (1). Types of humidifiers (2). Different systems of ventilation system used in textile mill (1). System of lighting used in textile mill (2)
Total of lectures required=7

Unit-3: Balancing of machines for spinning and weaving machines or balancing of machines for bleaching (4). Dyeing, Printing and Finishing department. Layout of different machines of spinning, weaving process house (3).
Total of lectures required=7

Unit-4: Costing (1), elements of cost, fixed and variable cost (1). Knowledge of cost cancellation for spinning (2), weaving and processing department. Viability evaluation of a project (1). Break even point (1) Types of staff organization (2).
Total of lectures required=8

Unit-5: Different system wages used in textile industry (2), working of sales and purchase system (2). Material handling used in a textile mill (2). Maintenance of machines and safety devices used in a textile mill (2). Inventory controls (2) Resource Planning.
Total of lectures required=10
Total of lectures required=41
Reference Books

Reference Books

1. Industrial Engineering, Organization & management by Tarachand
2. Industrial Economics & Principle of Management by T.M. Chabra
3. Industrial Economics & Principle of Management by S. K Sharma

4. Principles and Design of Spinning Machinery (EE702)

Unit 1: Transmission of motion through belts, ropes, chains, tapes etc. Tangential belt drive, Variable speed drive, Design of cone drums for Scutcher, principles of mixing, opening, cleaning in Blow Room line, difference in aerodynamics of lint and trash and its utilization in blow room machinery design. Redesigning of openers by using different principles for better performance.

Unit 2: Transmission of motion to various parts of carding machine, detail study of varieties of licker-in, cylinder and flat clothing and its influence on carding performance, different systems of carding web collection, sliver coiling system, Inertia of carding engine, latest carding machine developments

Unit 3: Concept of lap formation for comber, designing concept of comber cylinder, top comb, detaching system, concept of noil control, latest machinery developments in comber frame

Unit 4: Different drafting systems in draw frame, roller weighing system, roller setting, and different stop motion, Differential motions used in speed frame, design of cone drums for speed frame, Design of flyer, spindle vibration measurement and control, building motion of speed frame

Unit 5: Design and analysis of ring frame cames**,** designing concept of building motion of ring frame, design of ring spindle and bolster, designing concept of various rings and travelers,

Books

1. Spun Yarn Technology by Eric Oxtoby
2. The technology of short-shape staple spinning – W. Klein

5. Knitting Technology (ETEF- 703)

Unit 1: Difference between knits & woven (1), knitting terms & definition (2), knit technology (2), knitting industries (1)
Total= 6

Unit 2: Classification of weft knitting machine(2), elements of knitting machine(2), technology of loop formation(2), geometry of loop structure(2) relation between yarn count, machine gauge and stitch density(2).
Total =10

Unit 3: Classification of knit-structures(2), loop formation on single jersey, Rib machines and inter look machines(2), knitting notation(2), socks knitting technology (3), warp knitting introduction(2), classification of warp knitting(2).
Total = 13

Unit 4: Special knit fibre and mechanics (4), electronics in knitting (4).
Total =8

Unit 5: Modern developments in knitting industry (3), calculations regarding production gsm, stitch density etc, set-up of knitting mills (3).
Total = 6
Grand Total = 46

Reference and Text Book-

1. Knitting technology – chamberlin
2. Knitting technology – W.J. Spencer
3. International Textile Journel – Knitting
4. Knitting Calculation – Chamberlin
5. Wet Knitting Vol. 1&2 –Published by IIT New Delhi.
6. Knitting – NCUTE

8TH SEMESTER, B. TECH. TEXTILE ENGINEERING

1. Open Elective-III: Syllabus as provided by UPTU

1. EOE081 Non Conventional Energy Resources/
2. EOE082 Introduction to Soft Computing/
3. EOE083 Product Development/
4. EOE084 Automation and Robotics

2.1 Departmental Elective III: Technical Textiles (ETCEF 031)/ Theory of Textile Structure (ETEF-032)

2.1 Technical Textiles (ETCEF 031

Unit-1: Textiles in filtration, Filter fabric construction- Woven, needle felt & knitted filter fabric (2), Finishing treatment of filter fabric (2), Thermal and chemical properties of filter fabric (2), Essential requirements of good filter fabric (2).
Total = 8

Unit-2: Waterproof breathable fabrics- Densely woven fabric, membranes, coatings (3), Membranes- Micro porous membrane, Hydrophilic membrane (2), Coatings-Microporous coating, Hydrophilic coatings (2), Performance of waterproof breathable fabrics (2).
Total =9

Unit 3: Medical textiles, Fibres used (1), Classification of medical textiles – Nonimplantable materials-wound dressings, bandages, plasters etc. (2), Extracorporal devices – Artificial kidney, liver, lung (2), Implantable material- Sutures, Soft tissue implant, Orthopaedic implants, Cardiovascular implants.(2),Healthcare/ hygiene products (1).
Total =8

Unit 4: Smart textiles brief introduction of smart textiles, classification of smart textile(2), passive smart textiles(2) , active smart textiles , very smart or ultra smart textiles (2), application of phase change material in smart textiles(2), brief discussion of smart shirt, smart suit, musical jacket, space suit etc(2).
Total =10

Unit 5: Introduction of Geo Textile (2), Classification of Geo Textiles, Functions of Geo textiles-soil reinforcement, drainage (fluid transmission), filtration, Separation, erosion control/absorption (2), objects of geo textiles, manufacturing of geo textile (3).Essential properties of geo textiles- Mechanical determinants, Hydraulic determinants, durability determinants (2)
Total 7
Grand Total 42

Reference Books:-

1. Hand book of technical textiles – AR Harrocks, SC Anand (Textile Institute, wood head publication).

2. Smart fibre, fabrics and clothing Tao X (Textile Institute, wood head publication).
3. Shears Handbook of Industrial Textiles.

2.2 Department Elective III: Theory of Textile Structure (ETEF-032)

Unit (1): Classification of yarn (1), Yarn diameter & density (1), Twist in yarn (1), Geometry of twisted yarn (2), Contraction due to twist (2), Twist and fibre packing in yarn – open packing, hexagonal close packing, redial packing density in yarn (2)
Total lectures required = 09

Unit (2): Twist and strength relationship (1), Theoretical treatment of yarn irregularity (2), fibre extent & fibre migration (3), estimation of strength of blended yarn (3)
Total lectures required = 09

Unit (3): Spinnability of textile fibres (1), Elements of fabric geometry (3), Flexible & rigid threads model (2), Pierce's equation and latter modifications (4)
Total lectures required = 10

Unit (4): Relation of fabric properties to simple geometry (2), Crimp interchange in woven fabric (2), Cover factor (1),
Total lectures required = 05

Unit (5): Theoretical treatment of fabric deformation in tension, bending & shear (3) Cloth setting theories (1)

Reference Books: -

1. Textile yarns – B.C. Goswamoi, J.G. Martindle, F.L. Scardino.
2. Textile structure – J.W.S. Hearle, S.Backer, Grossberg.
3. Pierce Geometry – Textile institute

Department Elective IV: Process Control in Spinning (ETEF-041)

Unit-1: Importance and considerations for evolving a system for process control (1) Scope for process control in spinning, Key variable for process control, Establishing norms and standards, Taking corrective action. (Control of Mixing quality and Cost)-Instrumental evaluation of cotton, Control of mixing quality through fibre Characteristics, Simultaneous control of mixing cost and quality, Bale Management (1)

Unit-2: Control of waste and cleaning (1) in Blow room and carding-Determination of trash (1) control and cleaning efficiency, Cleaning and waste in Blow (1) room and card, Norms for waste and (1) cleaning in card & Blow room, Assessment of performance of Blow room (1). Optimizing cleaning at cards (1) Control of comber waste (1) Control of yarn realization (1)

Unit-3: Measurement and analysis of productivity definition of indices of productivity (1), Measurement of productivity (1), Productivity analysis, productivity and profitability (1), Means to Improve productivity (1) Maximizing machine efficiency in ring spinning (1) Controlling end breakage rate in ring spinning (1).
Reference book (1) process Control in spinning by ATIRA

Unit-4: Control of yarn quality(1)- Count, Strength and their variability(1), Study of with in bobbin and between bobbin count variation(1), Control of variability of lea strength(1), Single yarn strength and elongation(1) Control of yarn unevenness imperfection(1)

Unit-5: Control of yarn faults and package defects (1) - slubs, crackers, spinners doubles (2) bad piecing, slough of and hairiness (1). Machinery Audit- Impact of machine conditions on processing performance and yarn quality (1), test instruments for Machinery Audit (1)

Reference Books:
1. Process Control in Spinning by ATIRA
2. Process Control in Spinning by K.R. Salhotra

Department Elective IV: Process Control in Weaving (ETEF042)

Unit-1: Scope for process Control (1) Systems of process Control in wvg (2) Setting norms (2) and Schedule of Checks machinery audit (2)

Total lecture required=7

Unit-2: Process Control in wvg (2) optimizing quality of preparation. Control of quality of knot (2) producing good package (1) control of productivity (2)
Total lecture required=7

Unit-3: Process Control in warping (2) minimizing end wreak age in warping (2) quality of warping beam (2) control of productivity (2)
Total lecture required=8

Unit-4: Process Control in sizing (2) Choice of size recipe and size pick-up (2)
Control of size pick-up, Control of yarn stretch (2) quality of sized beams, Control of productivity (2)
Total lectures required=8

Unit-5: Care is use and Selection of healds (2) Care in use and selections of Reads (1)
Care in dressing of the beams for reducing incidence of cross ends (2) Causes and remedies of fabric defects (2)
Total lectures required=7
Grand total of lectures required=37

Reference Books

1. Wvg tablets by ATIRA
2. Machine catalogues of various machines manufacture
3. Process control in weaving by ATIRA

4. Principles and Design of Weaving Machinery (EE801)

Unit 1: Designing concepts of drum winding, designing of winding drum, traverse in different winding machines, Concepts of latest tensioners for winding, designing concept of slub catchers

Unit 2: Designing concepts of warping drums, sectional warping, Designing concepts of high efficiency creels, Transmission of motion in warping machines, concept of designing of stop motions in warping

Unit 3: Designing concept of saw box for sizing machine, designing of drying cylinder, heating system for drying cylinder, Concept of cold sizing for worsted warp sizing, latest machinery developments in sizing machines

Unit 4: Principles governing the design of looms. Mechanics of basic weaving operations, Kinetics of slay, slay eccentricity, designing concepts of tappet for different twill weaves, warp tension measurement and its control, Different picking system, weft insertion system of sulzer, rapier, airjet and waterjet systems

Unit 5: Brief introduction of electronic Dobby and Jacquard, Designing concept of multiphase weaving, mechanism of different stop motions on shuttleless looms.

U.P. TECHNICAL UNIVERSITY, LUCKNOW.
Study and Evaluation scheme
[Effective from the Session 2009-10]
MAN MADE FIBRE TECHNOLOGY

YEAR 2nd, SEMESTER- III

S. No.	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	ETCEF301	Natural Textile Fibre	2	1	0	15	10	25	50	75	3
2	EAS031/EOE031-EOE038	Mathematics III/ Science Based Open Elective	3	1	0	30	20	50	100	150	4
3	EHU301/ EHU302	Industrial Psychology/ Industrial Sociology	2	0	0	15	10	25	50	75	2
4	EF302	Yarn Technology – I	3	1	0	30	20	50	100	150	4
5	ECF303	Polymer Chemistry	3	1	0	30	20	50	100	150	4
6	ETEF304	Wet Processing of Textiles- I	3	1	0	30	20	50	100	150	4
7	EHU111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/DESIGN/DRAWING											
8	ETCEF301P	Identification of Fibres-I	0	0	3	0	20	20	30	50	1
9	EF302P	Yarn Technology-I	0	0	3	0	20	20	30	50	1
10	EF303P	Polymer Chemistry Lab	0	0	3	0	20	20	30	50	1
11	ETEF304P	Wet Processing of Textiles - I	0	0	3	0	20	20	30	50	1
12	GP 301	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								1000	26

Paper Code	**Science Based Open Elective**
EOE031/EOE041	Introduction to Soft Computing
EOE032/EOE042	Nano Sciences
EOE033/EOE043	Laser Systems and Applications
EOE034/EOE044	Space Sciences
EOE305/EoE405	Polymer Science & Technology
EOE036/EOE046	Nuclear Science
EOE037/EOE047	Material Science
EOE038/EOE0487	Discrete Mathematics

U.P. TECHNICAL UNIVERSITY, LUCKNOW.
Study and Evaluation scheme
[Effective from the Session 2009-10]
MAN MADE FIBRE TECHNOLOGY

YEAR 2nd, SEMESTER- IV

S. No.	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM					
			L	T	P	CT	TA	Total	ESE		
THEORY SUBJECTS											
1	EHU402/ EHU401	Industrial Sociology/ Psychology	2	0	0	15	10	25	50	75	2
2	EOE041-OE048/ EAS401	Science Based Open Elective/ Mathematics III	3	1	0	30	20	50	100	150	4
3	EF401	Yarn Technology -II	3	1	0	30	20	50	100	150	4
4	EF402	Principle of Chemical Engineering	3	1	0	20	10	30	70	100	4
5	ETEF403	Wet Processing of Textiles –II	3	1	0	30	20	50	100	150	4
6	ETEF404	Colour & Design	2	1	0	15	10	25	50	75	3
7	EHU111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/DESIGN/DRAWING											
8	ETCEF401P	Identification of Fibres –II	0	0	3	0	20	20	30	50	1
9	EF402P	Yarn Technology –II	0	0	3	0	20	20	30	50	1
10	ETCFE403P	Chemical Engineering -II	0	0	3	0	20	20	30	50	1
11	ETEF404P	Wet Processing of Textiles –II	0	0	3	0	20	20	30	50	1
12	GP 401	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								1000	26

U.P. TECHNICAL UNIVERSITY, LUCKNOW.
Study and Evaluation scheme
[Effective from the Session 2010-11]
MAN MADE FIBRE TECHNOLOGY

YEAR 3rd, SEMESTER- V

S. NO.	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU501	Engineering & Managerial Economics	3	1	0	30	20	50	100	150	3
2	ETCEF501	Textile Testing-I	3	1	0	30	20	50	100	150	4
3	EF502	Fabric Technology – I	3	1	0	30	20	50	100	150	4
4	EF503	Instrumentation & Automatic Control	2	1	0	15	10	25	50	75	3
5	ETEF504	Fabric Structure	2	1	0	15	10	25	50	75	3
6	ETCEF505	Chemistry & Production of Fibres	3	1	0	30	20	50	100	150	4
7	EHU111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/DESIGN/DRAWING											
8	ETCEF501P	Textile Testing –I	0	0	3	0	20	20	30	50	1
9	EF502P	Fabric Technology – I	0	0	3	0	20	20	30	50	1
10	EF503P	Fibre Production & Characterization	0	0	3	0	20	20	30	50	1
11	ETEF504P	Fabric Analysis	0	0	3	0	20	20	30	50	1
12	GP 501	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								**1000**	**26**

U.P. TECHNICAL UNIVERSITY, LUCKNOW.
Study and Evaluation scheme
[Effective from the Session 2010-11]
MAN MADE FIBRE TECHNOLOGY

YEAR 3rd, SEMESTER- VI

S. NO.	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EHU601	Industrial Management	3	0	0	30	20	50	100	150	3
2	ETEF011/EF012	Departmental Elective-I	3	1	0	30	20	50	100	150	4
3	ETCFE601	Textile Testing II	2	1	0	15	10	25	50	75	3
4	EF602	Fabric Tech- II	2	1	0	15	10	25	50	75	3
5	ETEF603/EC011	Structure & Properties of Fibre	3	1	0	30	20	50	100	150	4
6	EF604/EC012	Post Spinning Operation	3	1	0	30	20	50	100	150	4
7	EHU111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/DESIGN/DRAWING											
8	ETCEF601P	Textile Testing –II	0	0	3	0	20	20	30	50	1
9	EF602P	Fabric Technology – II	0	0	3	0	20	20	30	50	1
10	EF603P	Analytical Chemistry	0	0	3	0	20	20	30	50	1
11	EF604P	Post Spinning Operation	0	0	3	0	20	20	30	50	1
12	GP 601	General Proficiency	-	-	-	-	-	50	-	50	1
		Total								**1000**	**25**

Paper Code	**Department Elective-I**
ETEF011/EC602	Coating of Textiles
EF012/ ETC604	Technology of Finishing-I

U.P. TECHNICAL UNIVERSITY, LUCKNOW.
Study and Evaluation scheme
[Effective from the Session 2011-12]
MAN MADE FIBRE TECHNOLOGY

YEAR 4th, SEMESTER- VII

S. NO.	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EOE071-EOE074	Open Elective-I	3	1	0	30	20	50	100	150	4
2	ETCEF021-ETCEF022	Department Elective-II	3	1	0	30	20	50	100	150	4
3	ETFE701	Mill Planning & Organization	3	1	0	30	20	50	100	150	4
4	EF702	Fibre Manufacture & Process Control	3	1	0	30	20	50	100	150	4
5	ETEF703	Knitting Technology	3	1	0	30	20	50	100	150	4
6	EHU111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/DESIGN/DRAWING											
7	ETFE703P	Knitting Technology	0	0	3	0	20	20	30	50	1
8	ETF751	Project	0	0	3	0	50	50	-	50	2
9	ETF752	Seminar	0	0	2	0	50	50	-	50	1
10	ETF753	Industrial Training - Viva Voce	0	0	2	0	50	50	-	50	1
11	GP 701	General Proficiency	-	-	-	-	-	50	-	50	1
		Total	**15**	**5**	**10**	**150**	**270**	**470**	**530**	**1000**	**26**

Paper Code	**Open Elective-I**
EOE071	Entrepreneurship Development
EOE072	Quality Management
EOE073	Operation Research
EOE074	Introduction to Biotechnology

Paper Code	**Department Elective-II**
ETCEF021	Garment Manufacturing Technology
ETCEF022	Nonwoven

U.P. TECHNICAL UNIVERSITY, LUCKNOW.
Study and Evaluation scheme
[Effective from the Session 20011-12]

Man Made Fibre Technology

YEAR 4th, SEMESTER- VIII

S. NO.	Course Code	Subject	Periods			Evaluation Scheme				Subject Total	Credit
						SESSIONAL EXAM			ESE		
			L	T	P	CT	TA	Total			
THEORY SUBJECTS											
1	EOE801-EOE804	Open Elective-II	3	1	0	30	20	50	100	150	4
2	ETCEF031/ ETEF032	Dept Elective-III	3	1	0	30	20	50	100	150	4
3	ETEF041/ ETEF042	Dept Elective-IV	3	1	0	30	20	50	100	150	4
4	EF801	High Performance Fibres	3	0	0	30	20	50	100	150	3
5	EHU-111	Human Values & Professional Ethics	2	2	0	15	10	25	50	75	-
PRACTICAL/DESIGN/DRAWING											
6	ETF851	Project	0	0	12	-	100	100	250	350	8
7	GP-801	General Proficiency	-	-	-	-	-	50	-	50	1
8		**Total**	**12**	**3**	**12**	**120**	**180**	**350**	**650**	**1000**	**26**

Paper Code	**Open Elective-II**
EOE081	Non Conventional Energy Resources
EOE082	Nonlinear Dynamic Systems
EOE083	Product Development
EOE084	Automation & Robotics
Paper Code	**Department Elective-III**
ETCEF031	Technical Textiles
ETEF032	Theory of Textile Structure
Paper Code	**Department Elective-IV**
ETEF041	Process Control in Spinning
ETEF042	Process Control in Weaving

3TH SEMESTER, B. TECH. MANMADE FIBRE TECHNOLOGY

EAS-301/EAS-401: MATHEMATICS –III

L	T	P
3	1	0

Unit – I : Function of Complex variable

Analytic function, C-R equations, Cauchy's integral theorem, Cauchy's integral formula for derivatives of analytic function, Taylor's and Laurent's series, singularities, Residue theorem, Evaluation of real integrals of the type $\int_0^{2\pi} f(Cos\,\theta, Sin\,\theta)\,d\theta$ and $\int_{-\infty}^{\infty} f(x)\,dx$ **10**

Unit – II : Statistical Techniques - I

Moments, Moment generating functions, Skewness, Kurtosis, Curve fitting, Method of least squares, Fitting of straight lines, Polynomials, Exponential curves etc., Correlation, Linear, non –linear and multiple regression analysis, Probability theory. **08**

Unit – III : Statistical Techniques - II

Binomial, Poisson and Normal distributions, Sampling theory (small and large), Tests of significations: Chi-square test, t-test, Analysis of variance (one way) , Application to engineering, medicine, agriculture etc.

Time series and forecasting (moving and semi-averages), Statistical quality control methods, Control charts, $\bar{X}$, R, p, np, and c charts. **08**

Unit – IV : Numerical Techniques – I

Zeroes of transcendental and polynomial equation using Bisection method, Regula-falsi method and Newton-Raphson method, Rate of convergence of above methods.

Interpolation: Finite differences, difference tables, Newton's forward and backward interpolation , Lagrange's and Newton's divided difference formula for unequal intervals. **08**

Unit – V : Numerical Techniques –II

Solution of system of linear equations, Gauss- Seidal method, Crout method. Numerical differentiation, Numerical integration , Trapezoidal , Simpson's one third and three-eight rules, Solution of ordinary differential (first order, second order and simultaneous) equations by Euler's, Picard's and forth-order Runge- Kutta mehthods. **08**

Test Books :-

1. Peter V. O'Neil, Advance Engineering Mathematics Thomson (Cengage) Learning, 2007.
2. Jain, Iyenger & Jain, Numerical Methods for Scientific and Engineering Computation, New Age International, New Delhi , 2003.
3. J.N. Kapur, Mathematical Statistics, S. Chand & company Ltd.,2000

Reference Books :-

1. R.K. Jain & S.R.K. Iyenger, Advance Engineering Mathematics, Narosa Publication House, 2002.
2. Chandrika Prasad, Advanced Mathematics for Engineers, Prasad Mudralaya, 1996.
3. E. Kreysig, Advanced Engineering Mathematics, John Wiley & Sons, 2005.
4. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 2005.
5. Devi Prasad, An introduction to Numerical Analysis, Narosa Publication house, New Delhi 2006.

6. T. Veerajan & T. Ramchandrandran, Theory & Problems in Numerical Methods, TMH, New Delhi, 2004.
7. S.P.Gupta, Statistical Methods, Sultan and Sons, New Delhi, 2004.
8. Devore, Probability and Statistics, Thomson(Cengage) Learning, 2007.
9. Walpole, Myers, Myers & Ye, Probability and Statistics for Engineers & Scientists, Pearson Education, 2003.

1. (Engineering Core: Interdisciplinary) Natural Textile Fibres: (ETCFE-301)

Unit (1) : Fibre, textile fibre (1). Classification of natural textile fibres (1), Essential and desirable properties of textile fibres (2), Grading of cotton (1), Physical properties of cotton fibre (1), Effect of acid & Alkalies on cotton fibre (1).
Total Lectures required = 07

Unit (2): Production, morphological structure, properties and end uses of: Flax (2), Jute (2), Ramie (2). Introduction to specialty natural fibres like banana, pineapple, spider etc
Total Lectures required = 08

Unit (3): Production of silk (raw) (2), Morphological structure of silk (1), Production of waste silk yarn (2), Chemical composition and physical properties of silk (1), Effect of acid and Alkalies (1), Various varieties of silk with brief description (1).
Total Lectures required =08

Unit (4): Morphological structure of wool (2), Composition of wool fibres (1), Properties of wool fibres (1), Varieties of wool fibre with brief description (2).
Total Lecture required = 06
Grand total of Lectures = 36

Books:

1. W.E. Morton & J.W.S. Hearle, Physical properties of textile fibres, Textile Institute, U.K.
2. Progress in textiles: Science and technology Vol.-2
   By Dr. V.K. Kothari, I.I.T. Delhi.
3. Hand book of textile fibres by J.Gordon Cook

2. Mathematics III/ Science Based Open Elective (EAS301/ EOE031-EOE038)
Syllabus to be decided by U.P.T.U.

3. Industrial Psychology/ Industrial Sociology (EHU301/EHU302)
Syllabus to be decided by UPTU

4. Yarn Technology-I (EF302)

Unit (1): Picking and cotton harvesting. Ginning, pre-cleaning of cotton .Pre and post ginning. Description and working of knife roller, mecarthy and saw gins, kinds of mixing. Principles of

selection of cotton for mixing hand, bin, stack mixing, auto mixer, different cotton varieties advantage and disadvantage of mixing.
Total lectures required = 06
Unit (2): Object of blow room. Types of opening beaters. Lap forming mechanism. Lap rejection and lap defects. Production and efficiency of different m/c. Chute feed system. Latest openers and beaters.
Total lectures required = 05
Unit (3): Objects, feeding, carding and doffing of material. Specification of different part, card clothing. Types of card clothing. Card setting, stripping grinding, assessment of neps, basic card calculation. Auto levelers in card.
Total lectures required = 07
Unit (4): Objects of draw frame. Different drafting systems. Roller slip and drafting waves, roller weighting, various stop motion, different weighting systems.
Total lectures required = 06
Unit (5): Working of draw frame. Drafting and its types. Concept of high speed draw frame. Calculation on draft and production.
Total lectures required = 06
Grand Total Lectures Required = 30
Books:

1. Man made fibre and their processing by W.Kiein
2. Spinning of man made and blends on cotton systems by K.R.Solholore
3. Manual of cotton spinning by Frank Fharnley
4. Technology of carding by R.Chattopadhyay
5. Spinning blow room and carding by Prof. K.R. Solholore

5. Polymer Chemistry (ECF303)

Unit (1): Scope of Polymer Chemistry, (a) definition of monomers, homopolymers and copolymers (2), (b) Chemistry of important monomers (1), (c) Thermo sets (d) Thermoplastic (e) Elastomers (f) Tg, Tm, and solution properties of polymers (3)
Total lectures required = 06

Unit (2): Condensation polymerization (1), (b) Mechanism, kinetic and molecular weight build up (3), (c) Bulk, Solution, Emulsion and suspension polymerization (3) (d) comparison of condensation and addition polymerization (1).
Total lectures required = 08

Unit (3): Addition polymerization (1), (b) Type of addition polymerization (1), (c) Radical, cationic and anionic polymerization (2) (d) Mechanism and kinetics of polymerization (2), (e) Copolymerization, Stereo-Regular polymerization, Block and Graft polymerization (2).
Total lectures required = 08

Unit (4): Characterization of molecular weight polymers (1), (b) End-Group analysis method (1), (c) Viscosity of measurement method (1), (d) G.P.C. Techniques for measurement (1).
Total lectures required = 04

Unit (5); Application of polymer to textiles (1): Fiber forming synthetic polymer (3) Properties of fiber forming synthetic polymer (3)
Total lectures required = 07

Grand Total Lectures required = 33

BOOKS:

1. *Text Book of Polymer science by* ***F.W. Bill Meyer***
2. *Text Book of polymer Vol. I, II, III by* ***M.S. Bhatnagar***
3. *Polymer science by* ***Gawaskar, Vishwanathan, sreedhar and Jaydev***
4. *Polymer chemistry by* ***B.K. Sharma***

6. Wet Processing of Textiles -I (ETEF304)

UNIT-1: Introduction to simple preparatory process, grey cloth singeing, hot plate, roller and gas singeing m/c. Desizing – its object and various types of desizing methods. Scouring – its objects and process involved. Various types of kier -its description and working. (8)

UNIT-2: Bleaching-its objects and chemistry involved, various types of bleaching agents – their preparation and uses ,bleaching of cotton fabrics, various faults in bleaching and their remedies and removal.(8)

UNIT-3: Merecrisation objects, physical and chemical aspects of mercerization , yarn mercerization and various methods of cloth mercerization ,various methods to test the efficiency of mercerization. (7)

UNIT-4: Finishing- objects, various types of physical and chemical finishes and their applications by various finishing equipments like padding mangle ,drying m/c's ,stenter ,calendar etc.(8)

UNIT-5: Introduction to various finishes like water proofing and water repellency, flame proofing and flame retardancy ,heat setting ,softening and stiffening. (7)
Grand Total Lectures Required =38

Books:

1. Chemical processing of cotton and p/c blends – ATIRA
2. A glimpse on the chemical technology and textile fibres by R.R. Chackrawartty
3. Technology of bleaching and mercerization by V.A. Shenai
4. Technology of finishing by V.A. Shenai

7 *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U.

4TH SEMESTER, B. TECH. MANMADE FIBRE TECHNOLOGY

1. Industrial Psychology/ Industrial Sociology (EHU402/EHU401)
Syllabus to be decided by UPTU

2. Science Based Open Elective/ Mathematics III (EOE041-EOE048/EAS401)

Syllabus to be decided by U.P.T.U.

3. Yarn Technology-II (EF401)

Unit 1: Combing: Object (1), lap preparation m/c for combing noils percentage (2), comber efficiency, cylinder and needle description (2). Role of lap preparation m/c's (1).
Total lectures required = 6

Unit 2: Description of one latest comber (2), timing and settling (2), controlling of comber waste and recent development (1). Combing cycle, types of combing (1).
Total lectures required = 6

Unit 3: Speed frame: Objects, mechanism of drafting (3), twisting and winding (2), building motion (3), differential motion (2).
Total lectures required = 10

Unit 4: Description of one latest speed frame(2) and related calculation(2),ring frame objects, mechanism of drafting (3),mechanism of twisting.
Total lectures required = 07

Unit 5: Twisting and winding(2),yarn twist levels (2),twist contraction and fibre migration during twisting(2),different ring and travelers and related calculations(3)
Total lectures required = 09
Grand Total Lectures Required = 38

Books:

1. Man made fibre and their processing by W.Kiein
2. Spinning of man made and blends on cotton systems by K.R.Solholore
3. Manual of cotton spinning by Frank Fharnley
4. Technology of carding by R.Chattopadhyay

4. Principle of Chemical Engineering (EF402)

Unit-1: Industrial calculation (2), material balance (4), energy balance (2)
Flow of fluids (1), mechanism of fluid flow (1), viscosity measure (1),flow meters(2), reciprocating pumps (1), centrifugal pumps (1)
Total of lectures required=10

Unit-2: Evaporation, types of Evaporation(2),drying tunnel and rotary driers (1)distillation(1),steam distillation(1)distillation method (1),mixing type of mixers(1),paddle stirrers, kneeding m/c(1)
Total of lectures required=8

Unit-3: Gas absorption (1),gas absorption equipment (1), theory of Gas absorption (1), extraction, type of extraction (1) theory of extraction (1)
Total of lectures required=5

Unit-4: Crystallization, crystal forms (1) Crystallization theory (1) rate of growth of crystal (1), Crystallization apparatus (1), filtration (1), filtration apparatus (2)
Total of lectures required=7

Total of lectures required=30

5. Wet Processing of Textiles II (ETEF403)

UNIT -1: Classification of dyes according to there mode of application. Elementary knowledge of cotton dying with direct, vat, sulphur, azoic reactive and Solublised vat dyes. (9)

UNIT-2: dyeing of wool and silk with acid and basic and metal complex dyes (7)

UNIT-3: Dyeing of popular synthetic textile material like nylon, acrylic and polyester Fastness properties of dyeing material. (7)

UNIT-4: Printing –an introduction ,classification of methods of printing .Outline of various method like hand block printing .screen printing .roller printing ,transfer printing rotary screen printing etc. (9)

UNIT-5: Thickeners and their application .Various style of printing –direct style resist style and discharge style of printing. (8)
Grand Total Lectures Required =40

Books:
1. Chemical processing of cotton and p/c blends – ATIRA
2. A glimplse on the chemical tech. of textile fibres by R.K. Charkrawarty
3. Technology of dyeing byV A. Shanai 4. Technology of printing byV A. Shanai

6. Colour & Design (ETEF404)

Unit (1): Light and colour phenomena, physical basis of colour (2), Emission & absorption theory of light, Colour vision and light theory of colours (2), Complementary colours, Chromatic circle (2), Pigment theory of colours (2), Brewster circle (2), Attributes of the primary & secondary colours, Colour measurement, Primary, Secondary, Tertiary & compound colours (1).
Total Lectures required = 11

Unit (2): Biren's triangle (2), Modification of colours (2), Coloured greys, Colours in combination, Colour contrast, contrast in hue, contrast of tone, colour harmony (1), Relative spaces occupied by colours, divisional colours, Application of colours, Mixed colour effect (1), Composition of designs, Condition to be observed during ornamentation of fabrics (1), Mode shade, Harmony of succession, gradation of hue, Different stages of colouring of textile materials (1), Colour and weave effect and its classification (1).
Total Lectures required =09

Unit (3): Bases of Textile design (2), One third and one fourth drop design (2), Half drop and drop reverse design, Unit repeating design, Geometric ornamentation (2), Construction of symmetrical designs (2), Stripe and check effect designs (2), Sari border / vertical border design (2), Factor affecting the woven designs, reversing inclined figure, Diamond, Ogee, & diagonal waved line base (2).
Total Lecture required = 14

Unit (4): A compulsory question based on drawing sheet and application of colours will be asked from any topic of unit 1 to 3.
Total lectures required for unit 4 are distributed in unit 1, 2, 3.

Grand total Lectures required = 34

Books:

1. W. Watson- Textile design and colour

7 *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U.

5TH SEMESTER, B. TECH. MANMADE FIBRE TECHNOLOGY

1. **Engineering and Managerial Economics (EHU501): Syllabus to be provided by U.P.T.U.**

2. Textile Testing-I (ETCEF-501)

Unit (1): Introduction of textile testing (1), sampling, random sampling, biased sampling (1), sampling techniques, square, cut square, zoning technique (1), selection of sample for testing, grading of cotton fibre with respect to staple length (1), laboratory measurement of fibre length, span length, Baer sorter (1), Shirley photo electric staple, servo fibro graph (1), salient features of HVI (1).
Total Lectures Required = 7

Unit (2): Atmospheric conditions for testing, absolute & relative humidity, moisture regain & moisture contest (1), importance of moisture in textiles, measurement of moisture regain & content (1), effect of moisture on properties (physical & mechanical) of textile material, factors affecting the regain (2), correct numerical weight, oven dry weight (1), shrilly moisture meter (1). **Total Lectures Required = 6**

Unit (3): Dry and wet bulb hygrometer, sling assmann, hair hygrometers (2), control of testing room atmosphere (1), AFIS (1), Napping potential (1), Nep count (1), rating of neps, maturity coefficient measurement by NaOH method, fibre fineness by airflow meter & Sheffield micronaire (2).
Total Lectures Required = 9

Unit (4): Fibre bundle strength by Pressley, stelometer (2), fibre quality index, linear density of man made fibres and strength (1), spin finish, crimp (1), Trash, Shirley trash analyzer (1). Yarn numbering system (1), wrapping test for lap, sliver, roving (1),
Total Lectures Required = 7

Unit (5): determination of yarn count, diameter (1), average & resultant count of folded yarn, relation between Ne, D, T, Nm (1). Instruments used for determination of count, quadrant balance, Knowles balance, bees lay balance and physical balance (2), Twist, classification of twist, twist measurement, direct counting method, continuous twist tester, twist-untwist method, (2), Twist tester, (2), R.B. twist tester, level of twist (1).
Total Lectures Required = 9
Grand total of lectures required = 36Books: -

1. Quality control and testing management by Dr. V.K. Kothari
2. Principle of textile testing by J.E. Booth
3. Physical testing of textiles by B.P. Savile.

3. Fabric Technology-I (EF502)

Unit (1): Study of different types of knots(1),classification of different warp winding m/c's with detail specification(2),working principles of thread stop motion(1),different types of tensioners (1),different types of slub catchers(1),yarn travesty system(1),different winding faults with remedies (1)
Total lectures required = 08

Unit (2): Classification of weft winding m/c (1),working principle of various types of winders(2),yarn numbering system(1),average and resultant count for ordinary and fancy yarns(1).
Total lectures required = 05

Unit (3): Classification of various types of warping m/c with specification(2),working of stop motion, full beam stop motion used on warping motion(2),quality of good beam,warping beam defect with remedies(2).
Total lectures required = 06

Unit (4): Object of sizing(1),different ingredients with method of preparation of size-paste(2),classification of sizing m/c(1),working principles of various controls used on sizing m/c(2),size-receipe(1),size-beam,faults with remedies (1).
Total lectures required = 08

Unit (5): Calculation regarding efficiency,production inventory etc. for winding, warping and sizing m/c(10)
Total lectures required = 10
Grand Total Lectures Required = 37
Text Books:
1. Tablets(ATIRA) a. Winding b. Warping c. Sizing
2. Process control in warping ,winding and sizing(ATIRA,BTRA)
3. Yarn preparation Vol. I & II R.Sen Gupta
4. Warp sizing by Rame Bottom
5. Yarn calculation by M.R.Sen Gupta

4. Instrumentation and Automatic Control (EF503)

Unit (1): The principle of measurement(1),static and dynamic measurement(2),functional elements for measuring systems(2),transducers, amplifier and recording systems(3)
Total lectures required = 08

Unit (2); Instrumentation for measurement of motion(1), strain force(1),torque (1), temperature(1), pressure(1), flow, surface, texture etc(1), design consideration in optical mechanical and electrical instruments(2)
Total lectures required = 7

Unit (3): Numerical control basic concepts (1), point to point systems (2), straight line and controlling system (1), Programme control-electrical, magnetic&, tape control, optical and mechanical methods (4)
Total lectures required = 08

Unit (4): Numerical control components (1), manual part programming (2), computer aided programming (1) C.N.C. (1), D.N.C. (1)
Total lectures required = 06
Grand Total Lectures Required = 29

Text Books:
1. Industrial instruments &control-S.K.Singh
2. Mechanical measurements-B.S.Sirohi &H.C.
3. Computer control of manufacturing-Yoren Koren

5. Fabric Structure (ETEF-504)

Unit (1): Classification of various fabrics (1), construction of plain weave and its derivatives (2), ordinary twill right hand twill (1), warp faced, weft faced & balanced twills (2), Satin regular, irregular and their extension (1).
Total Lectures Required = 7

Unit (2): combined twills, end to end and pick-to-pick combination (1). Fancy twills-large diagonal shaded twills (1), sateen base diagonals and brained twills (1), elongated twill-steep and low twills (1), pointed, wave zig-zag, curved, broken, herring borne, transposed, corks screw twills (2), Diamond, mock leno, ordinary honeycomb, brighten honeycomb (2),
Total Lectures Required = 8

Unit (3): Huck-a-back and crepe weave (), derivatives of hopsack barley corn stitched hopsack and twilled hopsack (2), Simple and wadded bed ford cords (1), weft and piques (1), principle of figuring with extra material extra warp figuring (2) extra weft, limitation of extra thread (2).
Total Lectures Required = 9

Unit (4): Backed cloths (1), weft backed cloths, warp backed cloths with weeding threads (2), double cloths, center stitched, self stitched (2), inter changing double cloth, cut effect in interchanging double cloths (2).
Total Lectures Required = 7
Grand Total of lectures required = 31
Reference Book: -
1. W. Watson Textile Design & colour Longmans Greens Co. London.
2. Z.J Grosicki Watson's Textile design and colour Newnes Buter Worth, London.
3. Z.J. Grosicki, Advance Textile Design Newnes Butter Worth, London.
4. "Nishant" A Grammar of textile.

6 Chemistry and Production of Fibres (ETCEF-505)

Unit (1): Polyethylene Terephthalate Fibre – History of development, Polymer production by DMT & PTA route, Chips drying, Fibre manufacturing, Effect of process variable on

properties of polyester fibre, some dope additives for specialty polyester fibre, Properties of polyester fibre.

Total Lectures required = 08

Unit (2): Polyamide Fibres – History of development, Different types of polyamide fibres, Nylon polymer production by continuous polymerization in VK Tube, Manufacturing of Nylon 6 fibre by melt spinning, Properties of nylon 6 fibre, Polymer production of Nylon 66, Nylon 66- fibre formation by melt spinning, Properties of Nylon 66 fibre, Brief introduction to Aramid fibres.

Total Lectures required =08

Unit (3): Introduction of polyolefin fibres, Polymerization of polyethylene, Polyethylene (PE) fibre formation, Properties of polyethylene fibre, Different type of polypropylene (PP), Polymerization of polypropylene, Polypropylene fibre formation, Properties of polypropylene fibre.

Total Lectures required =08

Unit (4): Introduction of vinyl fibres, Polyacrylonitrile (PAN) fibre, Use of ionic and neutral co-monomers, Polymerization of PAN, Acrylic fibre- formation by dry spinning, Dry –jet-wet spinning process, Effect of process variables on properties of PAN fibre, Dope formation, Properties of PAN fibre, Introduction to polyurethane fibre.

Total Lectures required = 08

Unit (5): Introduction of regenerated fibre, Concepts of regeneration of fibre, Raw material for viscose rayon, Manufacturing sequence of viscose fibre, Steeping and pressing, Cutting and shredding, Ageing, Xanthation of sodium cellulose, Mixing and filtration, Ripening, Wet spinning of viscose rayon, Formation of serrated edge cross-section of viscose rayon, Viscose fibre properties, Introduction of cuprammonium rayon in brief, Introduction of cellulose acetate rayon in brief, Introduction of lyocell fibre in brief.

Total Lectures required =08

Grand total lectures required = 40

Books:

1. Manufactured fibre technology by V.B. Gupta & V.K. Kothari
2. Essential fibre chemistry by M.E. Cartor
3. Synthetic fibres by Fourne
4. New fibre by T.Hongu, G.O. Phillips, Woodhead publications.
5. Fibre chemistry by M. Lewin, E.M. Pearce, Marcel & Dekkan Inc.
6. Regenerated Cellulosic fibres by C. Wooding, Woodhead publishing Ltd.

7 *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U.

6TH SEMESTER, B. TECH. MANMADE FIBRE TECHNOLOGY

1. **Industrial Management (EHU 601) Syllabus as provided by U.P.T.U.**

2. **Departmental Elective-I:Coating of Textiles (ETEF011/EC601)/ 2.2 Technology of Finishing (EF012/EC604)**

2.1 Coating of Textiles (ETEF011/EC602)

Unit-1: Polymeric materials for coating- (Rubbers: natural and synthetic, (3)(Polyvinyl chloride, Polyurethane, Acrylic polymers, Adhesive treatment (4)
Total lecture required=7

Unit-2: Coating Methods: Knife coating (1), Roll coating (1) transfer coating (1), Rotary screen printing (2) calendaring hot melt coating (2)
Total lecture required=7

Unit-3: Physical properties of coated fabric (2) Rheology of coating (1), Rheological Behaviors of fluids (2) pastes (1) hydrodynamic analysis of coating (2)
Total lecture required=8

Unit-4: Fabric for foul weather protection- Clothing comfort (1) Impermeable coating (1) breathable fabric (1) Non Apparel coating (1), Fabrics for Chemical protection (1) Thermo chromic Fabric (1) Temperature Adaptable Fabrics (1) Camouflage nets (1) Metal and conducting polymer-coated fabrics (2) Radiation cured coating (1)
Total lecture required=8

Unit-5: Test methods, Coating per Unit area, Degree of fusion/curing of coating (1) blocking Abrasion resistance (1) Test for colour- Fastness to dry and wet rubbing, Resistance to water penetration (3) Air permeability (1) water vapour permeability (1) low temperature bend test (1) low temperature impact test (1)
Total lecture required=9
Grand Total lecture required=39

Reference Books:
References:

1. Coating & Laminated Textiles by Water Fung
2. Coated Textile by A.K. Sen
3. Coated Fabric technology Vol 1-3 Technomic publication
4. Coated & laminated Fabric by AATCC symposium

2.2 Technology of Finishing-I (EC604/EF012)

Unit - 1: Principle of finishing of cotton, wool, silk (2), linen & jute (2). Classification of various finishing – (a) Mechanical (3)
(b) Chemical (3), Temporary and durable/permanent finishes. Semi permanent.
Total lectures-10
Unit – 2: Finishing machines – Mangle &their function (3), drying machines-(2), Stentering (2), Calandering (2), Samprising-(2), Decalising
Total lecture required - 11
Unit – 3: Beetling of linen (1), Crimping of Silk &rayan, Finishing of wool1), Softening, Stiffining (3).
Total lecture required – 8
Unit – 4: Easy care finish (3) , minimum application technique including foam technology (5)
Total lecture required – 8
Unit – 5: Finishing of woolen fabrics Dry Wet decatising, Felting, Milling, Permanent Setting, Shrink Proofing.
Total lecture required – 8
Total no. of lectures =45

Reference & Text Books:
1. Introduction to textile finishing by J.T. marsh
2. Chemical Processing of Synthetic Fibres & Blends by Datye & Vaidhya
3. Textile Finishing by V.A. Shenai

3. Textile Testing-II (ETCEF-601)

Unit (1): Textile properties of yarn and fabric(2), stress-strain curve(1), various methods for finding of yield point(1), methods for finding of various modulus(1), destination of tenacity(1), stiffness of fabric(1).
Total Lectures Required = 7

Unit (2): Yarn testing m/cs- single yarn strength tester, lea strength tester (2), fabric strength tester- impact tester, Graph test, fabric B.S. Test (3), Scott serigraph, Instron testing m/c (2).
Total Lectures Required = 7

Unit (3): Fabric abrasion & resistance to wear (1), Bursting strength test (1), Tear test, Wear (1), Brief introduction of FAST & KAWABATA (1), estimation of fabric thickness, rigidity, air permeability (1), water repellency (1), drape handling (1), crimp, thermal transmission properties (1).
Total Lectures Required = 8

Unit (4): Cover factor of fabric (1), snagging-mace snagging test (1), weaver trials, advantages and disadvantages of weaver trial (1), lab test (1), measurement of lap, sliver, yarn irregularity (1) salient features of Uster evenness tester (2)
Total Lectures Required = 7

Grand Total of lectures required = 29

Reference Books:

1. Physical testing of textiles by B.P. Saville.
2. Quality control and testing management by Dr. V.K. Kothari.
3. Principles of textile testing by J.E. Booth.
4. Quality control by V.K. Kothari.

4. Fabric Technology –II (EF602)

Unit (1): Classification of loom (1), different primary motion of handloom &power looms (3) .Secondary motion of loom(2), Adjustment and setting of loom for take up and let off with change of sorts(2)
Total of lectures required=8

Unit (2): Importance of auxiliary motions of power loom (2), various auxiliary motion and their working principles (6).
Total of lectures required=8

Unit (3): Working principle of dobby (2), classification of various shedding arrangements (2), working principles of jacquard (3), and working principle of drop box motion (2).
Total of lectures required=9

Unit (4): Classification of automatic loom(2),study of various auxiliary motion of automatic loom(2),calculation regarding cloth production ,efficiency, yarn inventory etc(2) ,fabric defects and remedies(2)
Total of lectures required=8.
Grand total lectures required=33

Recommended Book:-
1. Weaving mechanism Vol.1 & ll by N.N.Banarjee.
2. Weaving by ATIRA.
3. Automatic loom by ATIRA (tablet)
4. Weaving calculation by R.Sen Gupta.
5. Weaving Mechanism (Fox).
6. Mechanism/Weaving machine (TALUKDAR)

5. Structure and Properties of Fibres (ETEF602/EC011)

Unit (1): Moisture absorption, heat of absorption (2), differential heat of absorption, integral heat of absorption (2), Quantitative theory of heat moisture absorption (2), Rate of moisture absorption (1)
Total Lectures Required = 7

Unit (2): Mechanical properties of fibres (2), Relation between structure and mechanical properties of fibres (2), Basic mechanical properties (tenacity elongation, modulus, work of rupture (2), Elastic recovery, time effects (2).
Total Lectures Required = 8

Unit (3): Thermal behavior of textile fibres by DSC (2), TGA, thermal mechanical analysis (2), Density gradient column (2), Preparation of density gradient coloumn (2).
Total Lectures Required = 8

Unit (4): Birefringence behavior, dielectric properties (2), fibre friction, fibre friction measurement (2), fibre yarn to surface friction measurement and static charge measurement (5)
Total Lectures Required = 7

Unit (5): Creep behavior (2), Optical properties of fibres (2), concept of moisture absorption by fibres (2), relation between fibre structure and fibre properties (2).
Total Lectures Required = 8
Grand Total of lectures required = 38
Reference Book: -

1. Manufactured fibre technology by V.B. Gupta, V.K. Kothari
2. Physical properties of fibre by J.W.S. Hearle
3. Thermal behavior of material by Turi
4. Modern yarn production by Ray
5. Textile fibres by ATIRA
6. ASTM Standard books
7. Absorberey chartering
8. Polymers by fibre & textiles encyclopedia
9. Advances in fibre source by S.K. Mukhopadhyaya

Post spinning operation (ETCE012/EF604)

Unit (1): Concept of drawing ,drawing unit (3),Effect of drawing on orientation and crystallizations(3),Drawing of melt spun filaments, drawing of as-spun fibres through neck formation(3).
Total of lectures required=9

Unit (2); Heat setting ,nature of heat set(3),Influence of heat setting variables on structures &properties of fibres(3),Temporary, semi permanent and permanent heat setting, heat –setting equipments(3).
Total of lectures required=9

Unit (3): Method of producing bulk-false-twist texturing (3),effect of process variables, time, temperature, twist and tension on properties of false twist textured yarn,(3),edge crimping(3).
Total of lectures required=9

Unit (4): Air-jet texturing process(3),stuffer-box texturing process(3),effect of process variables on yarn properties of air jet & stuffer box textured yarns, Role of water in air jet texturing(4).

Total of lectures required=10

Unit (5): Tow to top conversion, stretch breaking (2),cutting method(1), Sydel stretch breaking(2),Pacific tow to top cutting system(2),Turbo stapler(2)
.**Total of lectures required=9**
Grand total lectures required =46.
Reference book:-
1. Manufactured fibre technology by V.B.Gupta & V.K.Kothari.
2. Synthetic fibre by Vaidhya
3. Textile fibre part-ll by V.K.Kothari.
4. Textured yarn technology by J.W.S.Hearle.

7 *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U.

7TH SEMESTER, B. TECH. MANMADE FIBRE TECHNOLOGY

1. **Open Elective-I: Syllabus as provided by U.P.T.U.**

2. **Open Elective-I: (EOE071-074) Syllabus as provided by U.P.T.U.**

2. **Department Elective-II: Garment Manufacturing Technology (ETCEF021)/ Nonwoven (ETCEF022)**

2.1 Garment Manufacturing Technology (ETCEF021)

Unit (1): Introduction to Garment Manufacturing Technology, Sample Cutting, Fusing Technology, Sewing Technology, Pressing Technology, Finishing and Inspection, Sample Room, Communication, and organization.
Total Lectures required =6

Unit (2): Introduction to Measurement of Fabric Dimensional Properties, Fabric Comfort, Thermal Comfort, Objective Evaluation of Fabric, Low Stress Fabric Properties, Kawabawa System, Fabric Assurance by Sample Testing, Fabric Defects, Fabric Inspection and Feedback to back process.
Total lectures required =8

Unit (3): Introduction to Garment Cutting, Marker Planning, Efficiency of Marker Planning, Methods of Marker Planning and Marker Use, Spreading of the Fabric to form a lay, Spreading Requirements, Methods of Spreading, Fabric Packages, objective of Cuttings, Methods of Cuttings,
Total lectures required =8

Unit (4): Introduction to Seam, Stitch, Stitch Classification, Stitch Structure, Seam Formation, Joining Material, Surface Characteristics, Seam Appearance, Damages (Thermal and Mechanical), Seam Performance, Seam Degradation, Seam Failure and Seam Testings, and Seaming Machine.
Total lectures required =8

Unit (5): Importance of Garment Processing and Finishes, Types of Garments, Processing of Garments and Specialty Garment Finishes.
Total lectures required =6
Grand Total Lectures required = 36

Reference Books:

1. Introduction to Garment Manufacturing Technology By T Ramchandran
2. Garment Manufacturing Technology by By T Ramchandran
3. Practical Clothing Construction Part I & II by Mary Methews

2.2 Non Woven (ETCEF022)

Unit 1: Concept about felts and non woven (3), application of non-woven fabrics (2), fibre-characteristics needed for the manufacture of non-woven (2).
Total =7
Unit 2: Manufacturing passage for the felt preparation (4), web chrematistic and their influence properties (3)
Total =7

Unit 3: Process variable (2), and their effect on properties of non woven (4).
Total = 6

Unit 4: Various system of binding the web (5), and their merits (2).
Total = 7

Unit 5: Flocked fabric (2), Laminates (2), latest development in non-woven industry (3).
Total = 7
Grand total = 34

Reference Book & Text Book –

1. Non Woven – N.N. Banarjee
2. Non woven – NCUTE
3. Fibre to fabric – Corbman Bernard P.

3. Mill Planning & Organization (ETEF701)

Unit-1: Selection of site for a textile mill (2). Preparation of project report (2). Construction of building of a textile mill (2). Types of buildings, single and multistoried buildings. Fire hazards and their control (1). Prevention tours accidents (1)
Total of lectures required=9

Unit-2: Humidification of a textile mill (1). Humidifiers and dehumidifiers (1). Types of humidifiers (2). Different systems of ventilation system used in textile mill (1). System of lighting used in textile mill (2)
Total of lectures required=7

Unit-3: Balancing of machines for spinning and weaving machines or balancing of machines for bleaching (4). Dyeing, Printing and Finishing department. Layout of different machines of spinning, weaving process house (3).
Total of lectures required=7

Unit-4: Costing (1), elements of cost, fixed and variable cost (1). Knowledge of cost cancellation for spinning (2), weaving and processing department. Viability evaluation of a project (1). Break even point (1) Types of staff organization (2).
Total of lectures required=8

Unit-5: Different system wages used in textile industry (2), working of sales and purchase system (2). Material handling used in a textile mill (2). Maintenance of machines and safety devices used in a textile mill (2). Inventory controls (2) Resource Planning.
Total of lectures required=10
Total of lectures required=41

Reference Books

1. Industrial Engineering, Organization & management by Tarachand
2. Industrial Economics & Principle of Management by T.M. Chabra
3. Industrial Economics & Principle of Management by S. K Sharma

4. Fibre Manufacturing & Process Control (EF702)

Unit-1: Melt spinning line-extruder, design features of extruder screws (2) mixing of additives,Continous polymer filter, spin pack and spinneret(2)spinning pump.disassemmly and cleaning of spinnerets, inspection of spinnerets (2) spinning variables and conditions for continuous spinning, special features of high speed spinning(3)
Total of lectures required=9
Unit-2: Quenching system and quench chamber, Different quenching system (2).High speed winder, automatic winder, metering pump (1)spin finish application, chips drying. (1)Effects of variable throughout rate. (2)Consequence of crystallization in chips on fibre spinning (2) structure formation during spinning (1)
Total of lectures required=9

Unit-3: Introduction to solution spinning classification of solution spinning (2)wet spinning, coagulation in wet spinning, effect of coagulation conditions on fibre properties(2) dry spinning, spinning cell for dry spinning (2),Cross-section formation in dry-spinning, spin finish during dry-spinning, coagulation of viscose fibre in coagulation bath(4)
Total of lectures required=10

Unit-4: Dry-jet-wet spinning, coagulation process (2) development of structure & morphology during dry-jet-wet spinning (2) importance of dry-jet-wet spinning (2) process control in dry-jet-wet spinning (2)
Total of lectures required=8

Unit-5: Role of spin-finish (2) introduction to spin finishes components (2) spin finish application. Different techniques of spin finish application, dipping roller method, metered finish system, quench duct lubricating system, spray technique(4)
Total of lectures required=8
Total of lectures required=44

Referenceces:

1-Manufactured fibre technology-V.B.Gupta&V.K.Kothari,Chapmann&Hall
2-Textile Fibre- V.K.Kothari(vol.2) IAFL Publication
3-High speed fibre spinning-A.Ziabickey-john willey
4-Essential fibre chemistry-ME CarterMarcel Dekker mc.N.york
5-Handbook of fibre science &technology-M.Leoineli and M.Pearse

5. Knitting Technology (ETEF- 703)

Unit 1: Difference between knits & woven (1), knitting terms & definition (2), knit technology (2), knitting industries (1)
Total= 6

Unit 2: Classification of weft knitting machine(2), elements of knitting machine(2), technology of loop formation(2), geometry of loop structure(2) relation between yarn count, machine gauge and stitch density(2).
Total =10

Unit 3: Classification of knit-structures(2), loop formation on single jersey, Rib machines and inter look machines(2), knitting notation(2), socks knitting technology (3), warp knitting introduction(2), classification of warp knitting(2).
Total = 13

Unit 4: Special knit fibre and mechanics (4), electronics in knitting (4).
Total =8

Unit 5: Modern developments in knitting industry (3), calculations regarding production gsm, stitch density etc, set-up of knitting mills(3).
Total = 6
Grand Total = 46

Reference and Text Book-

1. Knitting technology – chamberlin
2. Knitting technology – W.J. Spencer
3. International Textile Journel – Knitting
4. Knitting Calculation – Chamberlin
5. Wet Knitting Vol. 1&2 –Published by IIT New Delhi.
6. Knitting – NCUTE

6 *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U.

8TH SEMESTER, B. TECH. MANMADE FIBRE TECHNOLOGY

1. Open Elective-III: Syllabus as provided by UPTU

1. EOE081 Non Conventional Energy Resources/
2. EOE082 Introduction to Soft Computing/
3. EOE083 Product Development/
4. EOE084 Automation and Robotics

2. Departmental Elective III: Technical Textiles (ETCEF 031)/ Theory of Textile Structure (ETEF-032)

2.1 Technical Textiles (ETCEF 031)

Unit-1: Textiles in filtration, Filter fabric construction- Woven, needle felt & knitted filter fabric (2), Finishing treatment of filter fabric (2), Thermal and chemical properties of filter fabric (2), Essential requirements of good filter fabric (2).
Total = 8

Unit-2: Waterproof breathable fabrics- Densely woven fabric, membranes, coatings (3), Membranes- Micro porous membrane, Hydrophilic membrane (2), Coatings-Microporous coating, Hydrophilic coatings (2), Performance of waterproof breathable fabrics (2).
Total =9

Unit 3: Medical textiles, Fibres used (1), Classification of medical textiles – Nonimplantable materials-wound dressings, bandages, plasters etc. (2), Extracorporal devices – Artificial kidney, liver, lung (2), Implantable material- Sutures, Soft tissue implant, Orthopaedic implants, Cardiovascular implants.(2),Healthcare/ hygiene products (1).
Total =8

Unit 4: Smart textiles brief introduction of smart textiles, classification of smart textile(2), passive smart textiles(2) , active smart textiles , very smart or ultra smart textiles (2), application of phase change material in smart textiles(2), brief discussion of smart shirt, smart suit, musical jacket, space suit etc(2).
Total =10

Unit 5: Introduction of Geo Textile (2), Classification of Geo Textiles, Functions of Geo textiles-soil reinforcement, drainage (fluid transmission), filtration, Separation, erosion control/absorption (2), objects of geo textiles, manufacturing of geo textile (3).Essential properties of geo textiles- Mechanical determinants, Hydraulic determinants, durability determinants (2)
Total 7

Grand Total 42

Reference Books:-

1. Hand book of technical textiles – AR Harrocks, SC Anand (Textile Institute, wood head publication).
2. Smart fibre, fabrics and clothing Tao X (Textile Institute, wood head publication).
3. Shears Handbook of Industrial Textiles.

2.2 Theory of Textile Structure (ETEF-032)

Unit (1): Classification of yarn (1), Yarn diameter & density (1), Twist in yarn (1), Geometry of twisted yarn (2), Contraction due to twist (2), Twist and fibre packing in yarn – open packing, hexagonal close packing, redial packing density in yarn (2)
Total lectures required = 09

Unit (2): Twist and strength relationship (1), Theoretical treatment of yarn irregularity (2), fibre extent & fibre migration (3), estimation of strength of blended yarn (3)
Total lectures required = 09

Unit (3): Spinnability of textile fibres (1), Elements of fabric geometry (3), Flexible & rigid threads model (2), Pierce's equation and latter modifications (4)
Total lectures required = 10

Unit (4): Relation of fabric properties to simple geometry (2), Crimp interchange in woven fabric (2), Cover factor (1),
Total lectures required = 05

Unit (5): Theoretical treatment of fabric deformation in tension, bending & shear (3) Cloth setting theories (1)

Reference Books: -

1. Textile yarns – B.C. Goswamoi, J.G. Martindle, F.L. Scardino.
2. Textile structure – J.W.S. Hearle, S.Backer, Grossberg.
3. Pierce Geometry – Textile institute

3. Department Elective IV: Process Control in Spinning (ETEF-041)/ Process Control in Weaving (ETEF042)

Process Control in Spinning (ETEF-041)

Unit-1: Importance and considerations for evolving a system for process control (1) Scope for process control in spinning, Key variable for process control, Establishing norms and standards, Taking corrective action. (Control of Mixing quality and Cost)- Instrumental evaluation of cotton, Control of mixing quality through fibre Characteristics, Simultaneous control of mixing cost and quality, Bale Management (1)

Unit-2: Control of waste and cleaning (1) in Blow room and carding-Determination of trash (1) control and cleaning efficiency, Cleaning and waste in Blow (1) room and card, Norms for waste and (1) cleaning in card & Blow room, Assessment of performance of Blow room (1). Optimizing cleaning at cards (1) Control of comber waste (1) Control of yarn realization (1)

Unit-3: Measurement and analysis of productivity definition of indices of productivity (1), Measurement of productivity (1), Productivity analysis, productivity and profitability (1),

Means to Improve productivity (1) Maximizing machine efficiency in ring spinning (1) Controlling end breakage rate in ring spinning (1).
Reference book (1) process Control in spinning by ATIRA

Unit-4: Control of yarn quality(1)- Count, Strength and their variability(1), Study of with in bobbin and between bobbin count variation(1), Control of variability of lea strength(1), Single yarn strength and elongation(1) Control of yarn unevenness imperfection(1)

Unit-5: Control of yarn faults and package defects (1) - slubs, crackers, spinners doubles (2) bad piecing, slough of and hairiness (1). Machinery Audit- Impact of machine conditions on processing performance and yarn quality (1), test instruments for Machinery Audit (1)

Reference Books:
1. Process Control in Spinning by ATIRA
2. Process Control in Spinning by K.R. Salhotra

Process Control in Weaving (ETEF042)

Unit-1: Scope for process Control (1) Systems of process Control in wvg (2) Setting norms (2) and Schedule of Checks machinery audit (2)

Total lecture required=7

Unit-2: Process Control in wvg (2) optimizing quality of preparation. Control of quality of knot (2) producing good package (1) control of productivity (2)
Total lecture required=7

Unit-3: Process Control in warping (2) minimizing end wreak age in warping (2) quality of warping beam (2) control of productivity (2)
Total lecture required=8

Unit-4: Process Control in sizing (2) Choice of size recipe and size pick-up (2)
Control of size pick-up, Control of yarn stretch (2) quality of sized beams, Control of productivity (2)
Total lectures required=8

Unit-5: Care is use and Selection of healds (2) Care in use and selections of Reads (1) Care in dressing of the beams for reducing incidence of cross ends (2) Causes and remedies of fabric defects (2)
Total lectures required=7
Grand total of lectures required=37

Reference Books
1. Wvg tablets by ATIRA
2. Machine catalogues of various machines manufacture
3. Process control in weaving by ATIRA

4. High Performance Fibres (EF801)

Unit – I; Introduction- Definition, molecular dimensionality mechanical properties, Fibre markets, Hi -Performance Gelspun Polyethylene fibres- Manufacture, fibres characteristics, properties & applications.
Total lectures required=8

Unit- II: Aramids- Introduction, polymer preparation, Spinning, Structure & properties, applications, Fibres based on liquid crystalline polymer (PPTA fibre)
Total lectures required=7

Unit- III: Carbon Fibres- Physical properties, PAN bases Carbon fibres, Pitch based Carbon fibres, Vapour grown Carbon fibres, Applications
Total lectures required= 7

Unit-IV: Glass Fibres- Glass for Fibres, Fibre manufacture, fibre finish, fibre properties& application, optical fibres
Total lectures required=6

Unit –V: Vectran (Melt spun wholly aromatic polyester fibre), Fibre production, properties & application, PBO (Polyphenylene benzobisoxazole) fibres- Fibre production, properties & application. PEEK Fibres -Fibre production, properties & application
Total lectures required=8
Grand total of lectures required= 36

Reference Books
1. High Performance Fibree by J.W>S. Hearle

5. *Human Values & Professional Ethics (Audit course for all Semester) Syllabus as provided by U.P.T.U.